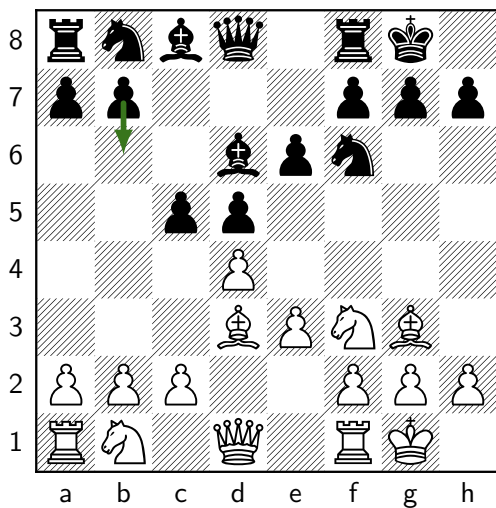


# Melanie-BOTS (CCRL engines) — London System

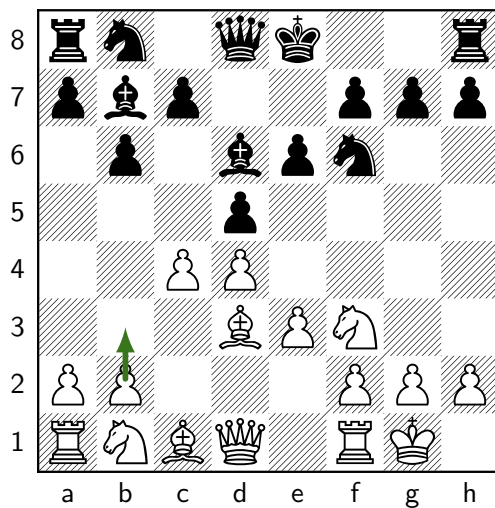
Melanie (Deep\_Vision)

## Abstract

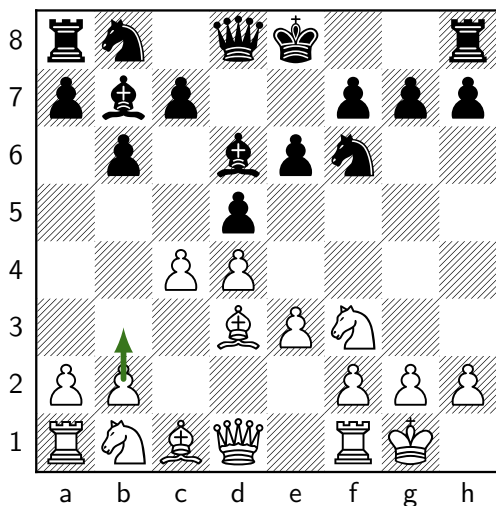
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



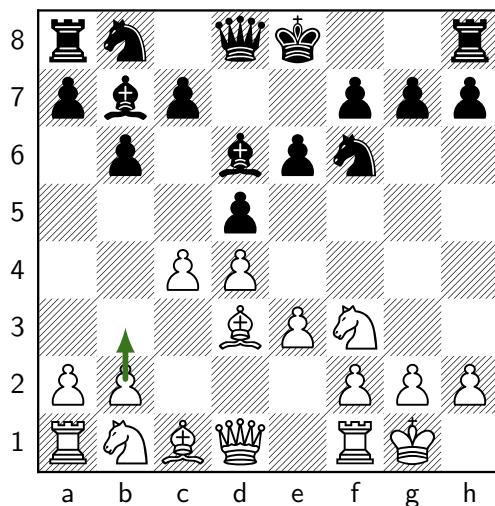
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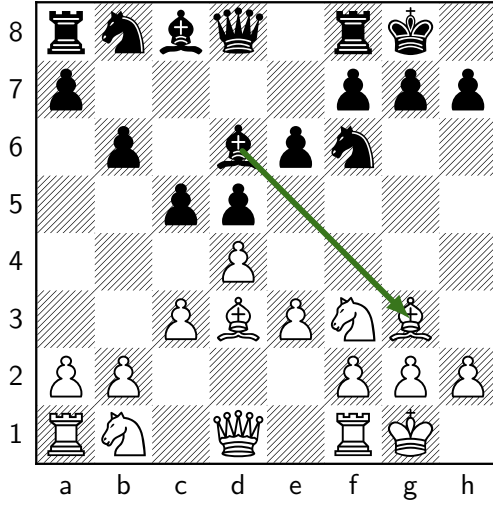


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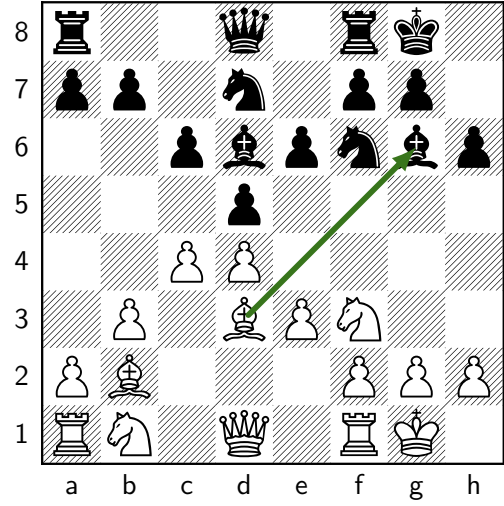


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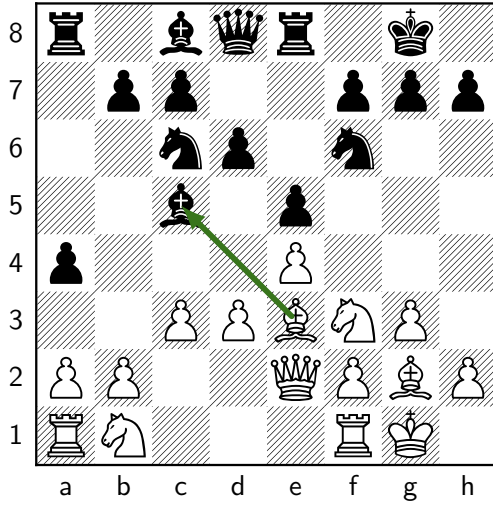
Figure 1: **Move 7, Black: b6.** (a) Our position; Melanie plays **b6** (**b7b6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2898` (ply 12) [structural distance  $d=48$ , QAP obj=54]: the master played **b3**; under  $\mu$  our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *rf1* as their *Nb1*, our *pg2* as their *Qd1*; transferred move **b7b6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2899` (ply 12) [structural distance  $d=48$ , QAP obj=54]: the master played **b3**; under  $\mu$  our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *rf1* as their *Nb1*, our *pg2* as their *Qd1*; transferred move **b7b6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1584` (ply 12) [structural distance  $d=48$ , QAP obj=54]: the master played **b3**; under  $\mu$  our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *rf1* as their *Nb1*, our *pg2* as their *Qd1*; transferred move **b7b6** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **b6** (summed QAP objective 163, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



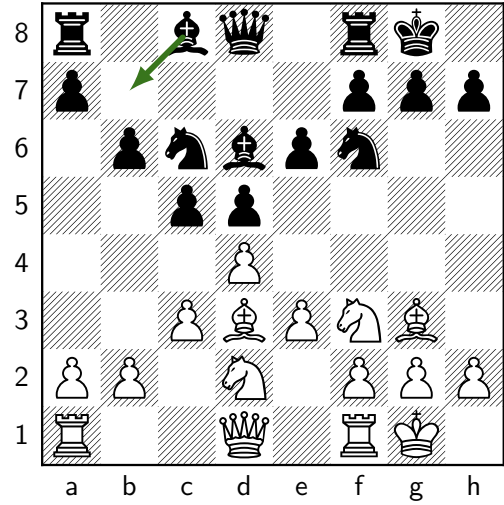
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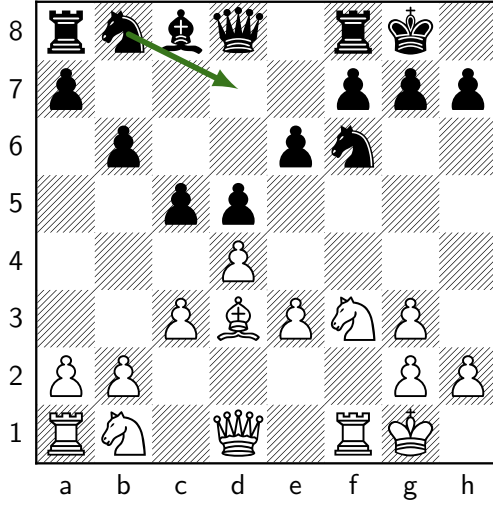


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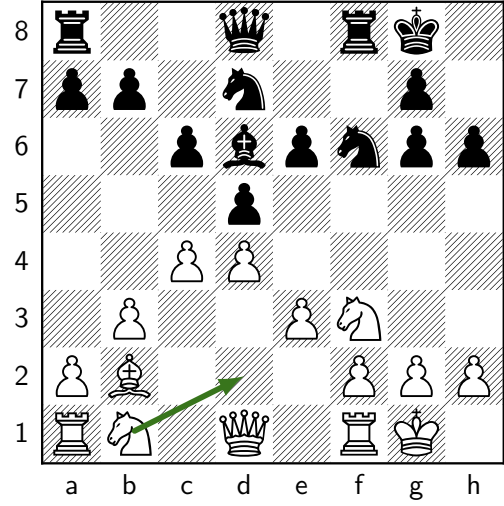


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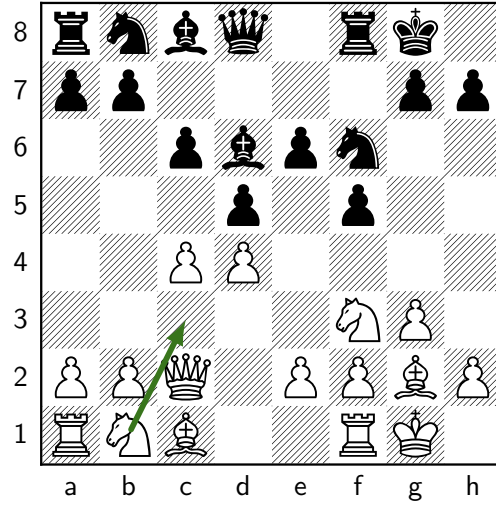
Figure 2: **Move 8, Black: Bxg3.** (a) Our position; Melanie plays Bxg3 (d6g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0001#621 (ply 20) [structural distance  $d=65$ , QAP obj=102]: the master played Bxg6; under  $\mu$  our Bd3 plays the role of that game’s Bd3; cross-type roles: our pa2 as their Ra1, our Ra8 as their Nb1; transferred move d6g3 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#321 (ply 18) [structural distance  $d=62$ , QAP obj=41]: the master played Bxc5; under  $\mu$  our Bd3 plays the role of that game’s Be3; cross-type roles: our nb1 as their Ra1, our Bc8 as their Nb1; transferred move d6g3 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0003#278 (ply 17) [structural distance  $d=61$ , QAP obj=111]: the master played Bb7; under  $\mu$  our Bc1 plays the role of that game’s Bc1; cross-type roles: our Nf6 as their Pb6, our Pc5 as their Nc6; transferred move c8b7. Synthesis: 2 retrieved analogs transferred to Bxg3 (summed QAP objective 143, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

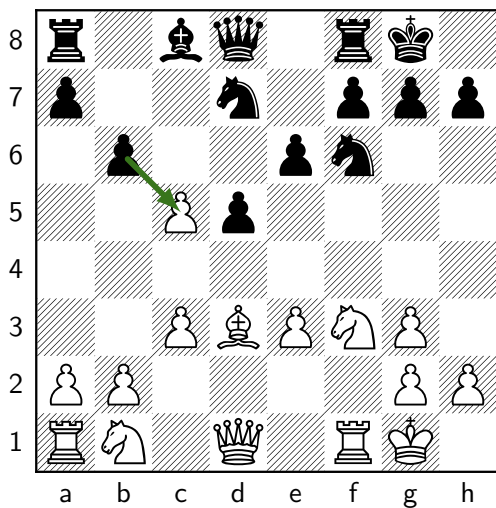


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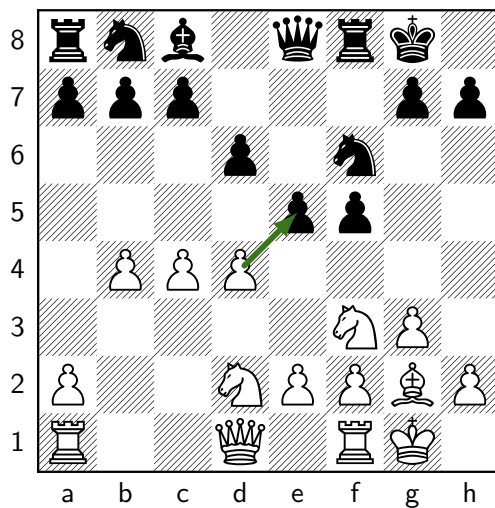


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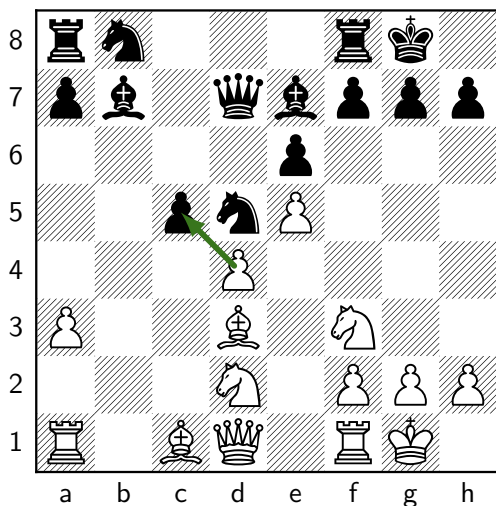
Figure 3: **Move 9, Black: Nbd7.** (a) Our position; Melanie plays Nbd7 (b8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#621` (ply 22) [structural distance  $d=50$ , QAP obj=57]: the master played Nbd2; under  $\mu$  our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *pg3* as their *Ra1*, our *Pd5* as their *Bb2*; transferred move b8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3673` (ply 16) [structural distance  $d=69$ , QAP obj=50]: the master played Nc3; under  $\mu$  our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *Nf6* as their *Bc1*, our *Qd8* as their *Pb2*; transferred move b8a6. Synthesis: 1 retrieved analog transferred to Nbd7 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



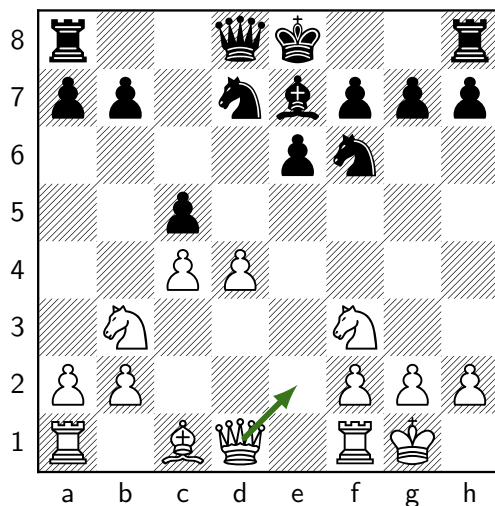
(a)



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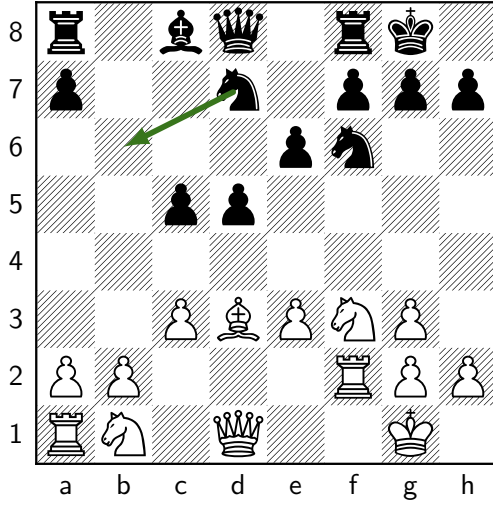


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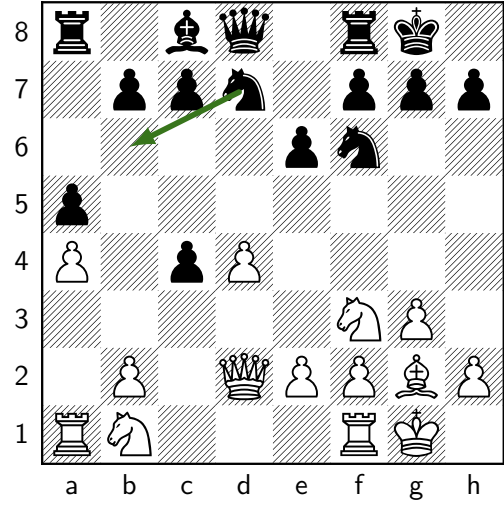


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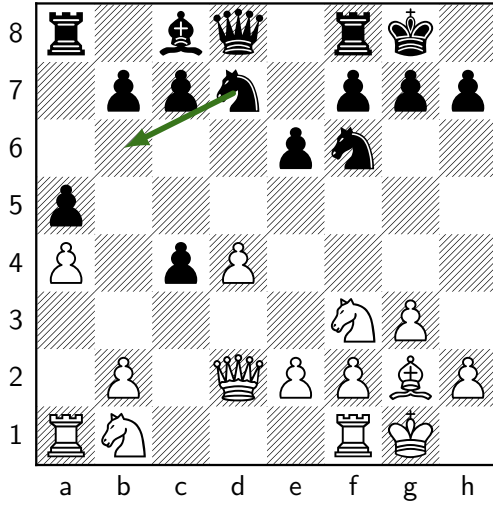
Figure 4: **Move 10, Black: bxc5.** (a) Our position; Melanie plays **bxc5** (**b6c5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1015` (ply 18) [structural distance  $d=78$ , QAP obj=68]: the master played **dxe5**; under  $\mu$  our *Pb3* plays the role of that game’s *Pd4*; cross-type roles: our *Nf6* as their *Ra1*, our *Pd5* as their *Qd1*; transferred move **b6c5** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3334` (ply 26) [structural distance  $d=87$ , QAP obj=48]: the master played **dxn5**; under  $\mu$  our *Pb3* plays the role of that game’s *Pd4*; cross-type roles: our *nf3* as their *Ra1*, our *Pd5* as their *Qd1*; transferred move **b6c5** (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1972` (ply 22) [structural distance  $d=88$ , QAP obj=74]: the master played **Qe2**; under  $\mu$  our *Qd1* plays the role of that game’s *Qd1*; cross-type roles: our *pg3* as their *Ra1*, our *ra1* as their *Bc1*; transferred move **d8c7**. Synthesis: 2 retrieved analogs transferred to **bxc5** (summed QAP objective 116, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



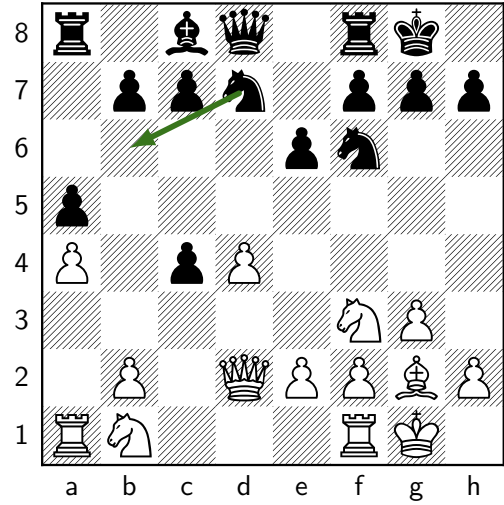
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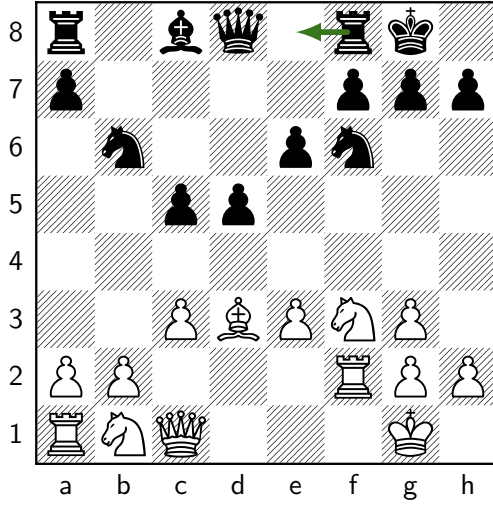


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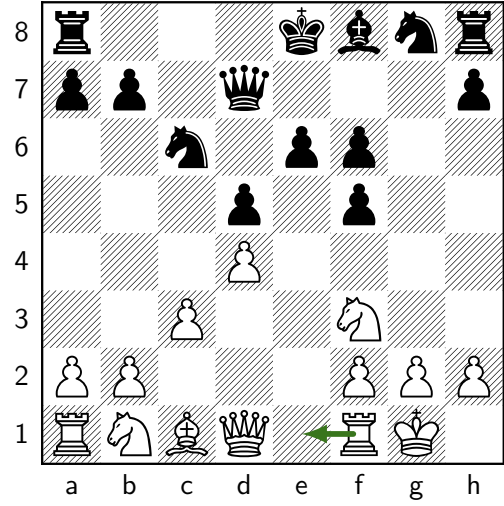


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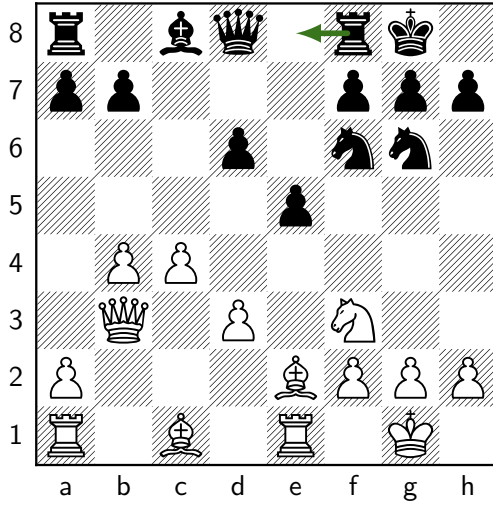
Figure 5: **Move 11, Black: Nb6.** (a) Our position; Melanie plays Nb6 (d7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3375` (ply 19) [structural distance  $d=78$ , QAP obj=21]: the master played Nb6; under  $\mu$  our Nd2 plays the role of that game’s Nd2; cross-type roles: our Nf6 as their Ra8, our Pc5 as their Bc8; transferred move d7b6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3376` (ply 19) [structural distance  $d=78$ , QAP obj=21]: the master played Nb6; under  $\mu$  our Nd2 plays the role of that game’s Nd2; cross-type roles: our Nf6 as their Ra8, our Pc5 as their Bc8; transferred move d7b6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#509` (ply 19) [structural distance  $d=78$ , QAP obj=21]: the master played Nb6; under  $\mu$  our Nd2 plays the role of that game’s Nd2; cross-type roles: our Nf6 as their Ra8, our Pc5 as their Bc8; transferred move d7b6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nb6 (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



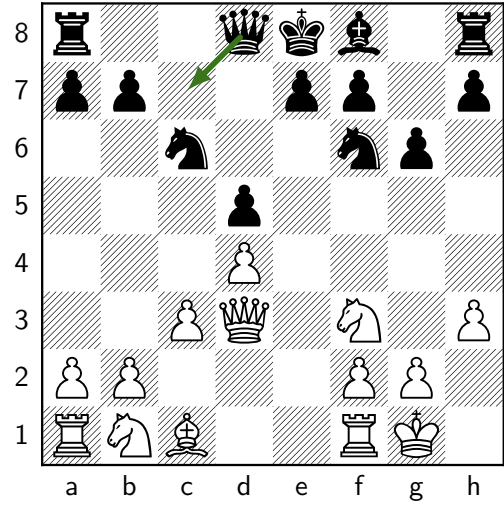
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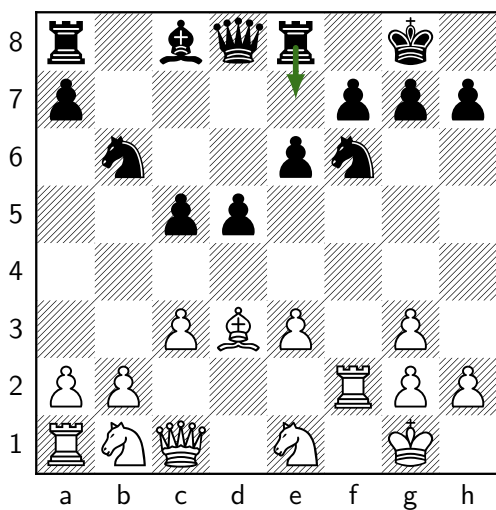


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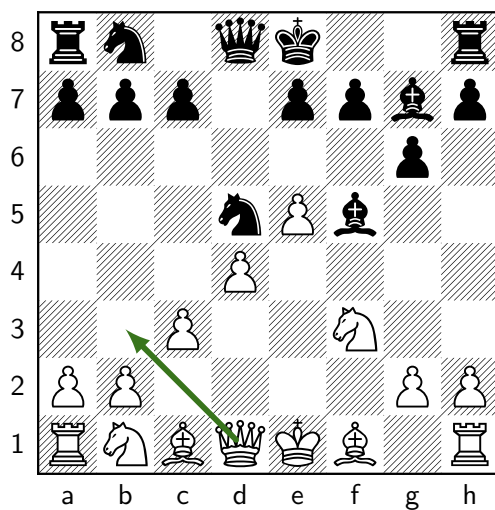


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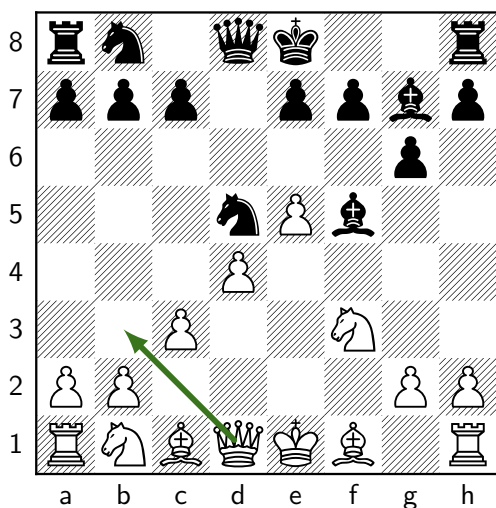
Figure 6: **Move 12, Black: Re8.** (a) Our position; Melanie plays **Re8** (**f8e8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#931` (ply 20) [structural distance  $d=71$ , QAP obj=15]: the master played **Re1**; under  $\mu$  our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *pc3* as their *Ra1*, our *Pd5* as their *Qd1*; transferred move **f8e8** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1210` (ply 27) [structural distance  $d=76$ , QAP obj=15]: the master played **Re8**; under  $\mu$  our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Pc5* as their *Ra8*, our *Pe6* as their *Bc8*; transferred move **f8e8** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1895` (ply 17) [structural distance  $d=75$ , QAP obj=14]: the master played **Qc7**; under  $\mu$  our *Rf1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Ra8*, our *Nf6* as their *Bf8*; transferred move **f8e8** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **Re8** (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



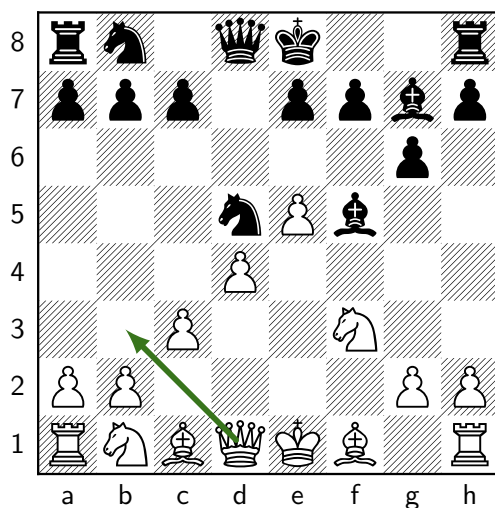
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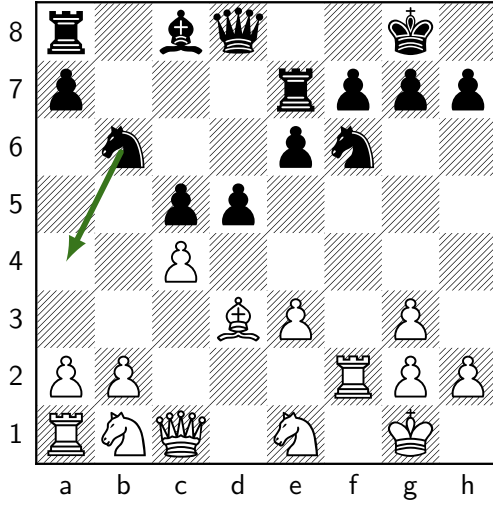
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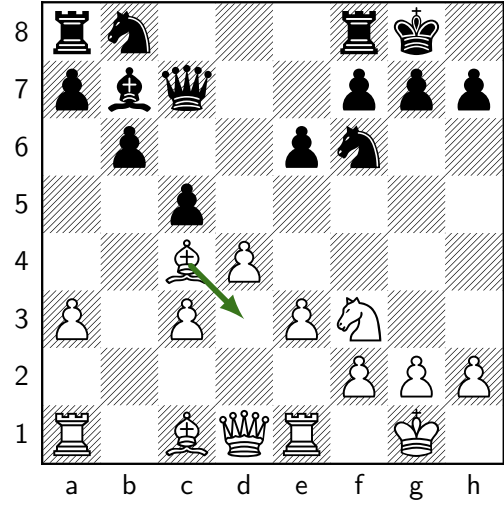
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Figure 7: **Move 13, Black: Re7**. (a) Our position; Melanie plays Re7 (e8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1304` (ply 14) [structural distance  $d=92$ , QAP obj=30]: the master played Qb3; under  $\mu$  our Re1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Pc5 as their Ra1, our Pd5 as their Bc1; transferred move e8e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1305` (ply 14) [structural distance  $d=92$ , QAP obj=30]: the master played Qb3; under  $\mu$  our Re1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Pc5 as their Ra1, our Pd5 as their Bc1; transferred move e8e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#81` (ply 14) [structural distance  $d=92$ , QAP obj=30]: the master played Qb3; under  $\mu$  our Re1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Pc5 as their Ra1, our Pd5 as their Bc1; transferred move e8e7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Re7 (summed QAP objective 120, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

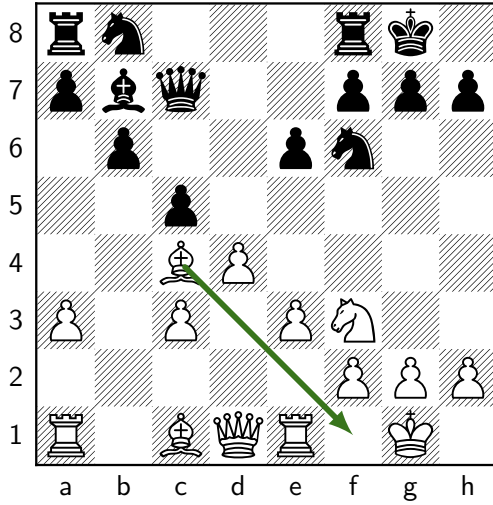




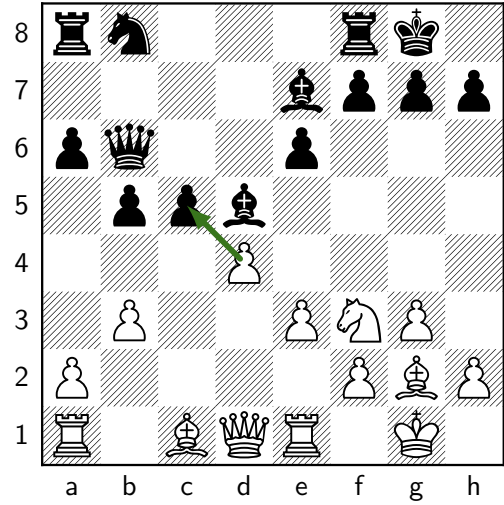
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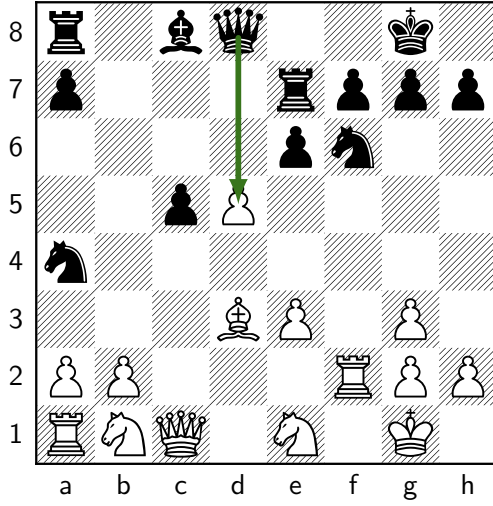


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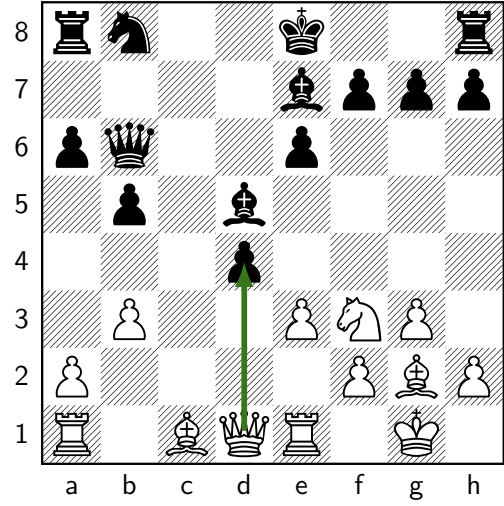


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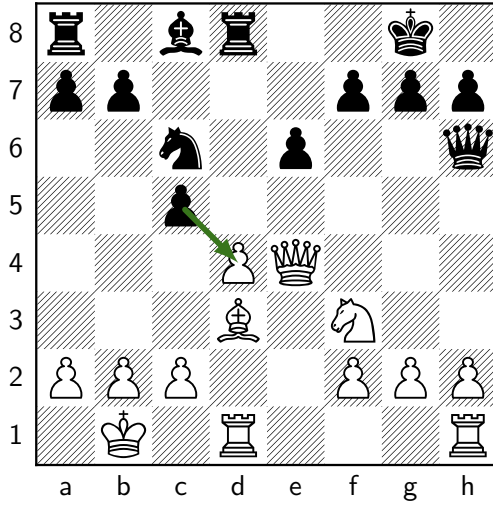
Figure 8: **Move 14, Black: Na4**. (a) Our position; Melanie plays Na4 (b6a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#650` (ply 22) [structural distance  $d=94$ , QAP obj=67]: the master played Bd3; under  $\mu$  our Nb3 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Ra1, our Re7 as their Bc1; transferred move b6a4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#651` (ply 22) [structural distance  $d=94$ , QAP obj=67]: the master played Bf1; under  $\mu$  our Nb3 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Ra1, our Re7 as their Bc1; transferred move b6a4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#228` (ply 24) [structural distance  $d=81$ , QAP obj=45]: the master played dxc5; under  $\mu$  our Pd4 plays the role of that game’s Pd4; cross-type roles: our pg2 as their Ra1, our Re7 as their Bc1; transferred move d5c4. Synthesis: 2 retrieved analogs transferred to Na4 (summed QAP objective 134, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



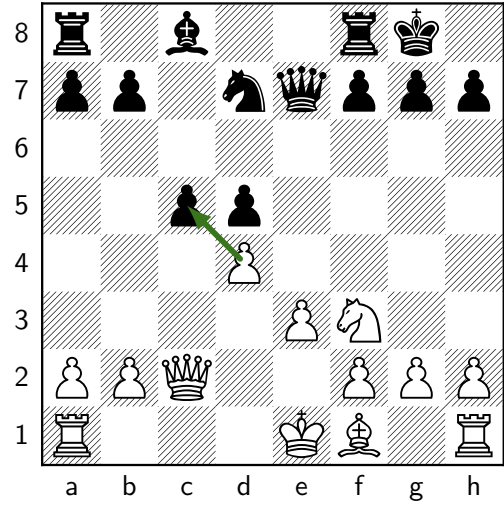
(a)



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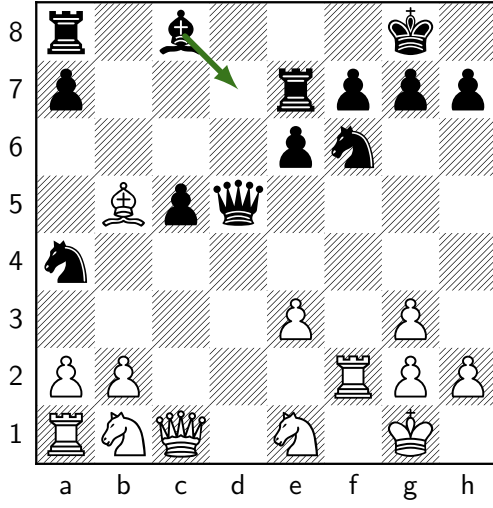


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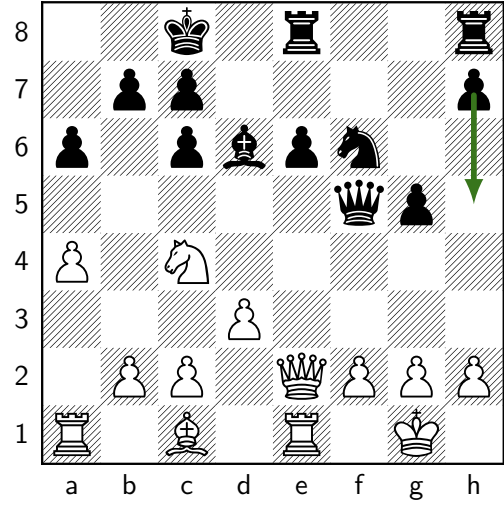


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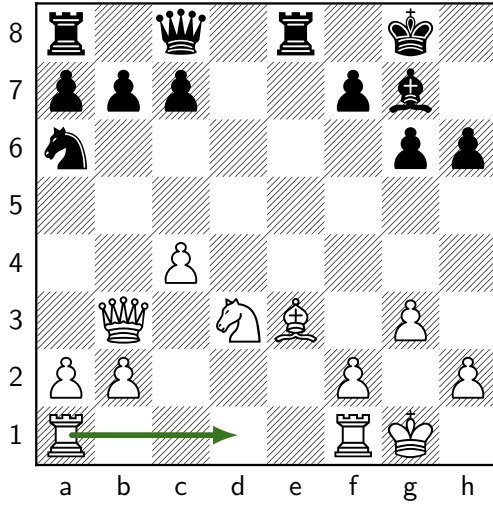
Figure 9: **Move 15, Black: Qxd5.** (a) Our position; Melanie plays Qxd5 (d8d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1444` (ply 24) [structural distance  $d=82$ , QAP obj=149]: the master played Qxd4; under  $\mu$  our Qd1 plays the role of that game’s Qd1; cross-type roles: our nb1 as their Ra1, our ne1 as their Pa2; transferred move d8d5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#321` (ply 25) [structural distance  $d=93$ , QAP obj=127]: the master played cxd4; under  $\mu$  our Pe3 plays the role of that game’s Pc4; cross-type roles: our Re7 as their Bc8, our Qd8 as their Rd8; transferred move e6d5. (d) Retrieved from `ccrl_top_p0001#1785` (ply 20) [structural distance  $d=92$ , QAP obj=121]: the master played dxc5; under  $\mu$  our Pe3 plays the role of that game’s Pd4; cross-type roles: our pg3 as their Ra1, our Bc8 as their Rh1; transferred move e6d5. Synthesis: 1 retrieved analog transferred to Qxd5 (summed QAP objective 149, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



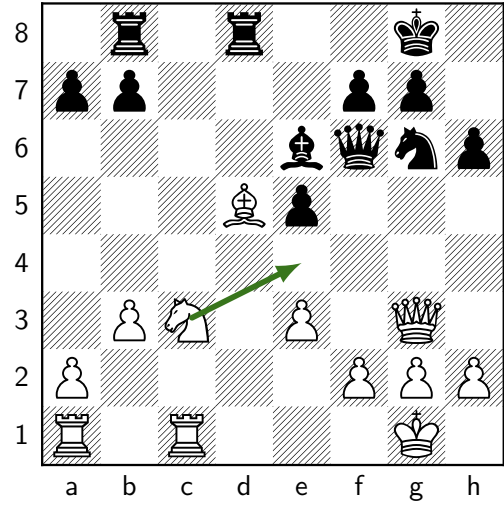
(a)



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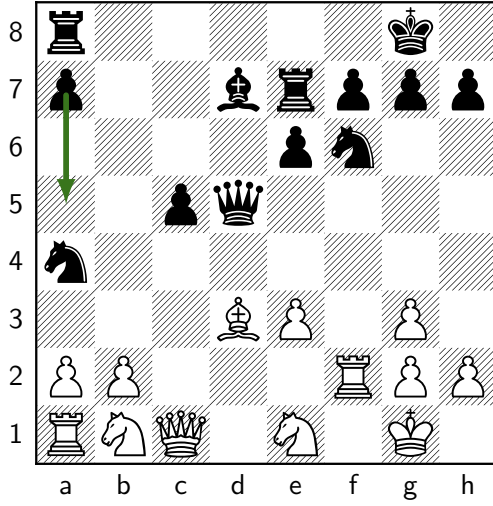


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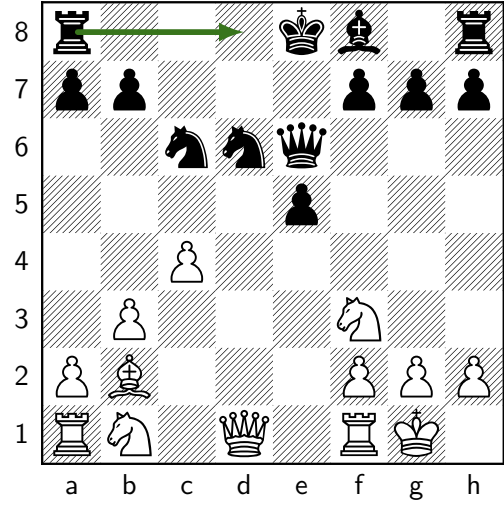


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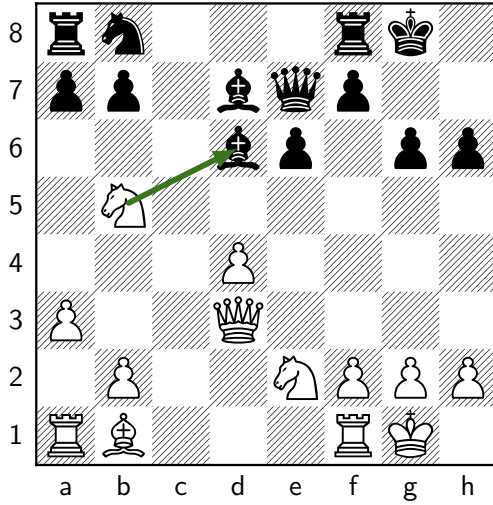
Figure 10: **Move 16, Black: Bd7.** (a) Our position; Melanie plays Bd7 (c8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3598` (ply 29) [structural distance  $d=108$ , QAP obj=50]: the master played h5; under  $\mu$  our *Re2* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *Nf6* as their *Re8*, our *Bc8* as their *Rh8*; transferred move e7c7. (c) Retrieved from `ccrl_top_p0000#1771` (ply 34) [structural distance  $d=104$ , QAP obj=48]: the master played Rad1; under  $\mu$  our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Nf6* as their *Rf1*, our *ne1* as their *Ph2*; transferred move a8b8. (d) Retrieved from `ccrl_top_p0000#21` (ply 44) [structural distance  $d=108$ , QAP obj=39]: the master played Ne4; under  $\mu$  our *Ph2* plays the role of that game’s *Nc3* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra1*, our *rf2* as their *Pa2*; transferred move h7h5. Synthesis: no retrieved analog transferred to Bd7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



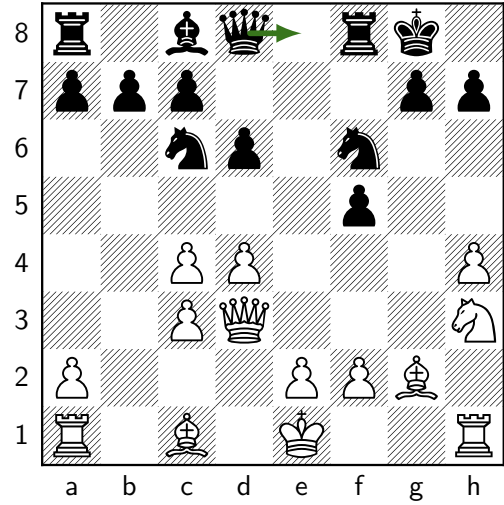
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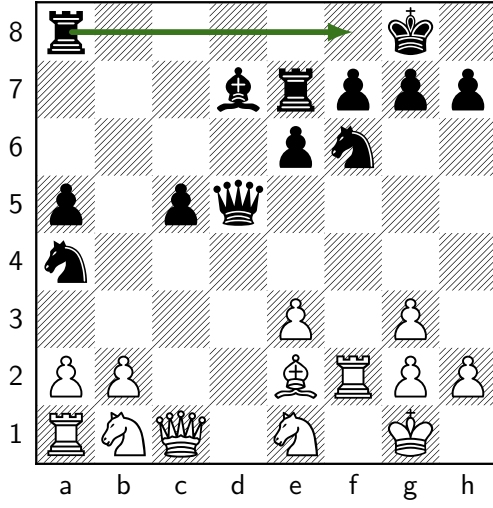


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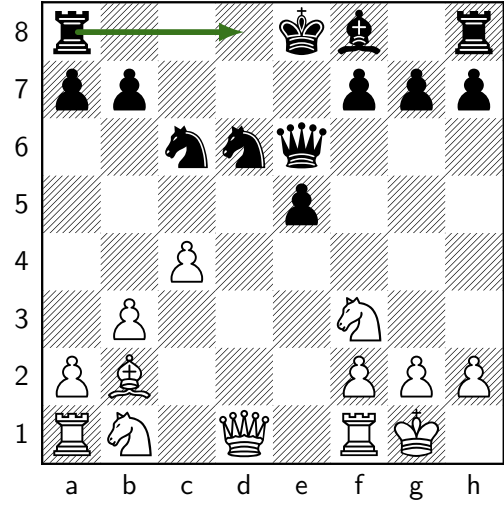


(d)

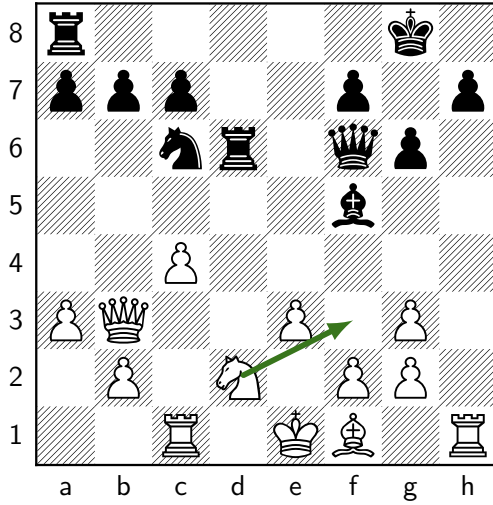
Figure 11: **Move 17, Black: a5**. (a) Our position; Melanie plays a5 (a7a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3661` (ply 21) [structural distance  $d=92$ , QAP obj=32]: the master played `Rd8`; under  $\mu$  our `Pa2` plays the role of that game’s `Ra1` (a **cross-type** correspondence); cross-type roles: our `ra1` as their `Bf8`, our `Bd7` as their `Pa7`; transferred move `a7a5` (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1077` (ply 30) [structural distance  $d=92$ , QAP obj=32]: the master played `Nxd6`; under  $\mu$  our `Qd4` plays the role of that game’s `Nb5` (a **cross-type** correspondence); cross-type roles: our `Pf7` as their `Ra1`, our `Pe6` as their `Rf1`; transferred move `d5d3`. (d) Retrieved from `ccrl_top_p0000#3749` (ply 19) [structural distance  $d=97$ , QAP obj=31]: the master played `Qe8`; under  $\mu$  our `Re2` plays the role of that game’s `Qd1` (a **cross-type** correspondence); cross-type roles: our `qc1` as their `Ra8`, our `ne1` as their `Bc8`; transferred move `e7e8`. Synthesis: 1 retrieved analog transferred to a5 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



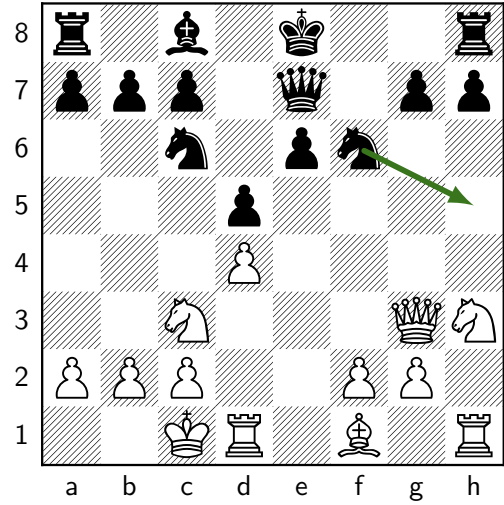
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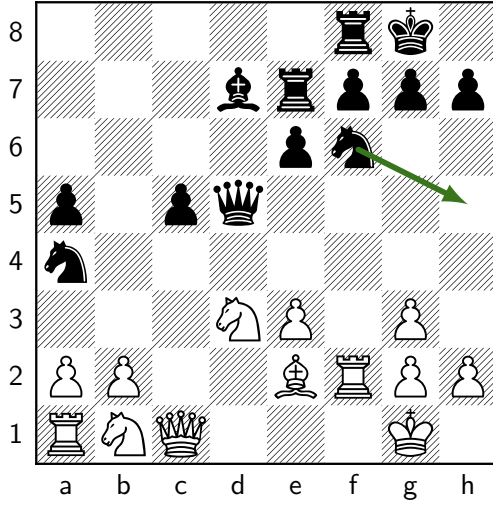


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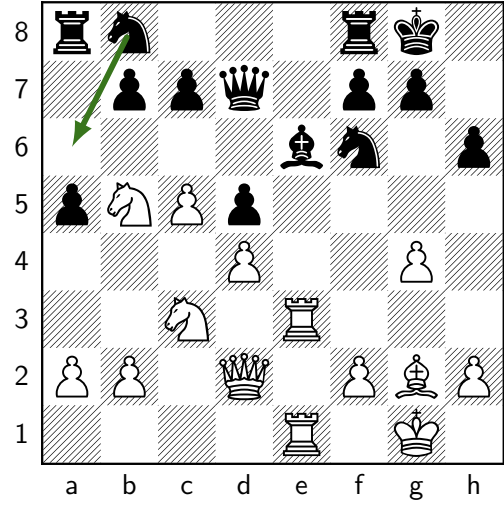


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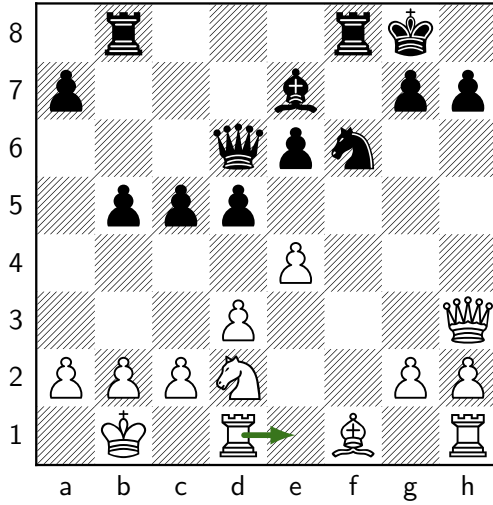
Figure 12: **Move 18, Black: Rf8**. (a) Our position; Melanie plays Rf8 (a8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3661` (ply 21) [structural distance  $d=84$ , QAP obj=46]: the master played Rd8; under  $\mu$  our Ra1 plays the role of that game’s Ra1; cross-type roles: our Pf7 as their Bf8, our Bd7 as their Pa7; transferred move a8f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3378` (ply 30) [structural distance  $d=103$ , QAP obj=41]: the master played Nf3; under  $\mu$  our Nf3 plays the role of that game’s Nd2; cross-type roles: our Ph7 as their Bf1, our Na4 as their Rh1; transferred move f6e4. (d) Retrieved from `ccrl_top_p0002#1464` (ply 23) [structural distance  $d=105$ , QAP obj=40]: the master played Nh5; under  $\mu$  our Re2 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Ra8, our ra1 as their Bc8; transferred move e7e8. Synthesis: 1 retrieved analog transferred to Rf8 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



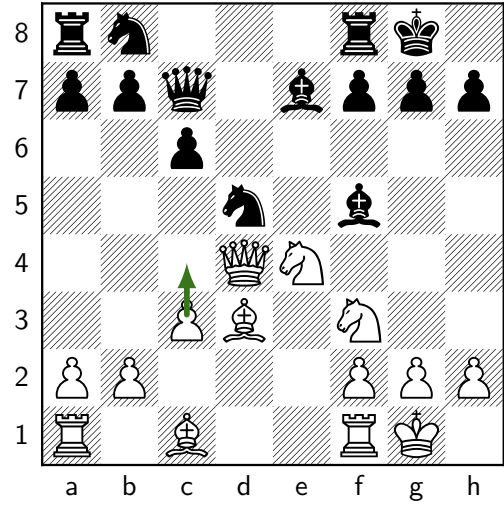
(a)



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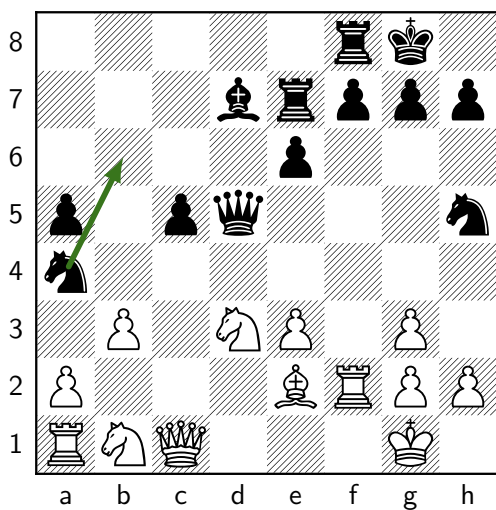


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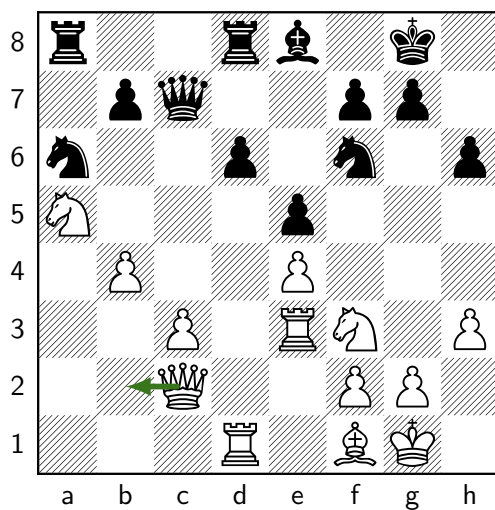


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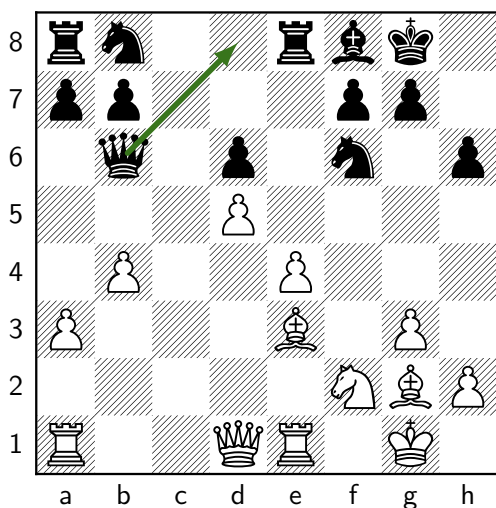
Figure 13: **Move 19, Black: Nh5**. (a) Our position; Melanie plays Nh5 (f6h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#31` (ply 35) [structural distance  $d=93$ , QAP obj=72]: the master played Na6; under  $\mu$  our Nf3 plays the role of that game’s Nb1; cross-type roles: our Pg7 as their Rf8, our nb1 as their Pb7; transferred move f6h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#696` (ply 26) [structural distance  $d=92$ , QAP obj=31]: the master played Re1; under  $\mu$  our Nf3 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our pa2 as their Bf1, our Bd7 as their Rh1; transferred move f6h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1345` (ply 24) [structural distance  $d=75$ , QAP obj=40]: the master played c4; under  $\mu$  our Pc4 plays the role of that game’s Pc3; cross-type roles: our Na4 as their Pa2, our be2 as their Pb2; transferred move c5c4. Synthesis: 2 retrieved analogs transferred to Nh5 (summed QAP objective 103, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



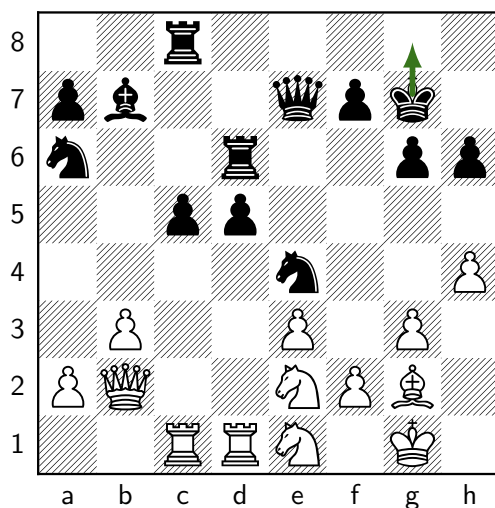
(a)



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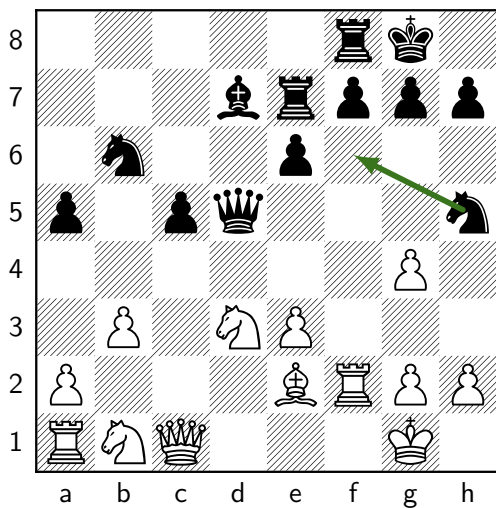
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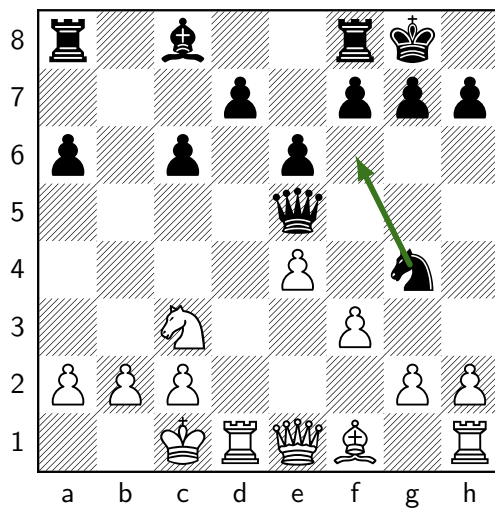
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Figure 14: **Move 20, Black: Nb6.** (a) Our position; Melanie plays Nb6 (a4b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#915` (ply 50) [structural distance  $d=102$ , QAP obj=57]: the master played Qb2; under  $\mu$  our Na5 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Rf8 as their Pg2, our Ph7 as their Re3; transferred move a4b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#138` (ply 35) [structural distance  $d=102$ , QAP obj=39]: the master played Qd8; under  $\mu$  our Na5 plays the role of that game’s Qb3 (a **cross-type** correspondence); cross-type roles: our pg3 as their Ra8, our Bd7 as their Nb8; transferred move a4b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2604` (ply 49) [structural distance  $d=98$ , QAP obj=52]: the master played Kg8; under  $\mu$  our Kg1 plays the role of that game’s Kg2; cross-type roles: our qc1 as their Bb7, our Nh5 as their Qe7; transferred move g8h8. Synthesis: 2 retrieved analogs transferred to Nb6 (summed QAP objective 96, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

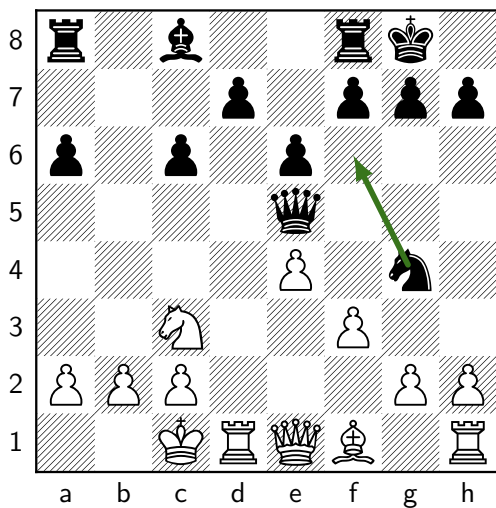




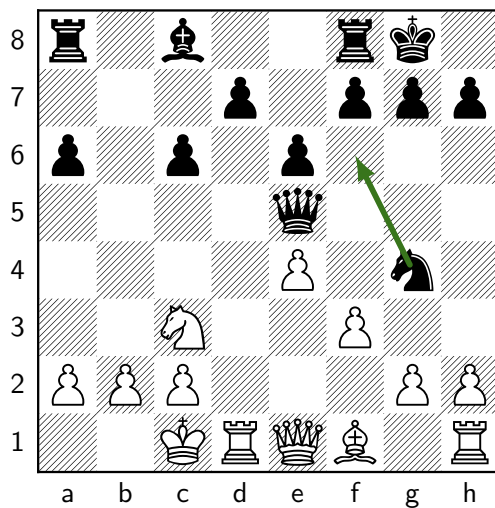
(a)



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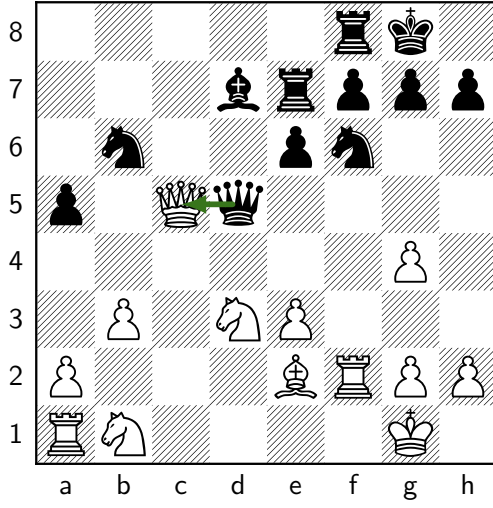
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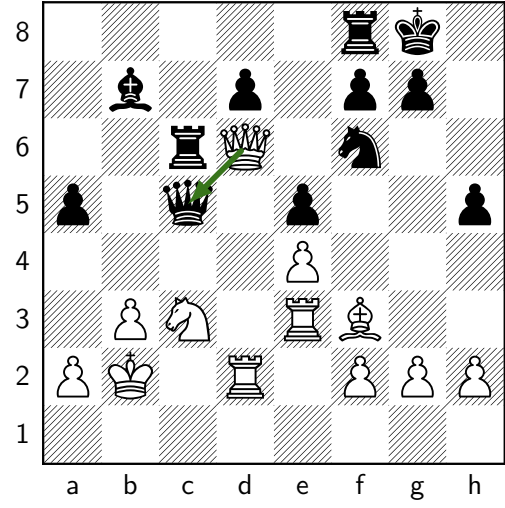
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Figure 15: **Move 21, Black: Nf6**. (a) Our position; Melanie plays Nf6 (h5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0001#2078 (ply 31) [structural distance  $d=79$ , QAP obj=37]: the master played Nf6; under  $\mu$  our Nh4 plays the role of that game’s Ng5; cross-type roles: our Bd7 as their Ra8, our Pf7 as their Bc8; transferred move h5f6 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0002#716 (ply 31) [structural distance  $d=79$ , QAP obj=37]: the master played Nf6; under  $\mu$  our Nh4 plays the role of that game’s Ng5; cross-type roles: our Bd7 as their Ra8, our Pf7 as their Bc8; transferred move h5f6 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#2077 (ply 27) [structural distance  $d=79$ , QAP obj=37]: the master played Nf6; under  $\mu$  our Nh4 plays the role of that game’s Ng5; cross-type roles: our Bd7 as their Ra8, our Pf7 as their Bc8; transferred move h5f6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Nf6 (summed QAP objective 181, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

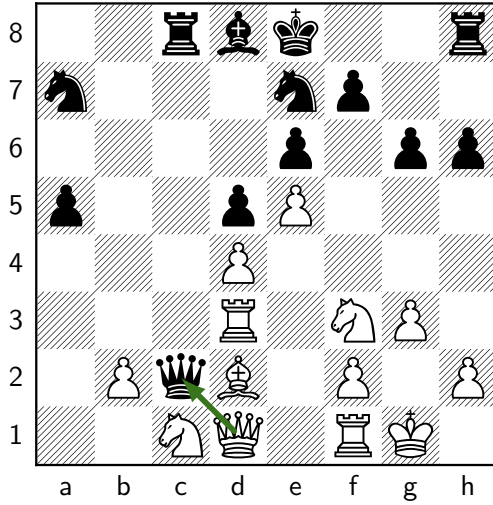




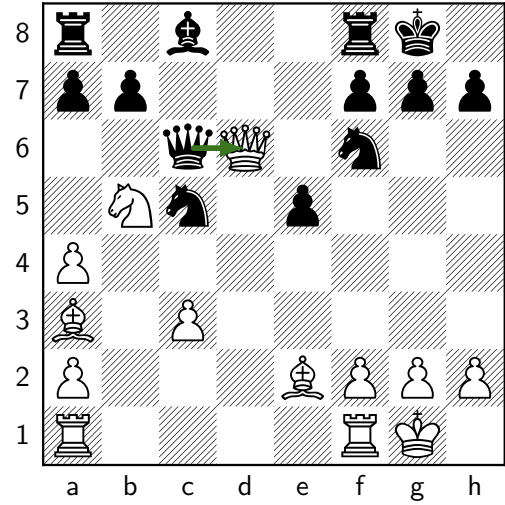
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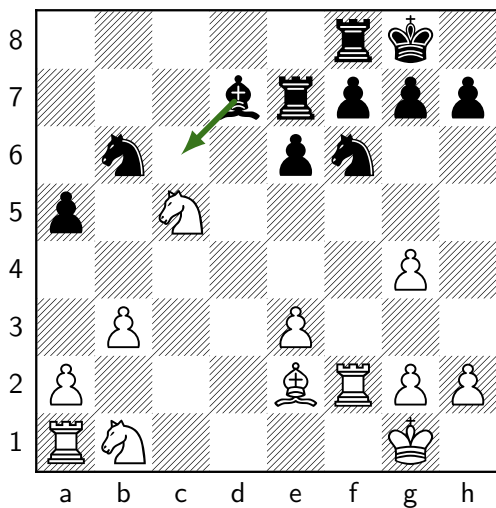


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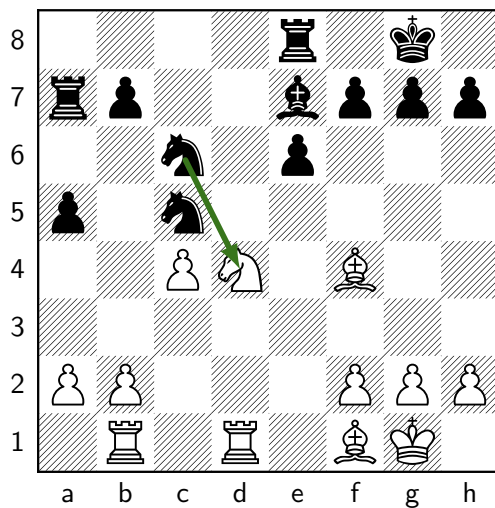


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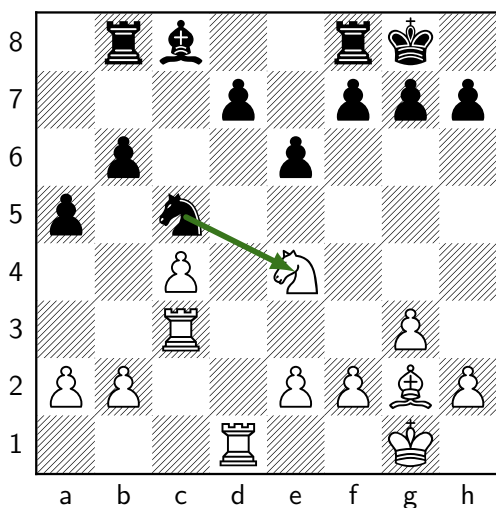
Figure 16: **Move 22, Black: Qxc5**. (a) Our position; Melanie plays Qxc5 (d5c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1083` (ply 46) [structural distance  $d=134$ , QAP obj=103]: the master played Qxc5; under  $\mu$  our Qd4 plays the role of that game’s Qd6; cross-type roles: our Nf6 as their Rd2, our Re7 as their Pg2; transferred move d5c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3828` (ply 48) [structural distance  $d=130$ , QAP obj=92]: the master played Qxc2; under  $\mu$  our Qd4 plays the role of that game’s Qd1; cross-type roles: our Re7 as their Bd2, our rf2 as their Ph2; transferred move d5c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2236` (ply 29) [structural distance  $d=90$ , QAP obj=78]: the master played Qxd6; under  $\mu$  our Qd4 plays the role of that game’s Qc3; cross-type roles: our pg4 as their Ra8, our pe3 as their Bc8; transferred move d5c5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qxc5 (summed QAP objective 365, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



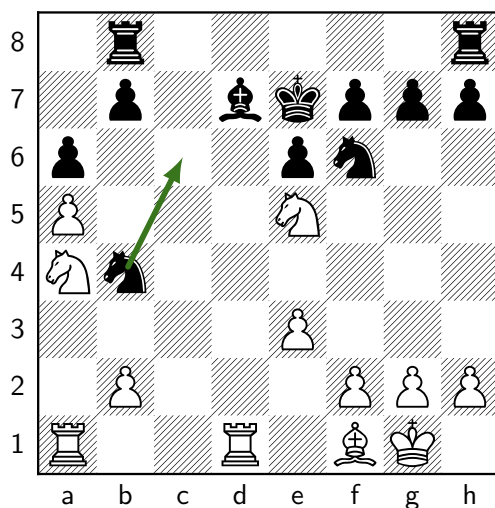
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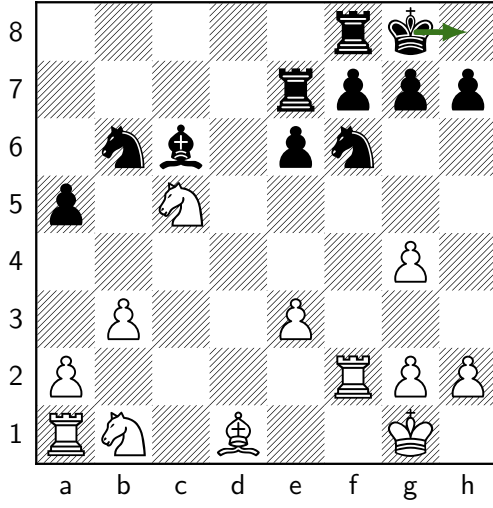


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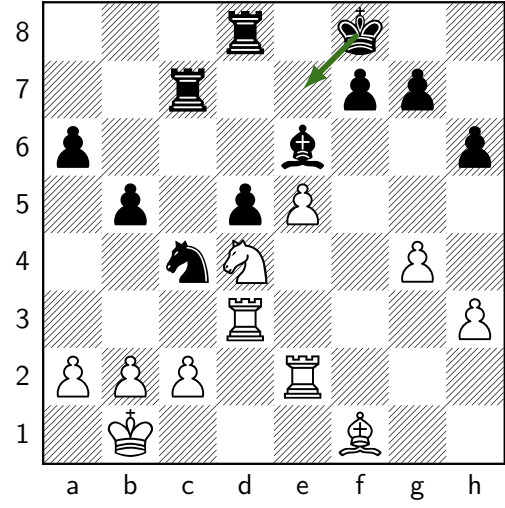


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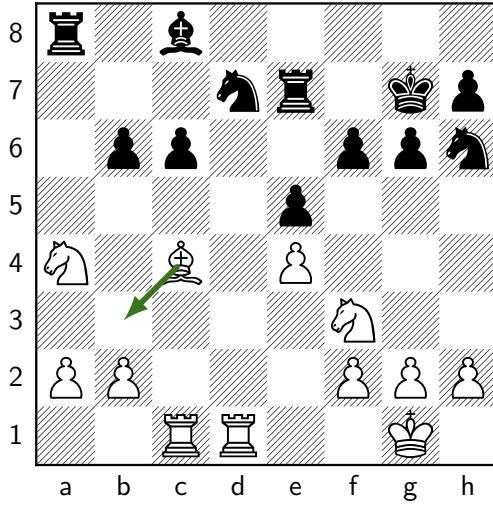
Figure 17: **Move 23, Black: Bc6.** (a) Our position; Melanie plays Bc6 (d7c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1124` (ply 43) [structural distance  $d=54$ , QAP obj=34]: the master played Nxd4; under  $\mu$  our Bd2 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Nf6 as their Re8, our nb1 as their Ra7; transferred move d7c6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1070` (ply 31) [structural distance  $d=60$ , QAP obj=23]: the master played Nxe4; under  $\mu$  our Bd2 plays the role of that game’s Nc4 (a **cross-type** correspondence); cross-type roles: our pb3 as their Rb8, our ra1 as their Bc8; transferred move d7c6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1676` (ply 33) [structural distance  $d=63$ , QAP obj=39]: the master played Nc6; under  $\mu$  our Nb3 plays the role of that game’s Nb5; cross-type roles: our Ph7 as their Rb8, our Re7 as their Pb7; transferred move b6d5. Synthesis: 2 retrieved analogs transferred to Bc6 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



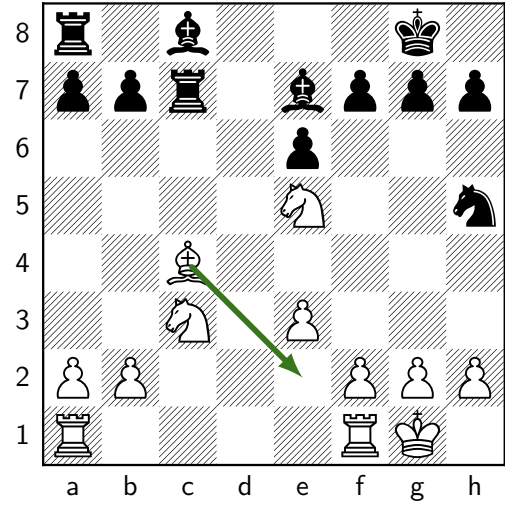
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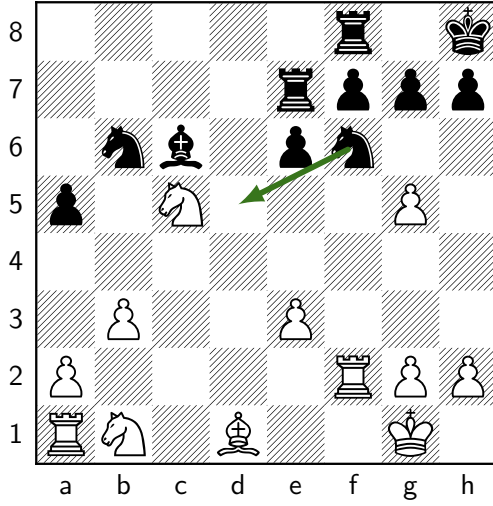


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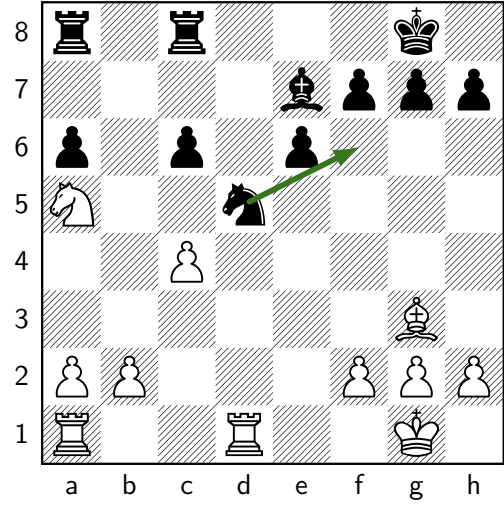


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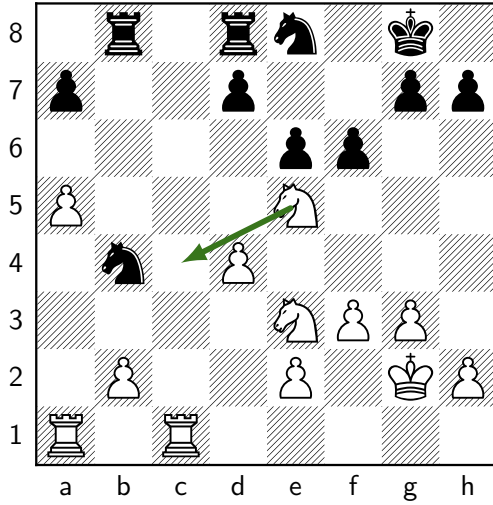
Figure 18: **Move 24, Black: Kh8**. (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1733` (ply 55) [structural distance  $d=72$ , QAP obj=43]: the master played Ke7; under  $\mu$  our Kg1 plays the role of that game’s Kf1; cross-type roles: our Pf7 as their Rc7, our Nf6 as their Be6; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2174` (ply 40) [structural distance  $d=75$ , QAP obj=67]: the master played Bb3; under  $\mu$  our Pa4 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Rc1, our Nf6 as their Rd1; transferred move a5a4. (d) Retrieved from `ccr1_top_p0001#3902` (ply 32) [structural distance  $d=72$ , QAP obj=52]: the master played Be2; under  $\mu$  our Nf3 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our pa2 as their Ra1, our nb1 as their Pf2; transferred move f6h5. Synthesis: 1 retrieved analog transferred to Kh8 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



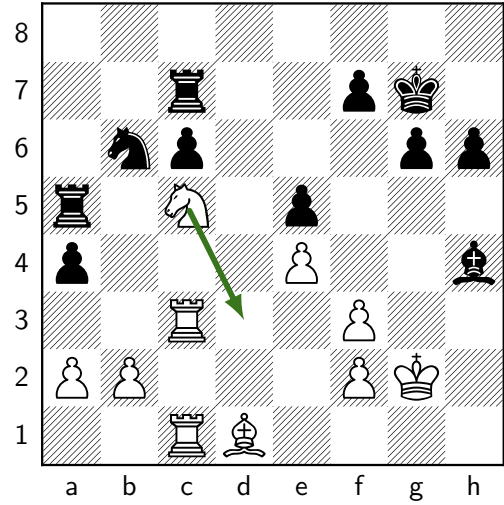
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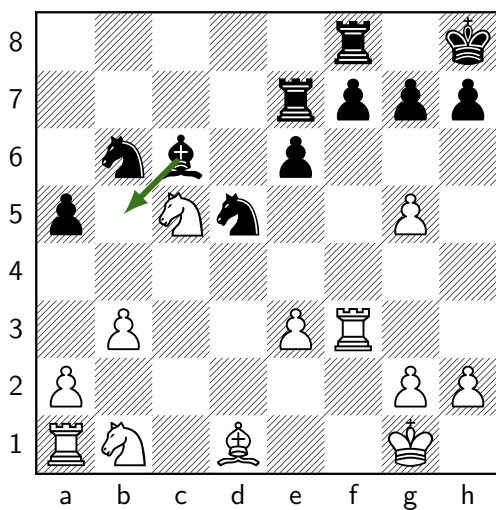


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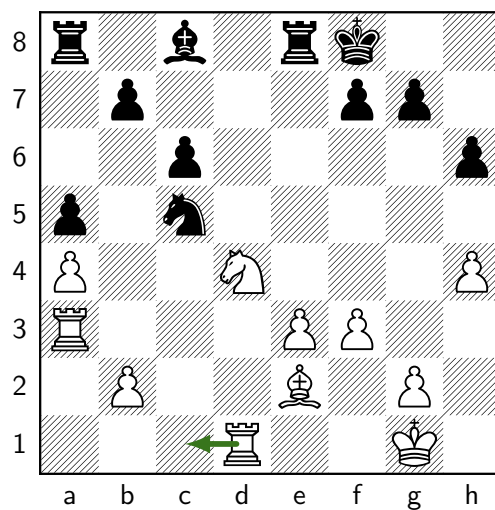


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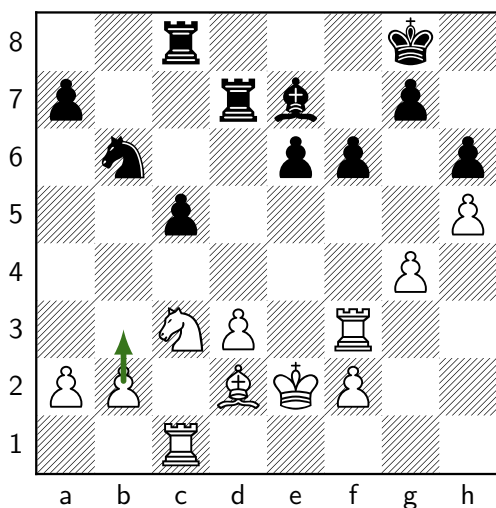
Figure 19: **Move 25, Black: Nfd5.** (a) Our position; Melanie plays Nfd5 (f6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#907` (ply 41) [structural distance  $d=57$ , QAP obj=32]: the master played Nf6; under  $\mu$  our Nf3 plays the role of that game’s Nd4; cross-type roles: our Bc6 as their Ra8, our Ph7 as their Be7; transferred move f6d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#177` (ply 50) [structural distance  $d=60$ , QAP obj=22]: the master played N5c4; under  $\mu$  our Nf3 plays the role of that game’s Ne5; cross-type roles: our Bc6 as their Rc1, our nb1 as their Pe2; transferred move f6d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2134` (ply 52) [structural distance  $d=60$ , QAP obj=12]: the master played Nd3; under  $\mu$  our Nf3 plays the role of that game’s Nc5; cross-type roles: our ph2 as their Rc1, our Nb6 as their Bd1; transferred move f6d5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nfd5 (summed QAP objective 66, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



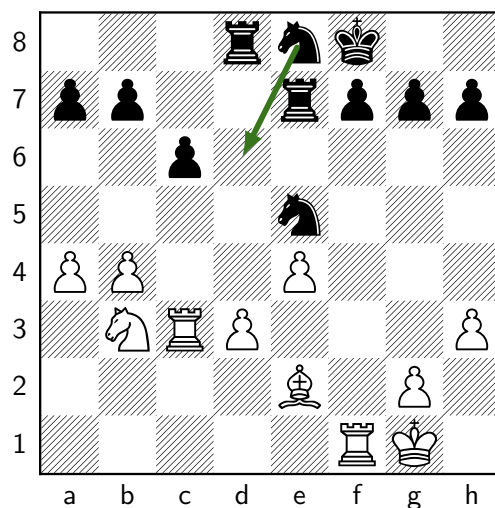
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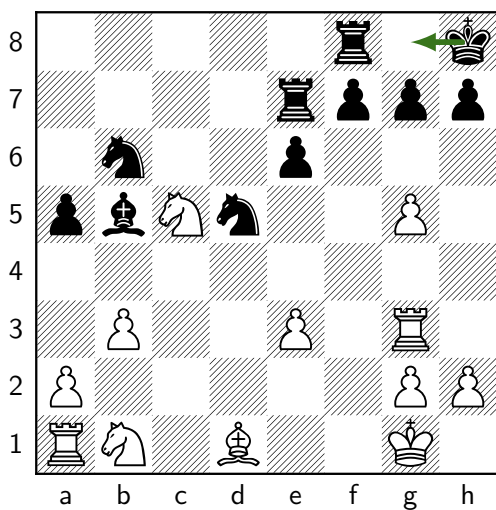


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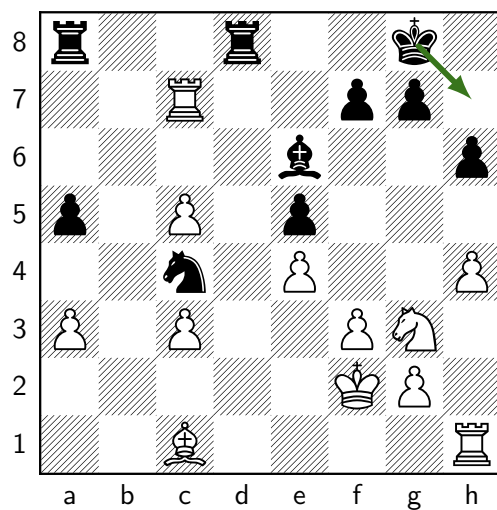


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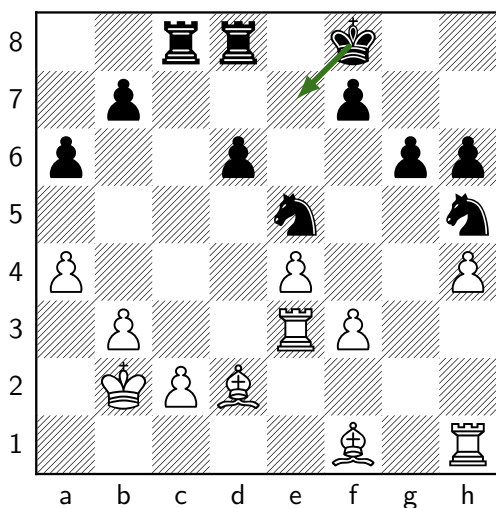
Figure 20: **Move 26, Black: Bb5**. (a) Our position; Melanie plays Bb5 (c6b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#876` (ply 42) [structural distance  $d=49$ , QAP obj=69]: the master played Rc1; under  $\mu$  our Bc3 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our ra1 as their Pb2, our Nb6 as their Be2; transferred move c6b5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#765` (ply 46) [structural distance  $d=59$ , QAP obj=56]: the master played b3; under  $\mu$  our Bc3 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Rc1, our Nd5 as their Bd2; transferred move c6b5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3038` (ply 47) [structural distance  $d=59$ , QAP obj=58]: the master played Nd6; under  $\mu$  our Pg2 plays the role of that game’s Ne1 (a **cross-type** correspondence); cross-type roles: our nc5 as their Rd8, our Rf8 as their Pa7; transferred move g7g6. Synthesis: 2 retrieved analogs transferred to Bb5 (summed QAP objective 125, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



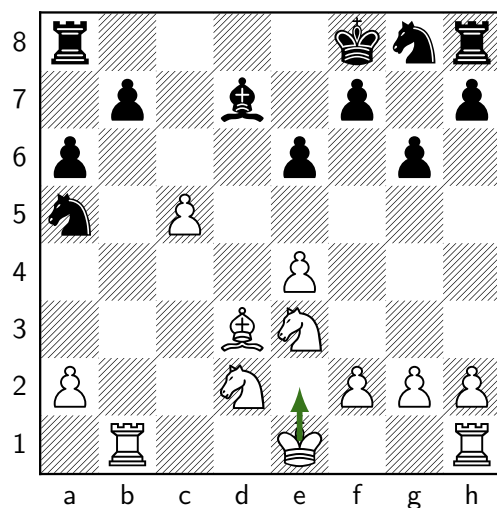
(a)



(b)



(c)



(d)

Figure 21: **Move 27, Black: Kg8.** (a) Our position; Melanie plays Kg8 (h8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1791` (ply 47) [structural distance  $d=65$ , QAP obj=36]: the master played Kh7; under  $\mu$  our Kh1 plays the role of that game’s Kg1; cross-type roles: our Nd5 as their rc7, our Nb6 as their pc5; transferred move h8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2631` (ply 39) [structural distance  $d=67$ , QAP obj=31]: the master played Ke7; under  $\mu$  our Kh1 plays the role of that game’s Kf1; cross-type roles: our Bb5 as their Pg6, our Re7 as their Nh5; transferred move h8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3787` (ply 36) [structural distance  $d=56$ , QAP obj=23]: the master played Ke2; under  $\mu$  our Kh1 plays the role of that game’s Ke1; cross-type roles: our Nd5 as their Rb1, our Pf7 as their Rh1; transferred move h8g8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg8 (summed QAP objective 89, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

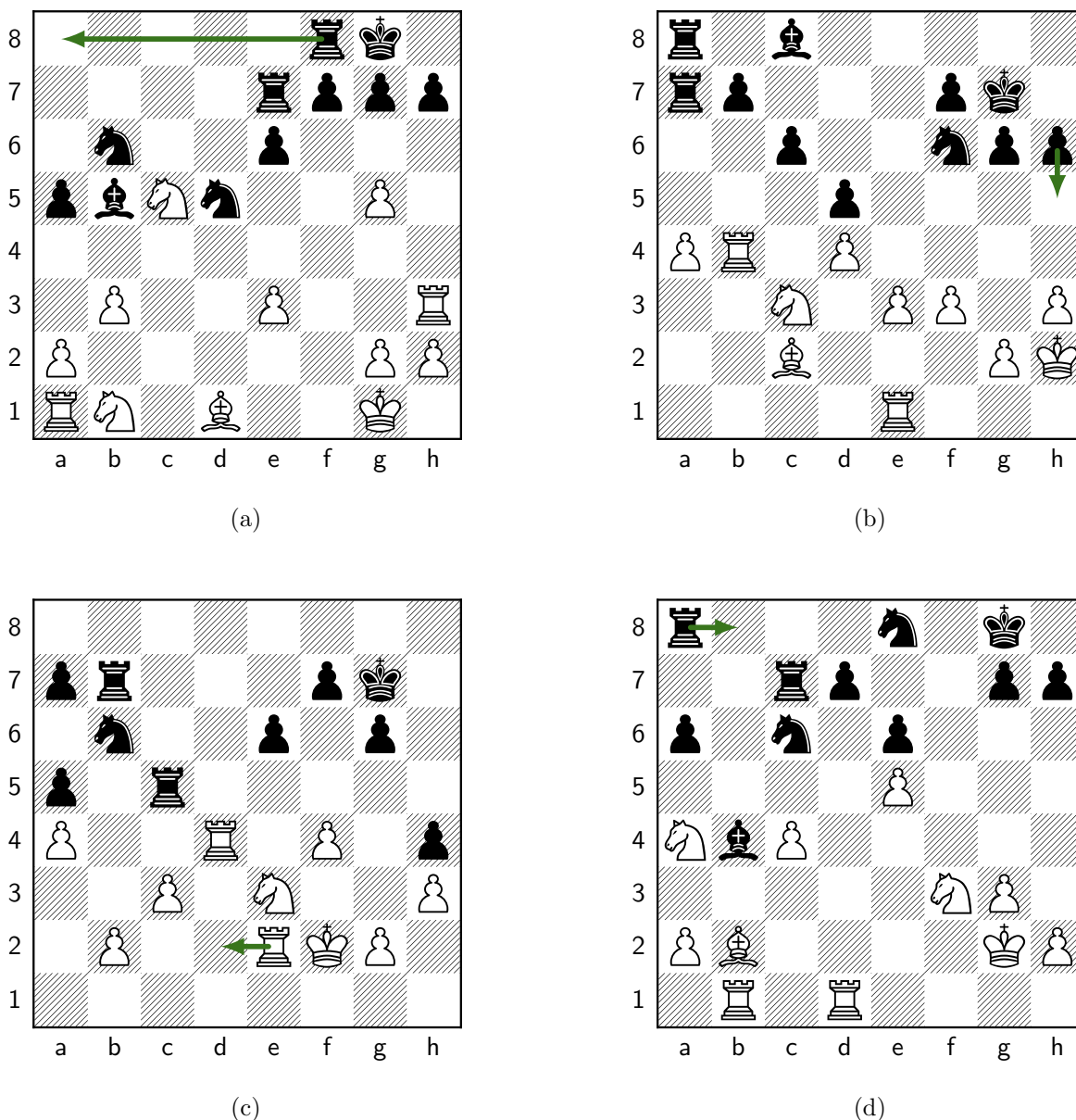
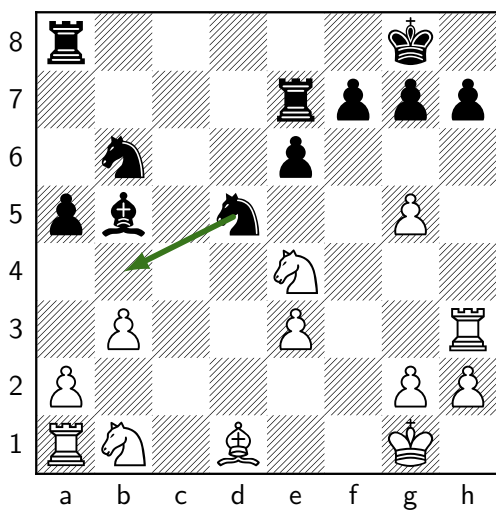
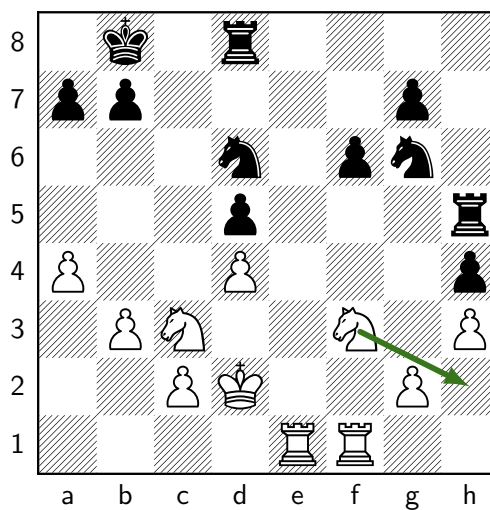


Figure 22: **Move 28, Black: Ra8 (f8a8)**. (a) Our position; Melanie plays Ra8 (f8a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2650` (ply 67) [structural distance  $d=64$ , QAP obj=50]: the master played h5; under  $\mu$  our *Rf1* plays the role of that game’s *Ph3* (a **cross-type** correspondence); cross-type roles: our *rh3* as their *Bc8*, our *Nd5* as their *Ra7*; transferred move f8a8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3626` (ply 60) [structural distance  $d=63$ , QAP obj=43]: the master played Red2; under  $\mu$  our *Rf1* plays the role of that game’s *Re2*; cross-type roles: our *Bb5* as their *Pg2*, our *Re7* as their *Ne3*; transferred move f8a8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1318` (ply 47) [structural distance  $d=57$ , QAP obj=30]: the master played Rb8; under  $\mu$  our *Rf1* plays the role of that game’s *Ra1*; cross-type roles: our *Pf7* as their *Ne8*, our *Nb6* as their *Rc7*; transferred move f8a8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ra8 (summed QAP objective 123, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.

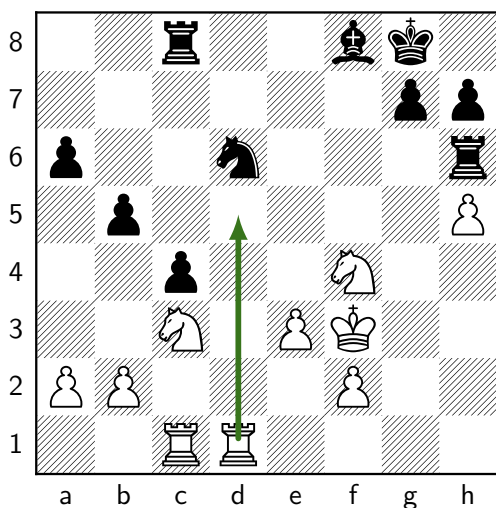




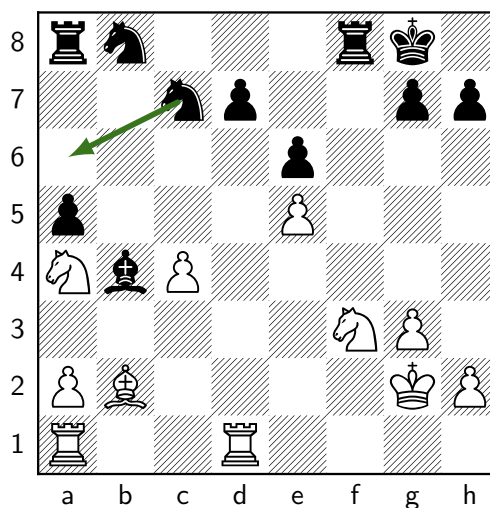
(a)



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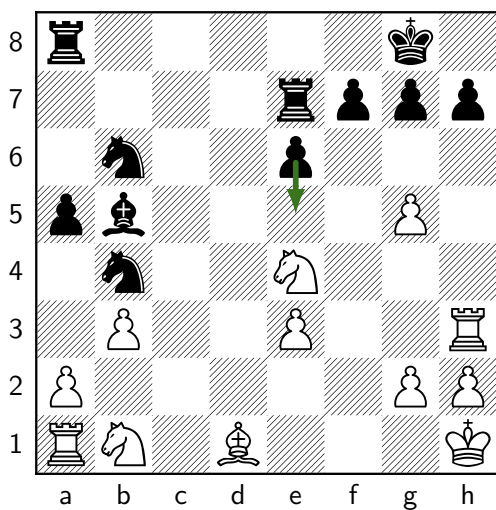


(c)

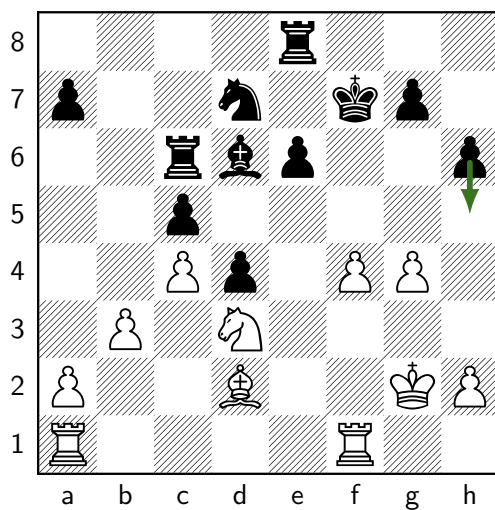


(d)

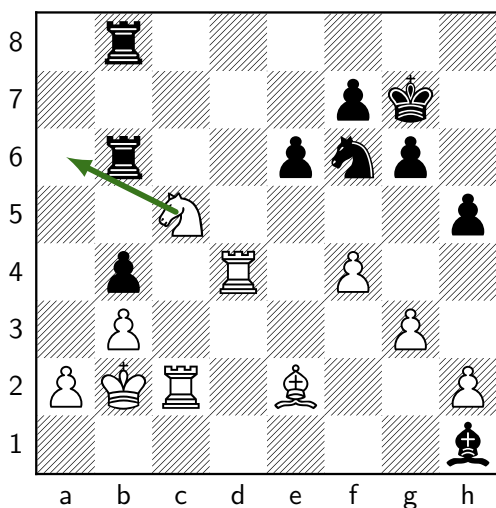
Figure 23: **Move 29, Black: Nb4.** (a) Our position; Melanie plays Nb4 (d5b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#975` (ply 54) [structural distance  $d=57$ , QAP obj=33]: the master played Nh2; under  $\mu$  our Nd4 plays the role of that game’s Nf3; cross-type roles: our Nb6 as their Re1, our pg2 as their Rf1; transferred move d5b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1747` (ply 62) [structural distance  $d=62$ , QAP obj=20]: the master played Rd5; under  $\mu$  our Nd4 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Nb6 as their Rc1, our Re7 as their Pa2; transferred move d5b4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1445` (ply 41) [structural distance  $d=62$ , QAP obj=25]: the master played Nca6; under  $\mu$  our Pa4 plays the role of that game’s Nc2 (a **cross-type** correspondence); cross-type roles: our Pe6 as their Ra8, our pg2 as their Nb8; transferred move a5a4. Synthesis: 2 retrieved analogs transferred to Nb4 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



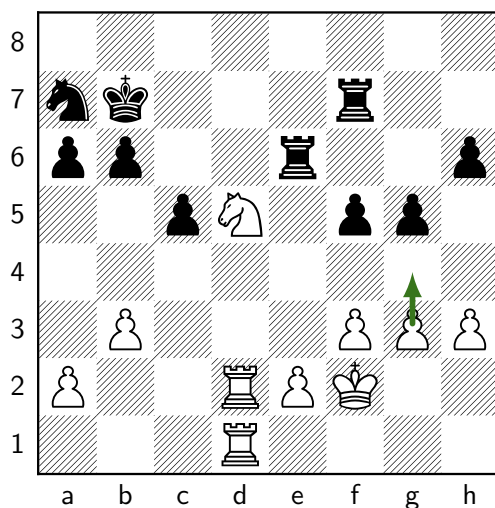
(a)



(b)

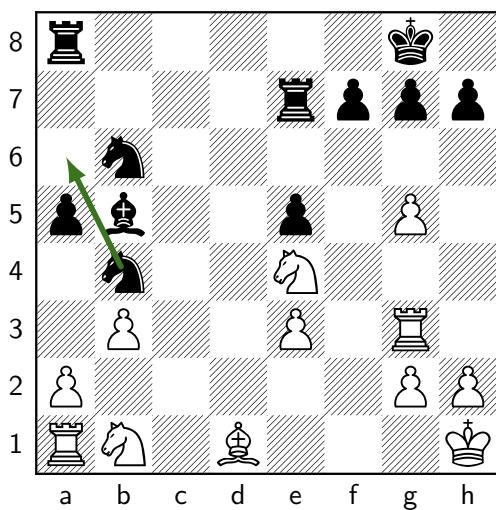


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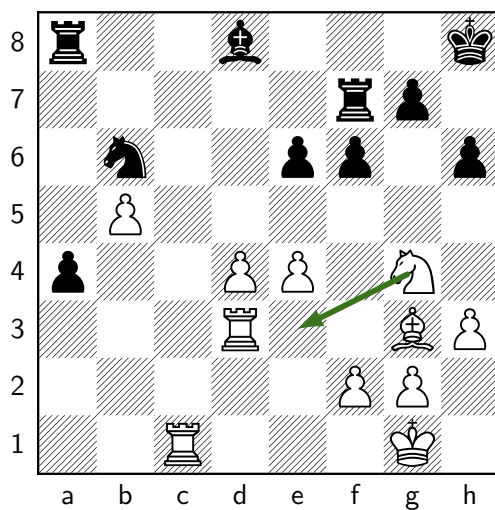


(d)

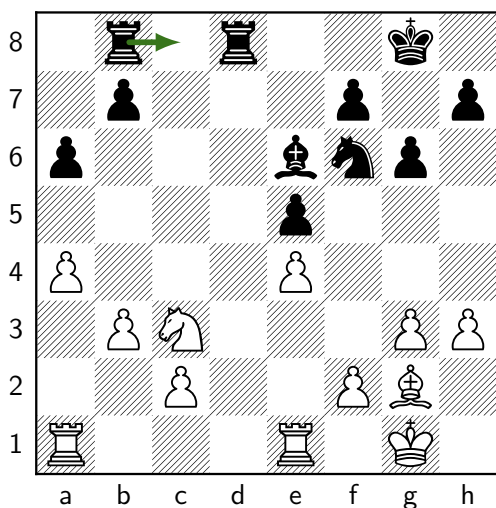
Figure 24: **Move 30, Black: e5**. (a) Our position; Melanie plays **e5** (**e6e5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1789` (ply 57) [structural distance  $d=54$ , QAP obj=32]: the master played **h5**; under  $\mu$  our *Pe3* plays the role of that game’s *Ph3*; cross-type roles: our *Ra8* as their *Pa7*, our *bd1* as their *Nd7*; transferred move **e6e5** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2808` (ply 64) [structural distance  $d=48$ , QAP obj=40]: the master played **Na6**; under  $\mu$  our *Nb5* plays the role of that game’s *Nc5*; cross-type roles: our *Ra8* as their *bh1*, our *ph2* as their *Rc2*; transferred move **b4c2**. (d) Retrieved from `ccrl_top_p0003#199` (ply 70) [structural distance  $d=55$ , QAP obj=36]: the master played **g4**; under  $\mu$  our *Pf2* plays the role of that game’s *Pg3*; cross-type roles: our *ph2* as their *Rd1*, our *bd1* as their *Pa2*; transferred move **f7f6**. Synthesis: 2 retrieved analogs transferred to **e5** (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



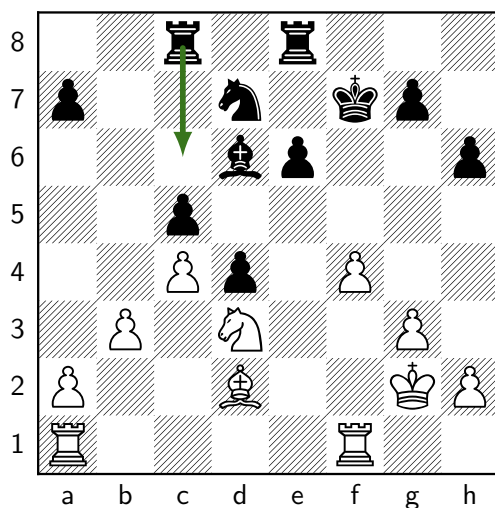
(a)



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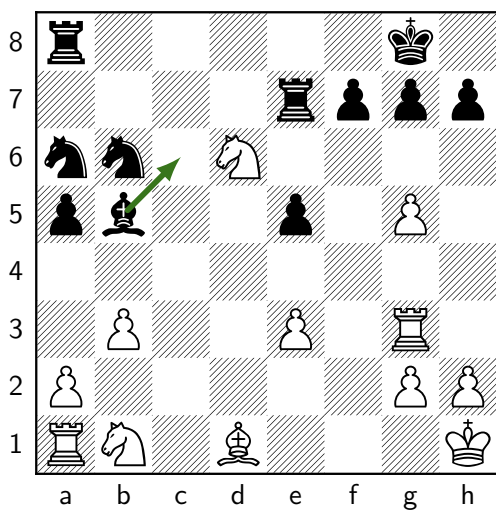


(c)

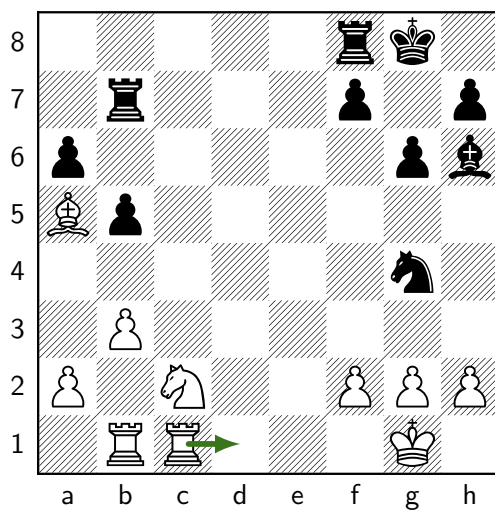


(d)

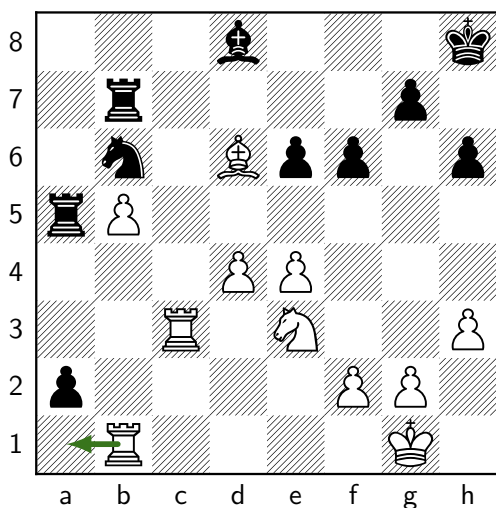
Figure 25: **Move 31, Black: Na6**. (a) Our position; Melanie plays Na6 (b4a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1458` (ply 58) [structural distance  $d=58$ , QAP obj=39]: the master played Ne3; under  $\mu$  our Nb5 plays the role of that game’s Ng4; cross-type roles: our pe3 as their ra8; transferred move b4a6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1388` (ply 45) [structural distance  $d=65$ , QAP obj=35]: the master played Rbc8; under  $\mu$  our Bb4 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our Re7 as their Pb7, our Nb4 as their Be6; transferred move b5d7. (d) Retrieved from `ccrl_top_p0002#1789` (ply 55) [structural distance  $d=60$ , QAP obj=31]: the master played Rc6; under  $\mu$  our Pf2 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Pe5 as their Re8, our ra1 as their Nd7; transferred move f7f5. Synthesis: 1 retrieved analog transferred to Na6 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



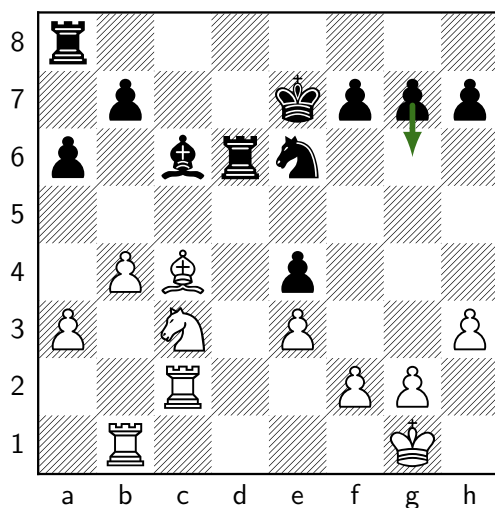
(a)



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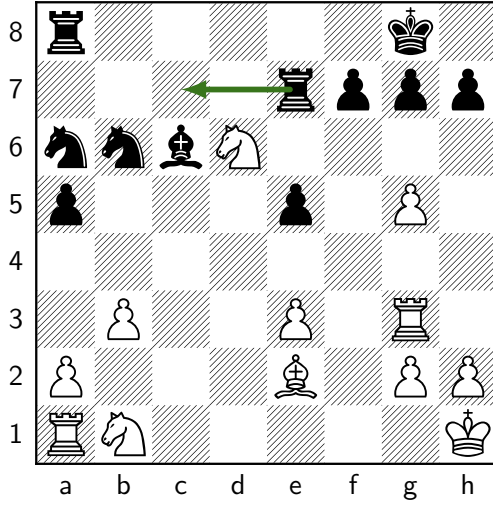


(c)

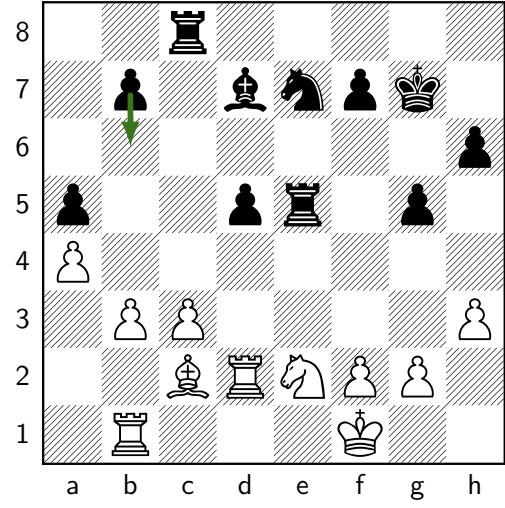


(d)

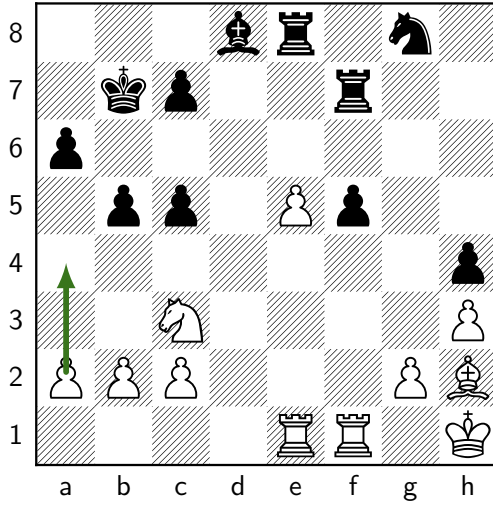
Figure 26: **Move 32, Black: Bc6**. (a) Our position; Melanie plays Bc6 (b5c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3818` (ply 54) [structural distance  $d=69$ , QAP obj=21]: the master played Rd1; under  $\mu$  our Bb4 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Re7 as their Pa2, our Bb5 as their Pf2; transferred move b5c6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1458` (ply 66) [structural distance  $d=66$ , QAP obj=7]: the master played Ra1; under  $\mu$  our Bb4 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our Nd6 as their pa2, our Ra1 as their Pg2; transferred move b5c6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2210` (ply 55) [structural distance  $d=68$ , QAP obj=31]: the master played g6; under  $\mu$  our Na3 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Nb6 as their Pf7, our Ra8 as their Ph7; transferred move a6b4. Synthesis: 2 retrieved analogs transferred to Bc6 (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



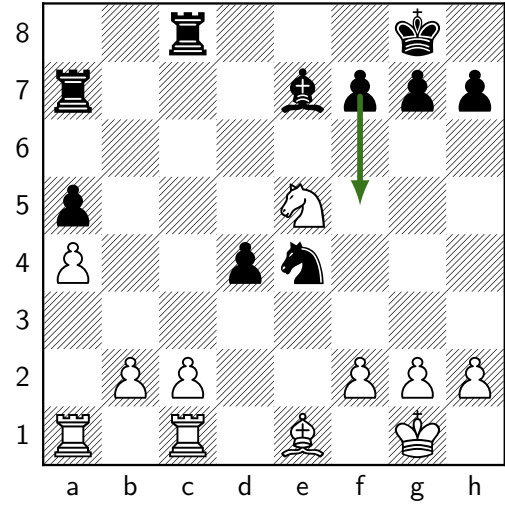
(a)



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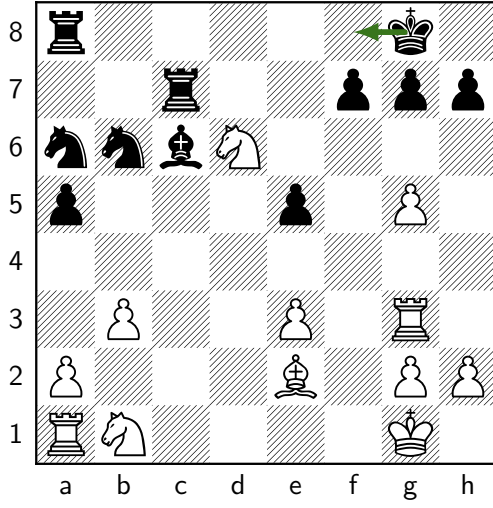


(c)

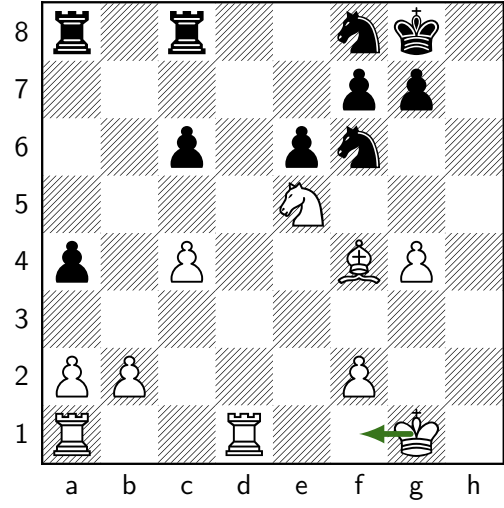


(d)

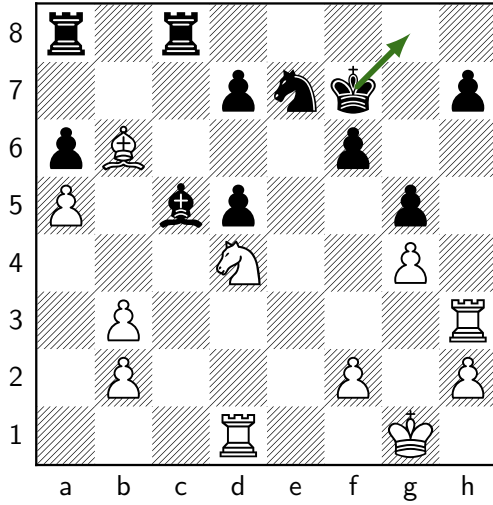
Figure 27: **Move 33, Black: Rc7**. (a) Our position; Melanie plays Rc7 (e7c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#193` (ply 71) [structural distance  $d=66$ , QAP obj=41]: the master played b6; under  $\mu$  our *Re2* plays the role of that game’s *Pb2* (a **cross-type** correspondence); cross-type roles: our *nd6* as their *Bd7*, our *pe3* as their *Ne7*; transferred move e7c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#87` (ply 74) [structural distance  $d=54$ , QAP obj=50]: the master played a4; under  $\mu$  our *Re2* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *nd6* as their *Re1*, our *Bc6* as their *Rf1*; transferred move e7b7. (d) Retrieved from `ccr1_top_p0000#3096` (ply 55) [structural distance  $d=66$ , QAP obj=48]: the master played f5; under  $\mu$  our *Pf2* plays the role of that game’s *Pf2*; cross-type roles: our *Bc6* as their *Rc8*, our *Pe5* as their *Ra7*; transferred move f7f5. Synthesis: 1 retrieved analog transferred to Rc7 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



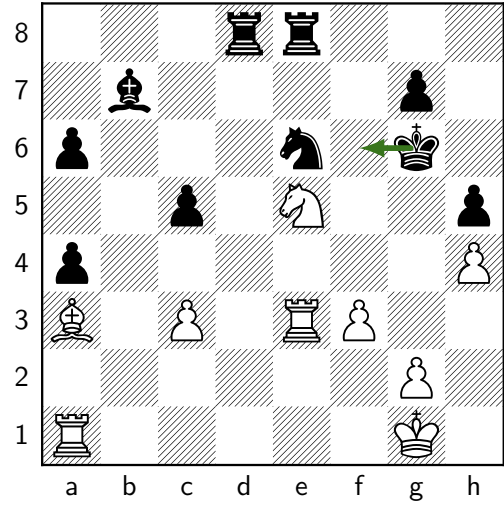
(a)



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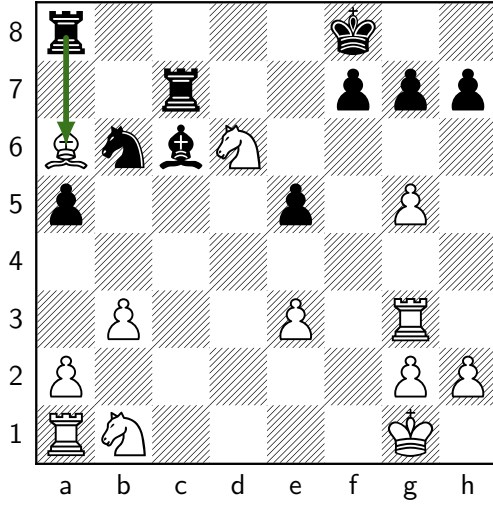


(c)

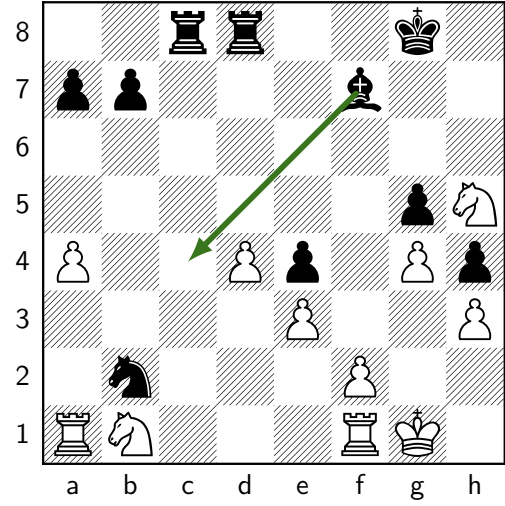


(d)

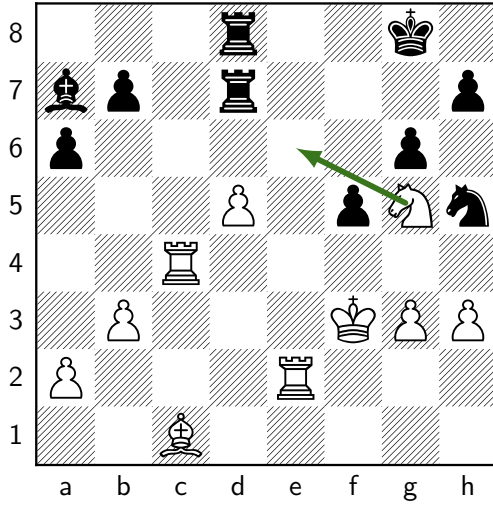
Figure 28: **Move 34, Black: Kf8.** (a) Our position; Melanie plays Kf8 (g8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#521` (ply 50) [structural distance  $d=60$ , QAP obj=36]: the master played Kf1; under  $\mu$  our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *ph2* as their *Ra1*, our *Nb6* as their *Pa2*; transferred move g8f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1095` (ply 65) [structural distance  $d=53$ , QAP obj=8]: the master played Kg8; under  $\mu$  our *Kg1* plays the role of that game’s *Kf2*; cross-type roles: our *pg5* as their *Ra8*, our *Bc6* as their *Rc8*; transferred move g8f8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3742` (ply 59) [structural distance  $d=64$ , QAP obj=3]: the master played Kf6; under  $\mu$  our *Kg1* plays the role of that game’s *Kg3*; cross-type roles: our *Pe5* as their *Rd8*, our *Pf7* as their *Bb7*; transferred move g8f8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf8 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



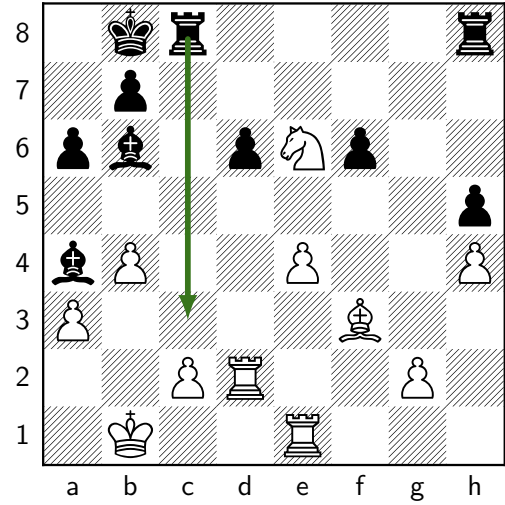
(a)



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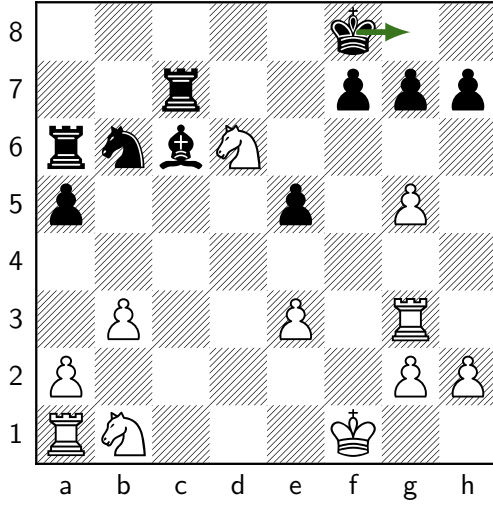
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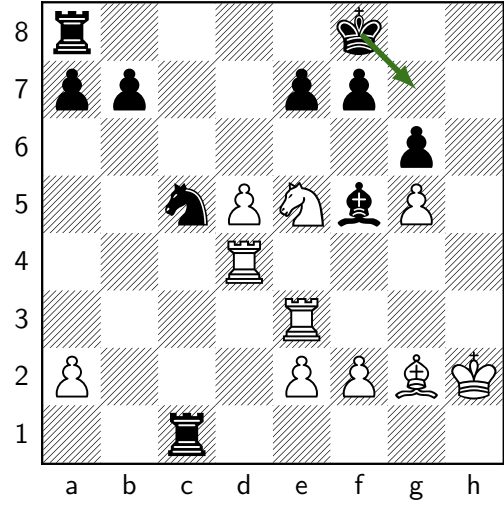
(d)

Figure 29: **Move 35, Black: Rxa6.** (a) Our position; Melanie plays Rxa6 (a8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3177` (ply 53) [structural distance  $d=49$ , QAP obj=28]: the master played Bc4; under  $\mu$  our Ra1 plays the role of that game’s Bf2 (a **cross-type** correspondence); cross-type roles: our Bc6 as their Rc8, our pg2 as their Rd8; transferred move a8a7. (c) Retrieved from `ccr1_top_p0000#3433` (ply 80) [structural distance  $d=57$ , QAP obj=17]: the master played Ne6; under  $\mu$  our Ra1 plays the role of that game’s Ng5 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Re2, our Nb6 as their Rc4; transferred move a8a7. (d) Retrieved from `ccr1_top_p0003#1350` (ply 59) [structural distance  $d=54$ , QAP obj=16]: the master played Rc3; under  $\mu$  our Pg2 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our pg2 as their Rh8, our Rc7 as their Pb7; transferred move g7g6. Synthesis: no retrieved analog transferred to Rxa6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

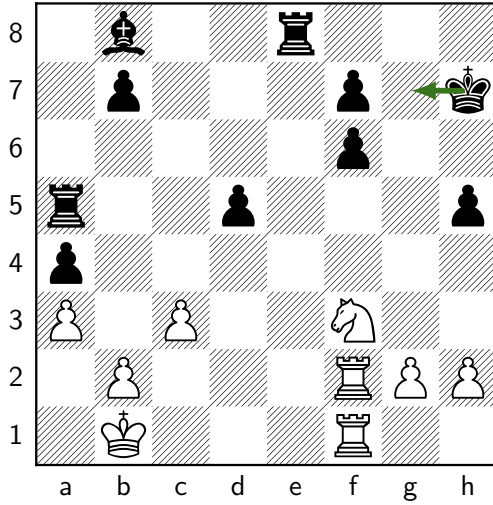




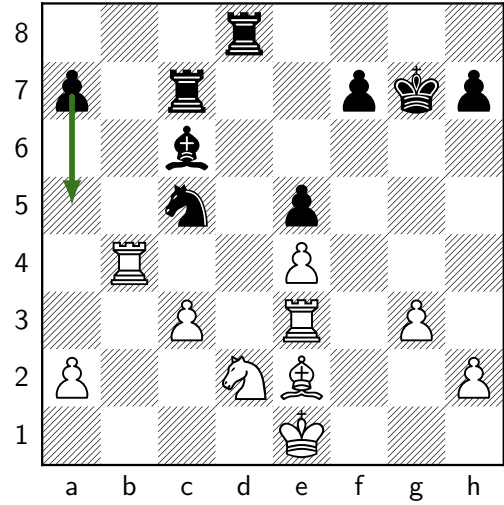
(a)



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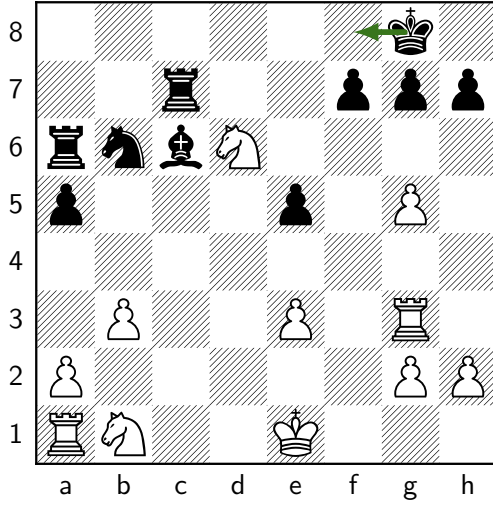


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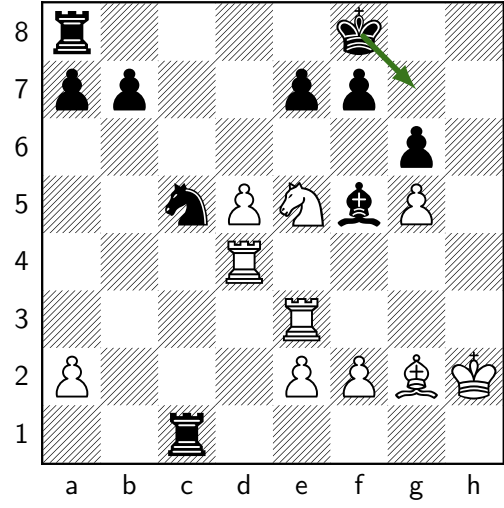


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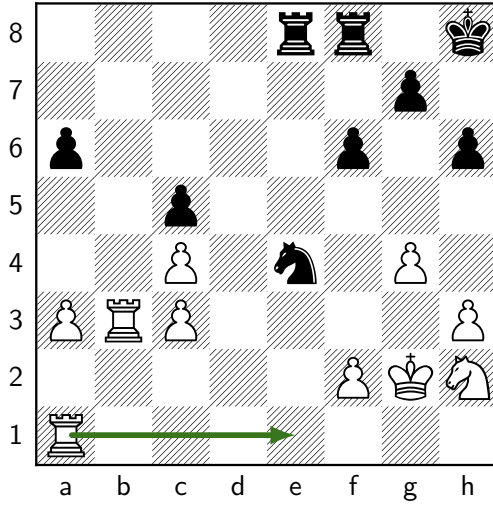
Figure 30: **Move 36, Black: Kg8.** (a) Our position; Melanie plays Kg8 (f8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#257` (ply 57) [structural distance  $d=61$ , QAP obj=38]: the master played Kg7; under  $\mu$  our *Kf1* plays the role of that game's *Kf1*; cross-type roles: our *Ph7* as their *Ra8*, our *Bc6* as their *Pb7*; transferred move f8g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#868` (ply 57) [structural distance  $d=50$ , QAP obj=11]: the master played Kg7; under  $\mu$  our *Kf1* plays the role of that game's *Kh2*; cross-type roles: our *Pg7* as their *Ra5*, our *Ra6* as their *Ph5*; transferred move f8g8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#962` (ply 55) [structural distance  $d=61$ , QAP obj=46]: the master played a5; under  $\mu$  our *Ph2* plays the role of that game's *Pa2*; cross-type roles: our *Bc6* as their *Rd8*, our *Pg7* as their *Rc7*; transferred move h7h5. Synthesis: 2 retrieved analogs transferred to Kg8 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



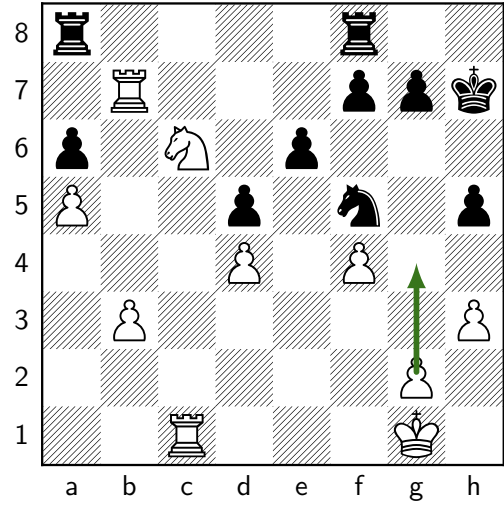
(a)



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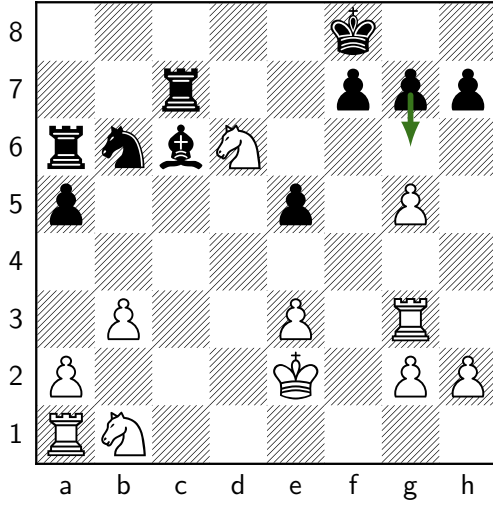


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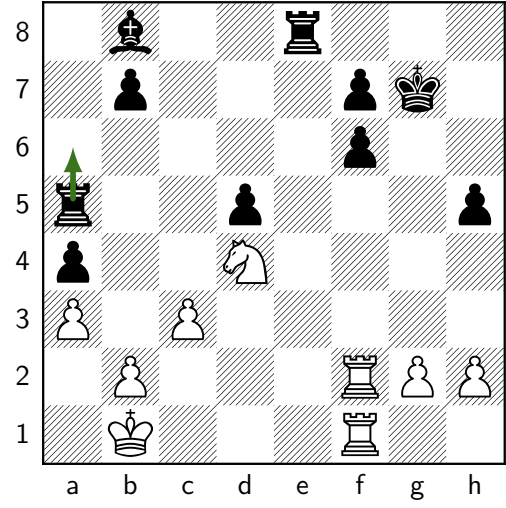


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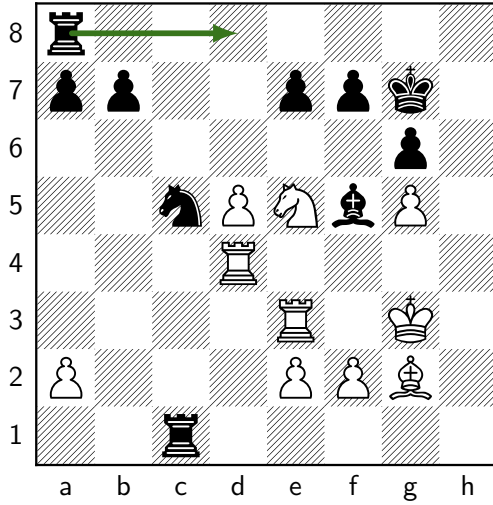
Figure 31: **Move 37, Black: Kf8.** (a) Our position; Melanie plays Kf8 (g8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0003#257` (ply 57) [structural distance  $d=55$ , QAP obj=38]: the master played Kg7; under  $\mu$  our Kg1 plays the role of that game's Kf1; cross-type roles: our ra1 as their Pe7, our Ra6 as their pg5; transferred move g8f8 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#488` (ply 64) [structural distance  $d=57$ , QAP obj=37]: the master played Re1; under  $\mu$  our Rc2 plays the role of that game's Ra1; cross-type roles: our pa2 as their Nh2, our Bc6 as their Rb3; transferred move c7e7. (d) Retrieved from `ccl1_top_p0003#1249` (ply 82) [structural distance  $d=45$ , QAP obj=31]: the master played g4; under  $\mu$  our Pg2 plays the role of that game's Pg2; cross-type roles: our Pa5 as their Rc1, our Ra6 as their Pa5; transferred move g7g6. Synthesis: 1 retrieved analog transferred to Kf8 (summed QAP objective 38, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



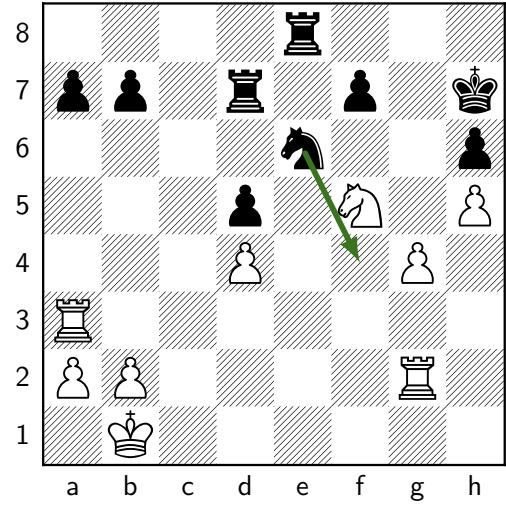
(a)



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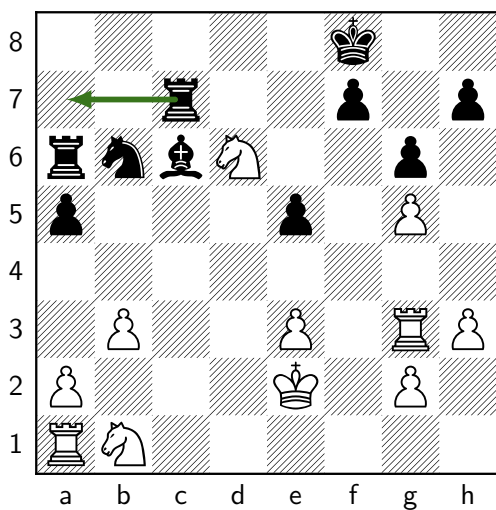


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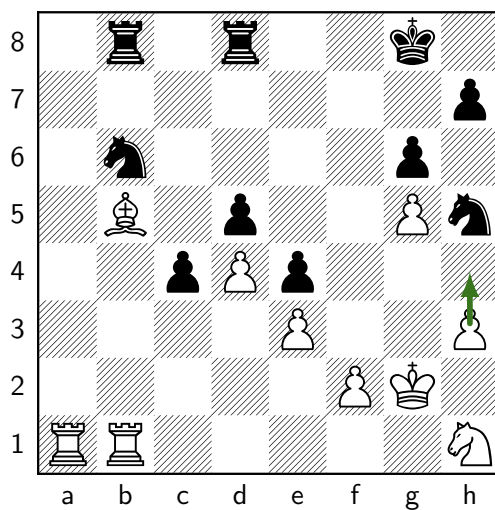


(d)

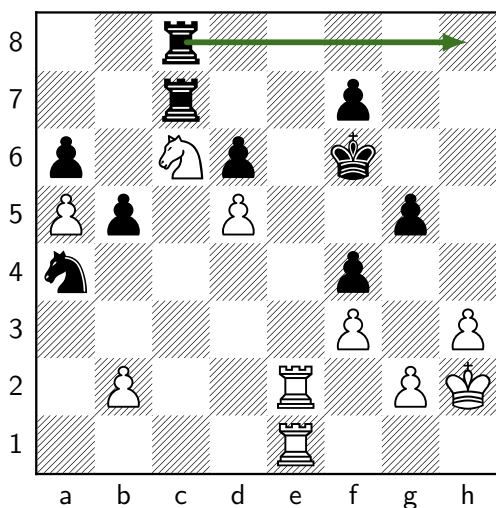
Figure 32: **Move 38, Black: g6**. (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#868 (ply 59) [structural distance  $d=41$ , QAP obj=20]: the master played Ra6; under  $\mu$  our Pg2 plays the role of that game’s Ra4 (a **cross-type** correspondence); cross-type roles: our Nb6 as their Pb7, our Ra6 as their Pd5; transferred move g7g6 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0003#257 (ply 59) [structural distance  $d=58$ , QAP obj=36]: the master played Rd8; under  $\mu$  our Bc3 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our ra1 as their Pe7, our Pa5 as their Nc5; transferred move c6b5. (d) Retrieved from ccr1\_top\_p0001#768 (ply 75) [structural distance  $d=49$ , QAP obj=33]: the master played Nf4; under  $\mu$  our Bc3 plays the role of that game’s Ne3 (a **cross-type** correspondence); cross-type roles: our Nb6 as their Pb7, our pb3 as their ra3; transferred move c6e4. Synthesis: 1 retrieved analog transferred to g6 (summed QAP objective 20, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



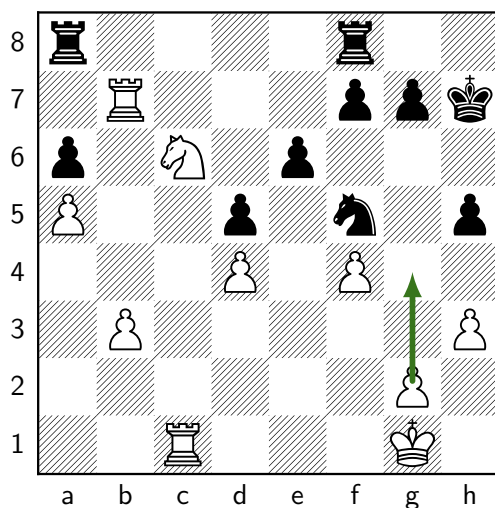
(a)



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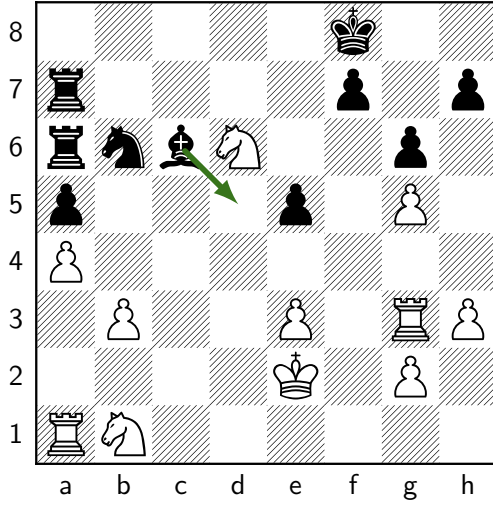


(c)

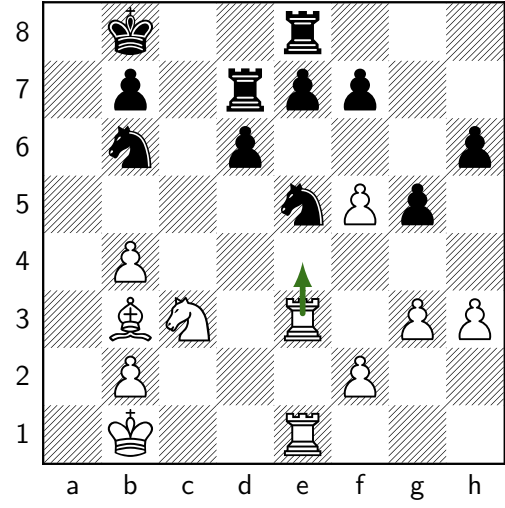


(d)

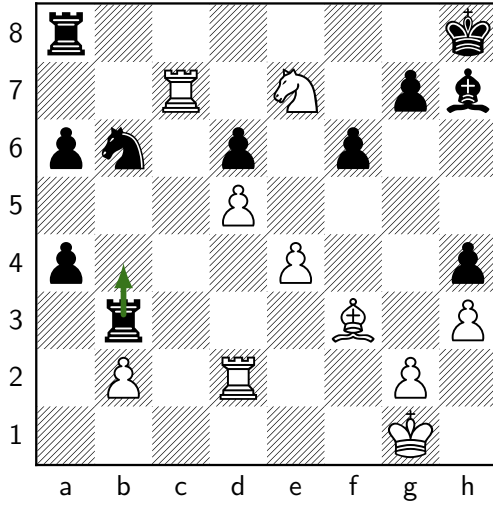
Figure 33: **Move 39, Black: Rca7.** (a) Our position; Melanie plays Rca7 (c7a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#2630 (ply 84) [structural distance  $d=63$ , QAP obj=12]: the master played h4; under  $\mu$  our Rc2 plays the role of that game’s Ph3 (a **cross-type** correspondence); cross-type roles: our Pe5 as their Rb1, our Pg6 as their Nh1; transferred move c7a7 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0002#1175 (ply 93) [structural distance  $d=69$ , QAP obj=6]: the master played Rh8; under  $\mu$  our Rc2 plays the role of that game’s Rc1; cross-type roles: our Bc6 as their Rc7, our Ra6 as their Pb5; transferred move c7a7 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0003#1249 (ply 82) [structural distance  $d=59$ , QAP obj=31]: the master played g4; under  $\mu$  our Nb3 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our pg5 as their Rc1, our Rc7 as their Pf4; transferred move b6a4. Synthesis: 2 retrieved analogs transferred to Rca7 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



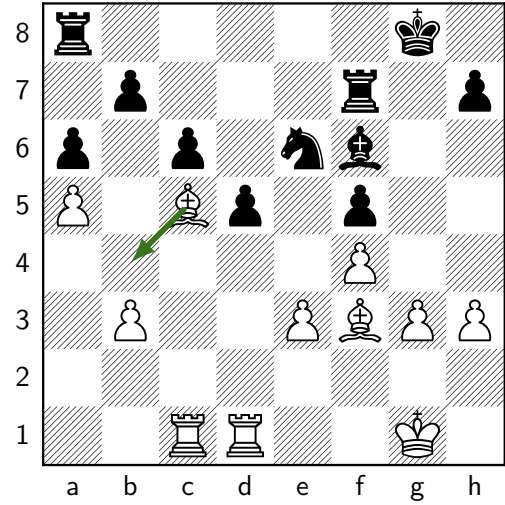
(a)



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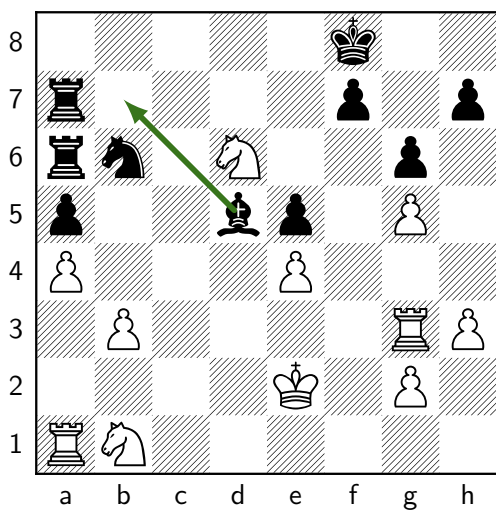


(c)

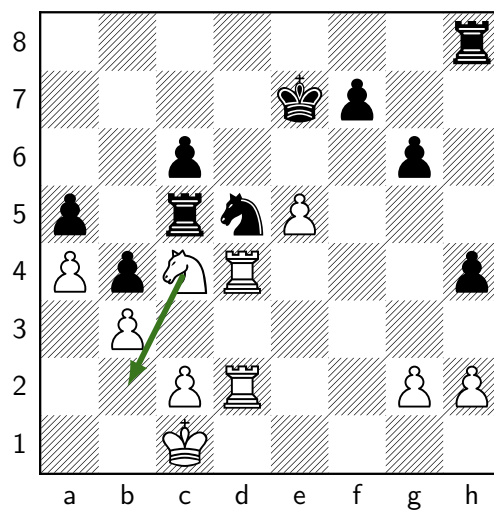


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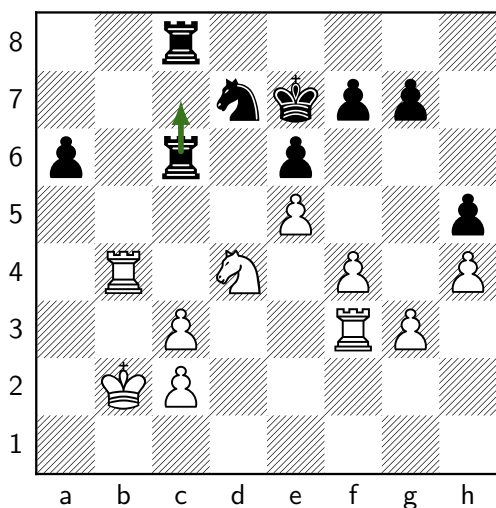
Figure 34: **Move 40, Black: Bd5.** (a) Our position; Melanie plays Bd5 (c6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2760` (ply 66) [structural distance  $d=73$ , QAP obj=11]: the master played Re4; under  $\mu$  our Bc3 plays the role of that game’s Re3 (a **cross-type** correspondence); cross-type roles: our nd6 as their Pb2, our pg2 as their Bb3; transferred move c6d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2409` (ply 81) [structural distance  $d=72$ , QAP obj=28]: the master played Rb4; under  $\mu$  our Bc3 plays the role of that game’s Rb6 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Ra8, our pg2 as their rc7; transferred move c6g2. (d) Retrieved from `ccr1_top_p0002#963` (ply 62) [structural distance  $d=75$ , QAP obj=25]: the master played Bb4; under  $\mu$  our Pf2 plays the role of that game’s Bc5 (a **cross-type** correspondence); cross-type roles: our Nb6 as their Rd1, our nb1 as their Ph3; transferred move f7f5. Synthesis: 1 retrieved analog transferred to Bd5 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



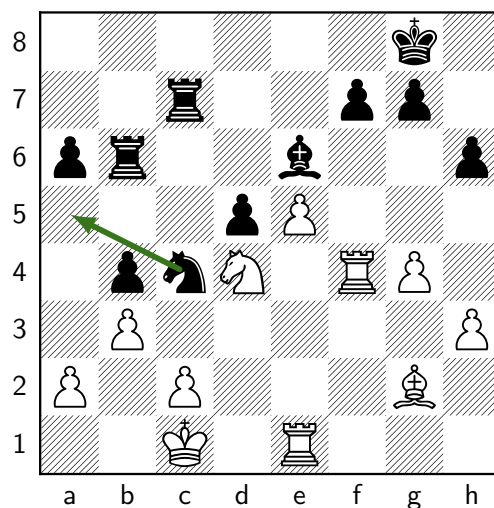
(a)



(b)

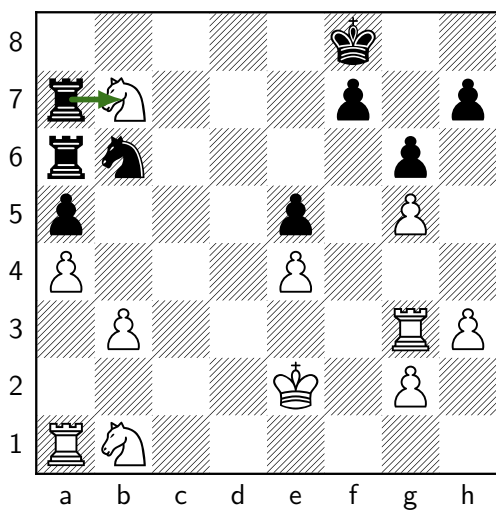


(c)

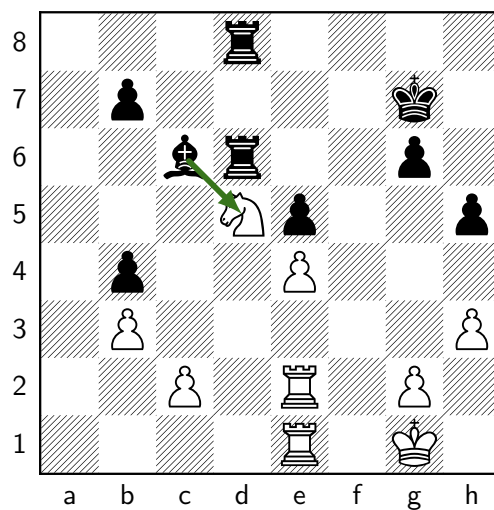


(d)

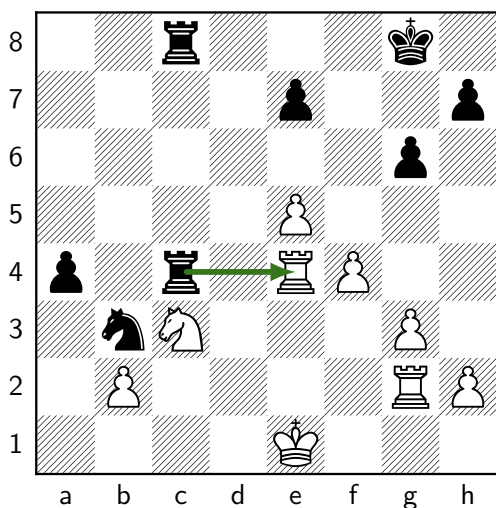
Figure 35: **Move 41, Black: Bb7**. (a) Our position; Melanie plays Bb7 (d5b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2063` (ply 74) [structural distance  $d=77$ , QAP obj=30]: the master played Nb2; under  $\mu$  our Bd4 plays the role of that game's Nc4 (a **cross-type** correspondence); cross-type roles: our nd6 as their Rd2, our pe4 as their Rd4; transferred move d5b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#955` (ply 67) [structural distance  $d=71$ , QAP obj=20]: the master played R6c7; under  $\mu$  our Bd4 plays the role of that game's Rc3 (a **cross-type** correspondence); cross-type roles: our Nb6 as their Rc8, our Ph7 as their Nd7; transferred move d5b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1733` (ply 85) [structural distance  $d=66$ , QAP obj=24]: the master played Na5; under  $\mu$  our Bd4 plays the role of that game's Nc5 (a **cross-type** correspondence); cross-type roles: our Nb6 as their Rc7, our Ra7 as their Pa6; transferred move d5c6. Synthesis: 2 retrieved analogs transferred to Bb7 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



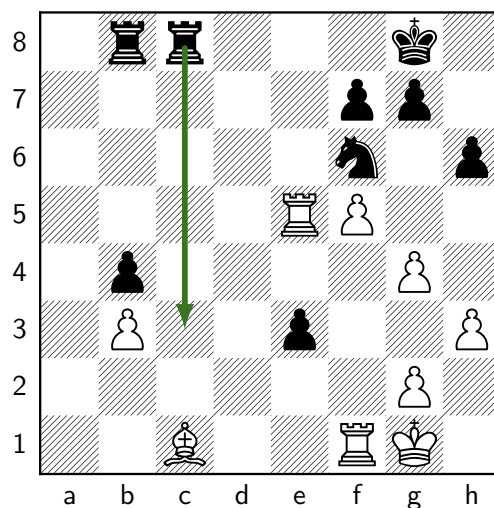
(a)



(b)



(c)



(d)

Figure 36: **Move 42, Black: Rxb7**. (a) Our position; Melanie plays Rxb7 (a7b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1524` (ply 67) [structural distance  $d=50$ , QAP obj=68]: the master played Bxd5; under  $\mu$  our *Ra2* plays the role of that game’s *Bc3* (a **cross-type** correspondence); cross-type roles: our *ra1* as their *Pb7*, our *rg3* as their *ph3*; transferred move a7b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2872` (ply 67) [structural distance  $d=49$ , QAP obj=33]: the master played Rxe4+; under  $\mu$  our *Ra2* plays the role of that game’s *Rc5*; cross-type roles: our *Nb6* as their *pe5*, our *nb7* as their *re4*; transferred move a7b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#699` (ply 69) [structural distance  $d=63$ , QAP obj=10]: the master played Rc3; under  $\mu$  our *Ra2* plays the role of that game’s *Rc1*; cross-type roles: our *ph3* as their *Rb8*, our *Pg6* as their *Nf6*; transferred move a7b7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rxb7 (summed QAP objective 119, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



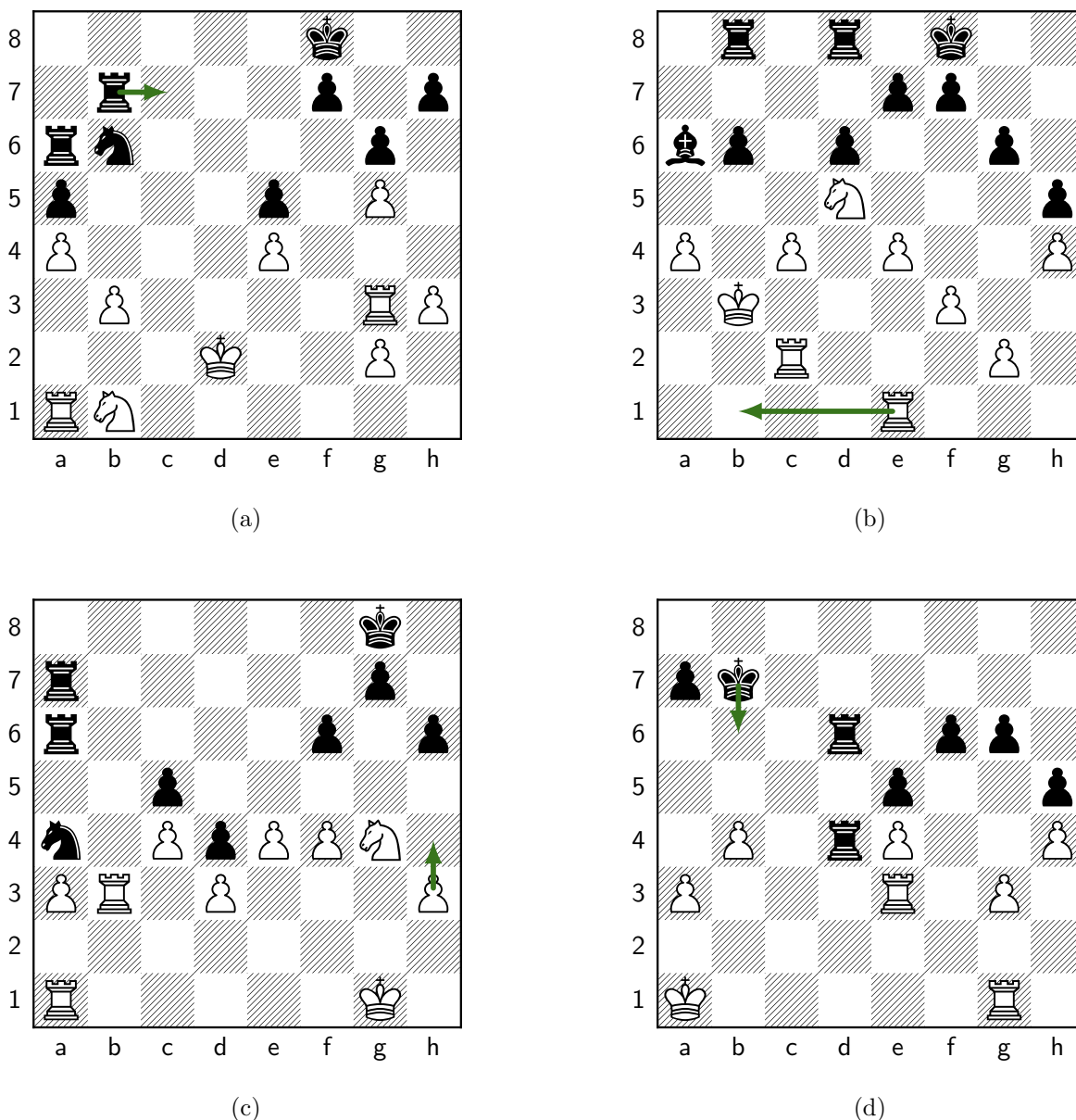
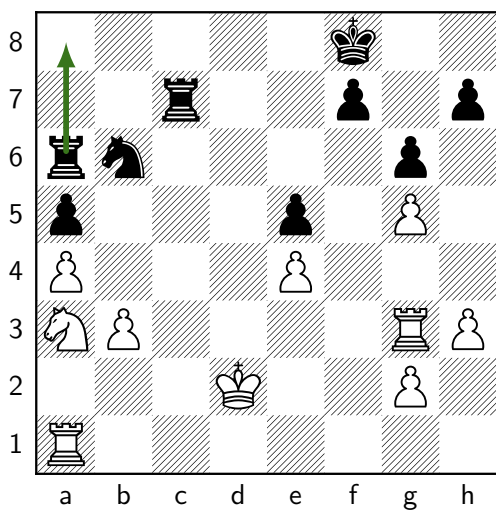
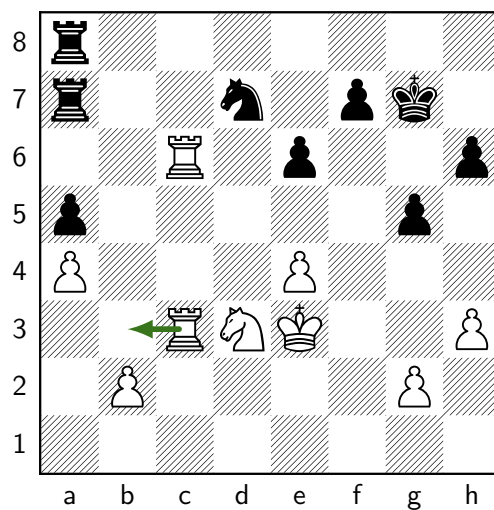


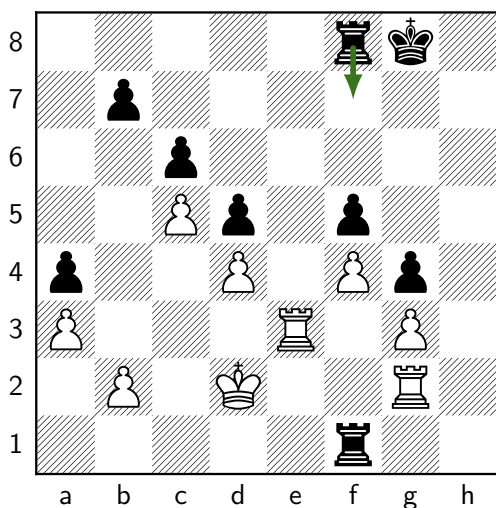
Figure 37: **Move 43, Black: Rc7.** (a) Our position; Melanie plays **Rc7** (**b7c7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1498` (ply 74) [structural distance  $d=63$ , QAP obj=46]: the master played **Rb1**; under  $\mu$  our **Rb2** plays the role of that game’s **Re1**; cross-type roles: our **Ra6** as their **Pa4**, our **Nb6** as their **ba6**; transferred move **b7c7** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3976` (ply 100) [structural distance  $d=37$ , QAP obj=39]: the master played **h4**; under  $\mu$  our **Ra3** plays the role of that game’s **Ph3** (a **cross-type** correspondence); cross-type roles: our **Pf7** as their **Ra1**, our **rg3** as their **na4**; transferred move **a6a8**. (d) Retrieved from `ccr1_top_p0000#3197` (ply 95) [structural distance  $d=60$ , QAP obj=21]: the master played **Kb6**; under  $\mu$  our **Kf1** plays the role of that game’s **Kb2**; cross-type roles: our **Rb7** as their **Pa7**, our **Nb6** as their **Rd4**; transferred move **f8e7**. Synthesis: 1 retrieved analog transferred to **Rc7** (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



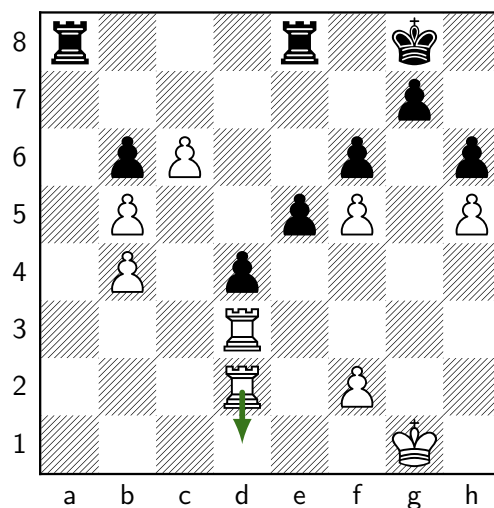
(a)



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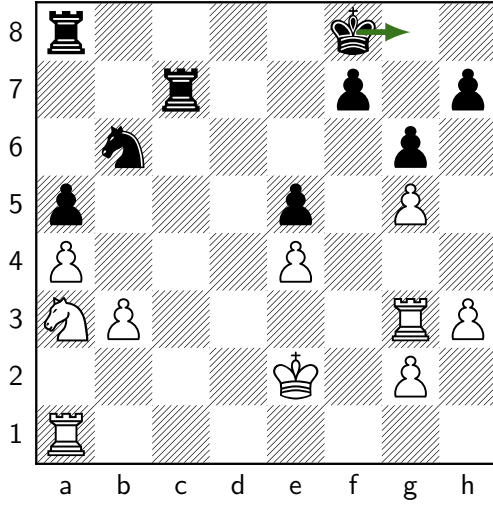


(c)

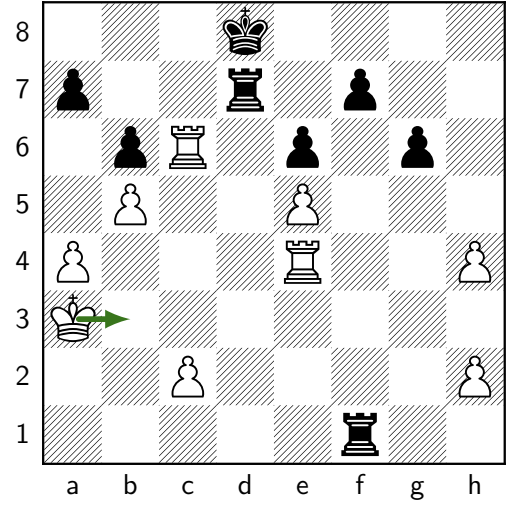


(d)

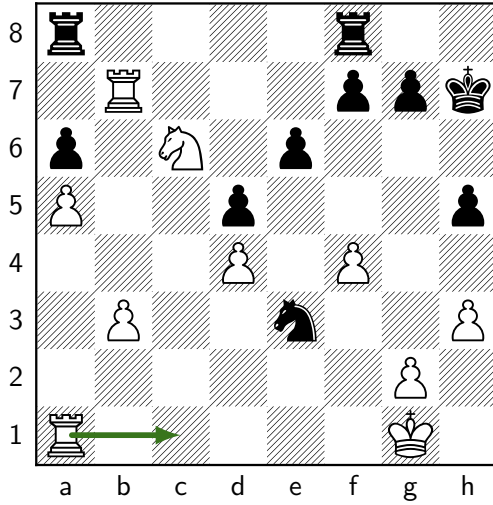
Figure 38: **Move 44, Black: Ra8.** (a) Our position; Melanie plays Ra8 (a6a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2599` (ply 72) [structural distance  $d=53$ , QAP obj=25]: the master played Rb3; under  $\mu$  our *Ra3* plays the role of that game's *Rc3*; cross-type roles: our *ra1* as their *Pb2*, our *Pe5* as their *Nd3*; transferred move a6a8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2103` (ply 97) [structural distance  $d=51$ , QAP obj=15]: the master played Rf7; under  $\mu$  our *Ra3* plays the role of that game's *Rf1*; cross-type roles: our *Rc7* as their *Pg4*, our *rg3* as their *pa3*; transferred move a6a8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1223` (ply 94) [structural distance  $d=49$ , QAP obj=14]: the master played Rd1; under  $\mu$  our *Ra3* plays the role of that game's *Rd2*; cross-type roles: our *Nb6* as their *Rd3*, our *Rc7* as their *Pb4*; transferred move a6a8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ra8 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



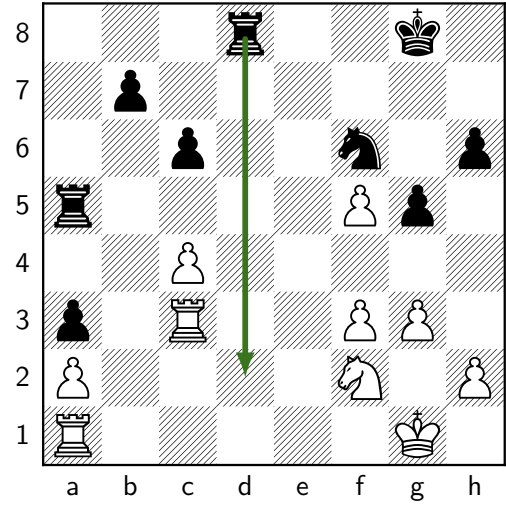
(a)



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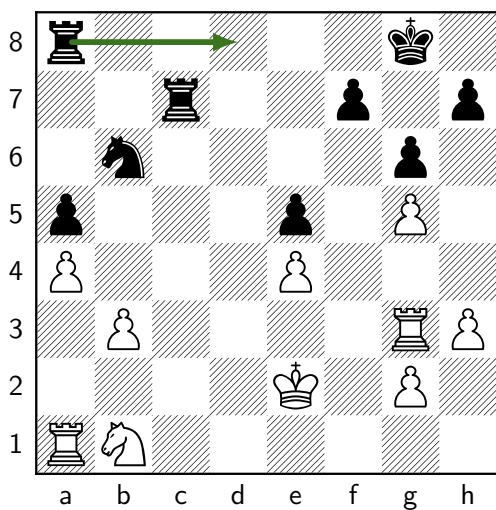


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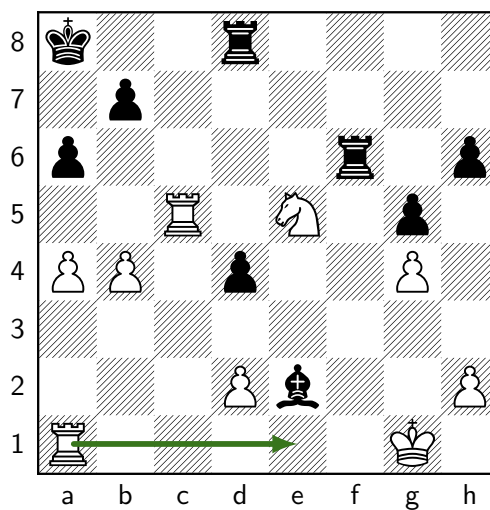


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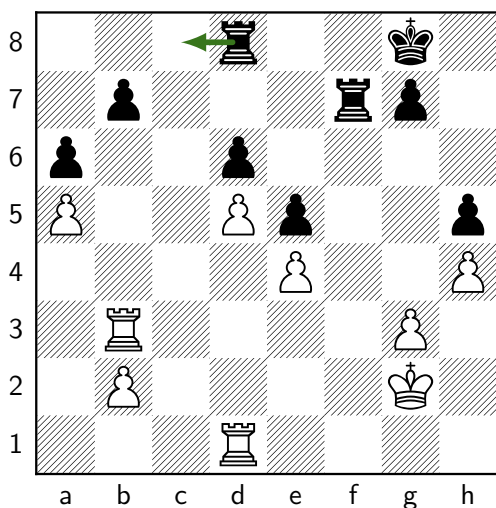
Figure 39: **Move 45, Black: Kg8**. (a) Our position; Melanie plays Kg8 (f8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#322` (ply 74) [structural distance  $d=52$ , QAP obj=20]: the master played Kb3; under  $\mu$  our Kf1 plays the role of that game’s Ka3; cross-type roles: our Pe5 as their rf1, our Ra8 as their Pc2; transferred move f8g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1249` (ply 80) [structural distance  $d=34$ , QAP obj=23]: the master played Rc1; under  $\mu$  our Ra1 plays the role of that game’s Ra1; cross-type roles: our na3 as their Pf4, our rg3 as their Pa5; transferred move a8b8. (d) Retrieved from `ccrl_top_p0002#3141` (ply 73) [structural distance  $d=41$ , QAP obj=21]: the master played Rd2; under  $\mu$  our Ra1 plays the role of that game’s Rd1; cross-type roles: our Rc7 as their Nf6, our pb3 as their Ra5; transferred move a8d8. Synthesis: 3 retrieved analogs transferred to Kg8 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



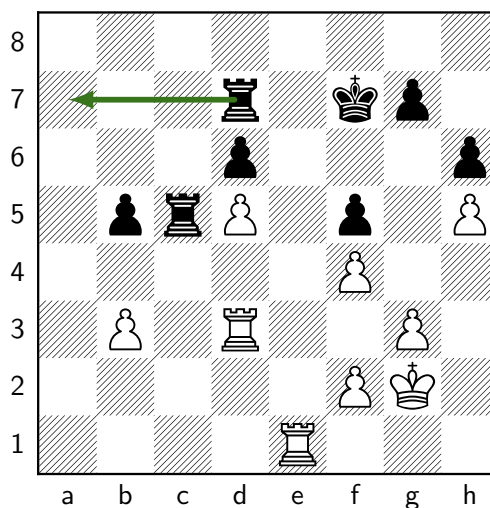
(a)



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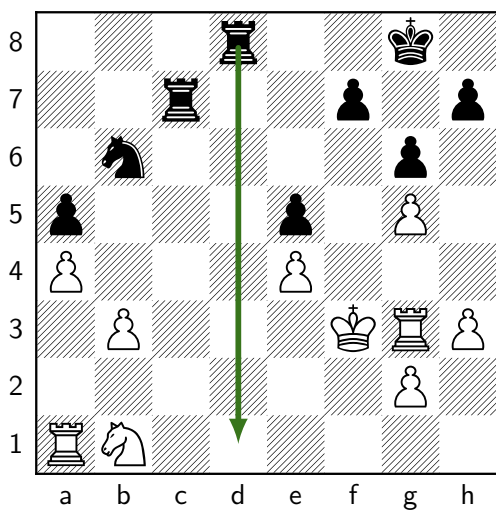


(c)

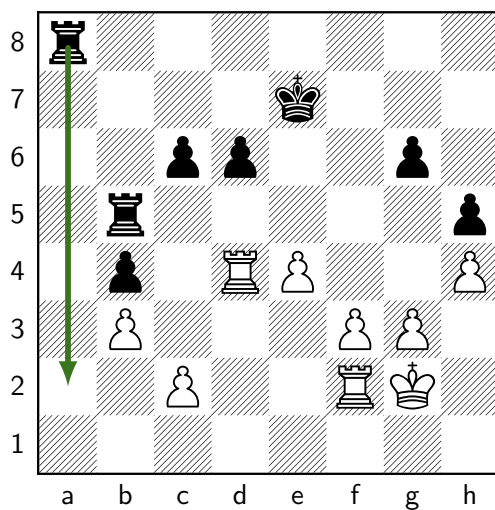


(d)

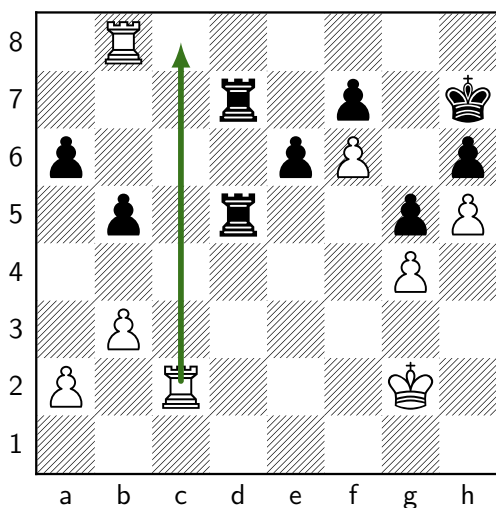
Figure 40: **Move 46, Black: Rd8.** (a) Our position; Melanie plays Rd8 (a8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0000#1728` (ply 74) [structural distance  $d=53$ , QAP obj=22]: the master played Re1; under  $\mu$  our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *Nb6* as their *be2*, our *Rc7* as their *Ph2*; transferred move a8d8 (**endorses this move**). (c) Retrieved from `cctl_top_p0001#708` (ply 91) [structural distance  $d=42$ , QAP obj=15]: the master played Rc8; under  $\mu$  our *Ra1* plays the role of that game’s *Rd1*; cross-type roles: our *Rc7* as their *Pe5*, our *nb1* as their *pe4*; transferred move a8d8 (**endorses this move**). (d) Retrieved from `cctl_top_p0003#22` (ply 77) [structural distance  $d=53$ , QAP obj=15]: the master played Ra7; under  $\mu$  our *Ra1* plays the role of that game’s *Rd2*; cross-type roles: our *rg3* as their *Pb5*, our *Nb6* as their *Rc5*; transferred move a8d8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rd8 (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



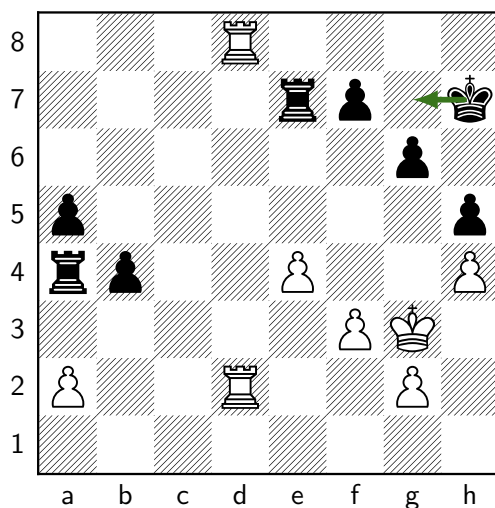
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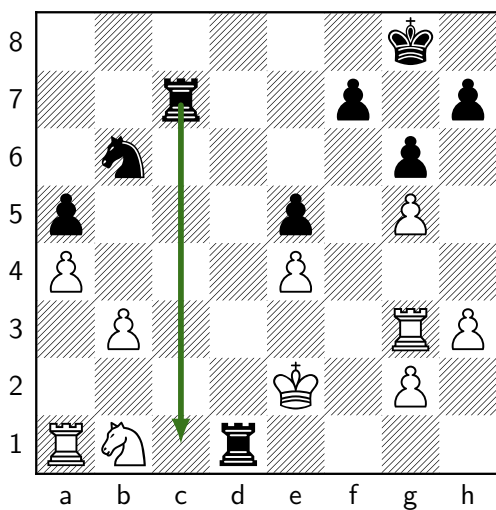


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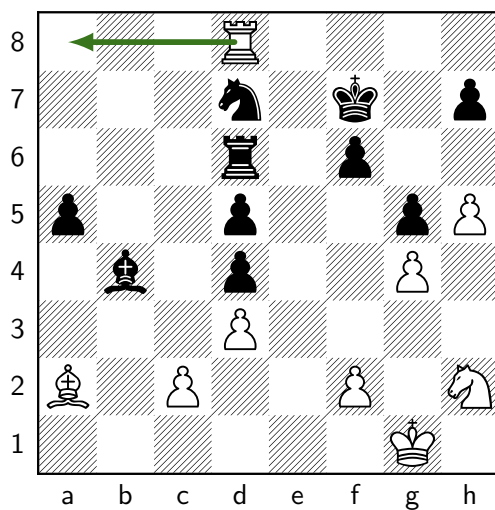


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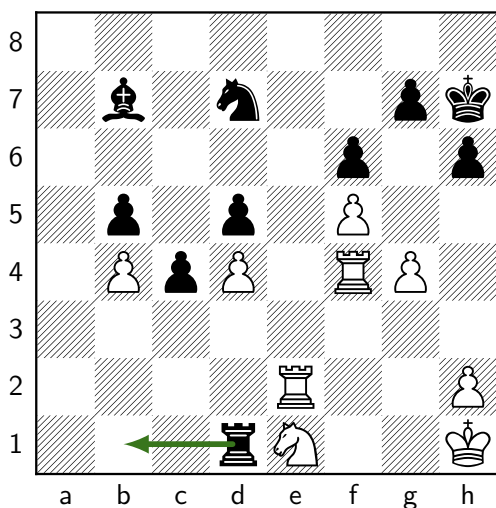
Figure 41: **Move 47, Black: Rd1.** (a) Our position; Melanie plays Rd1 (d8d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#905` (ply 87) [structural distance  $d=50$ , QAP obj=37]: the master played Ra2; under  $\mu$  our Rd1 plays the role of that game's Ra1; cross-type roles: our Nb6 as their rd4, our nb1 as their pb3; transferred move d8d1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1372` (ply 88) [structural distance  $d=58$ , QAP obj=17]: the master played Rcc8; under  $\mu$  our Rd1 plays the role of that game's Rc2; cross-type roles: our Rc7 as their Pa2, our Nb6 as their Rb8; transferred move d8d1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2820` (ply 83) [structural distance  $d=59$ , QAP obj=25]: the master played Kg7; under  $\mu$  our Kg1 plays the role of that game's Kh2; cross-type roles: our Nb6 as their Re7, our Nb6 as their Ra4; transferred move g8g7. Synthesis: 2 retrieved analogs transferred to Rd1 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



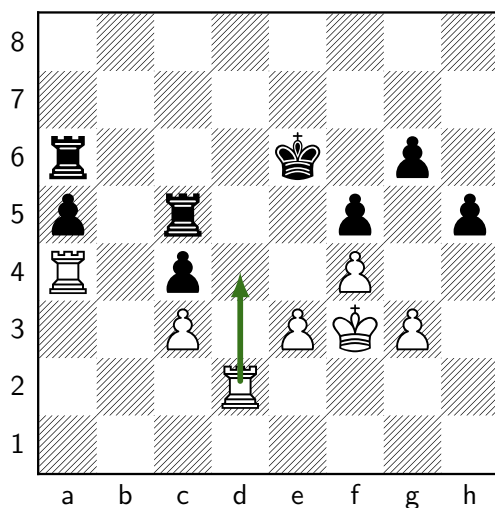
(a)



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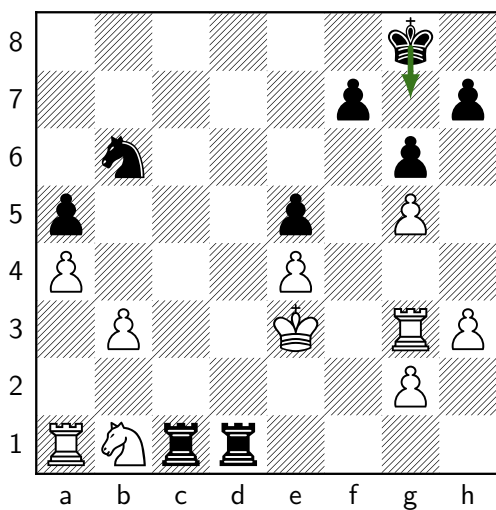


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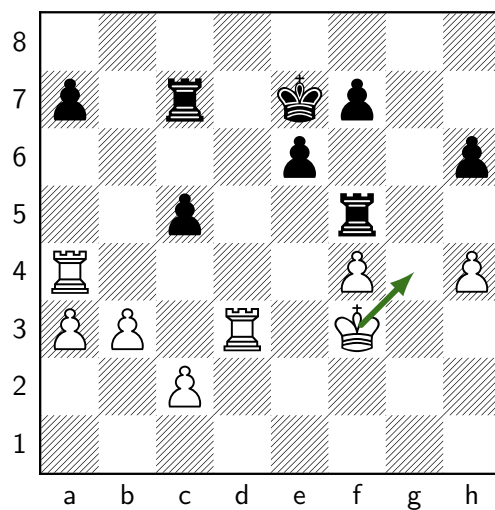


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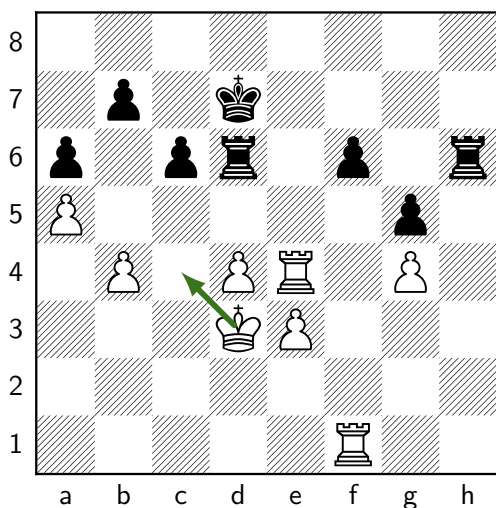
Figure 42: **Move 48, Black: Rcc1.** (a) Our position; Melanie plays Rcc1 (c7c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0001#1107 (ply 102) [structural distance  $d=63$ , QAP obj=25]: the master played Ra8; under  $\mu$  our Rd8 plays the role of that game’s Rd8; cross-type roles: our Nb6 as their Ba2, our Ph7 as their Nh2; transferred move d1h1. (c) Retrieved from ccr1\_top\_p0002#666 (ply 105) [structural distance  $d=49$ , QAP obj=23]: the master played Rb1; under  $\mu$  our Rd8 plays the role of that game’s Rd8; cross-type roles: our Ph7 as their Bb7, our Pf7 as their Nd7; transferred move d1h1. (d) Retrieved from ccr1\_top\_p0001#2692 (ply 84) [structural distance  $d=49$ , QAP obj=18]: the master played Rd4; under  $\mu$  our Pf2 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our nb1 as their pa5, our Nb6 as their ph5; transferred move f7f5. Synthesis: no retrieved analog transferred to Rcc1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



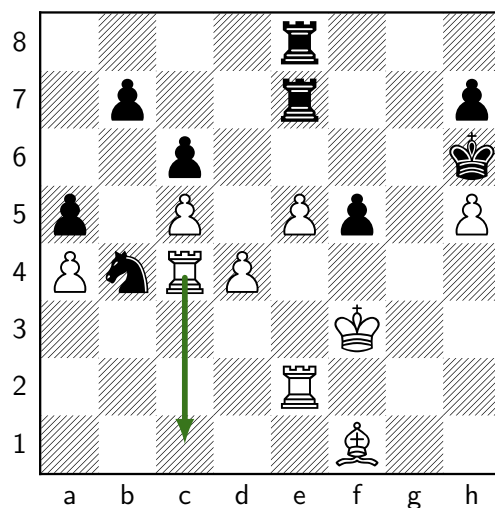
(a)



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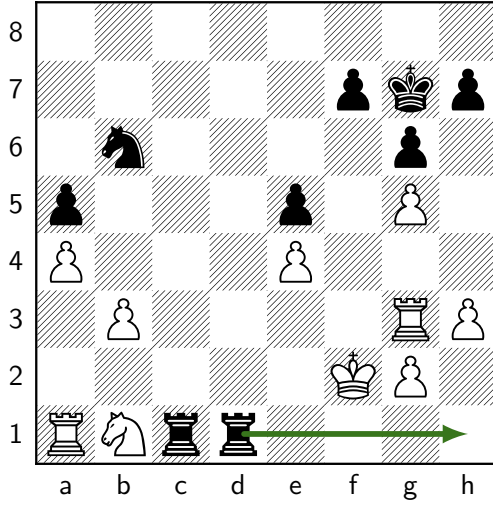
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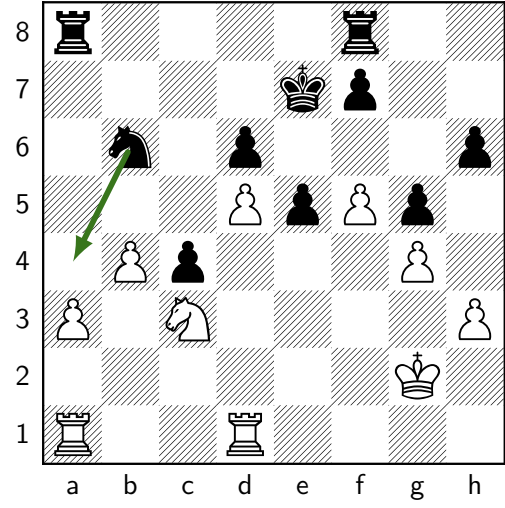
(d)

Figure 43: **Move 49, Black: Kg7.** (a) Our position; Melanie plays Kg7 (g8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3406` (ply 84) [structural distance  $d=59$ , QAP obj=19]: the master played Kg4; under  $\mu$  our Kg1 plays the role of that game’s Kf3; cross-type roles: our Pe5 as their Rd3, our nb1 as their Pf4; transferred move g8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2585` (ply 112) [structural distance  $d=59$ , QAP obj=6]: the master played Kc4; under  $\mu$  our Kg1 plays the role of that game’s Kd3; cross-type roles: our ph3 as their Rf1, our nb1 as their Pd4; transferred move g8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3038` (ply 100) [structural distance  $d=55$ , QAP obj=22]: the master played Rc1; under  $\mu$  our Rc8 plays the role of that game’s Rc4; cross-type roles: our Ph7 as their Bf1, our Pa5 as their Re2; transferred move c1c7. Synthesis: 2 retrieved analogs transferred to Kg7 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

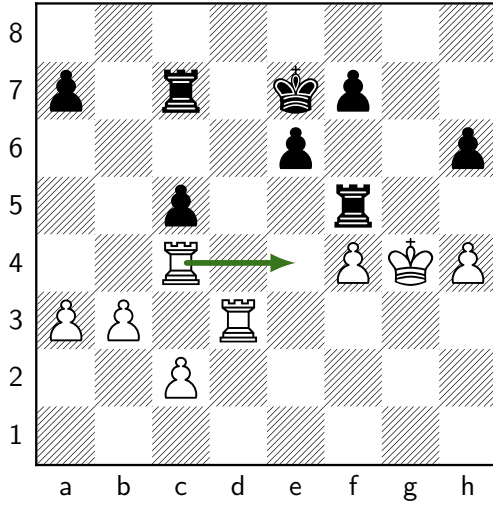




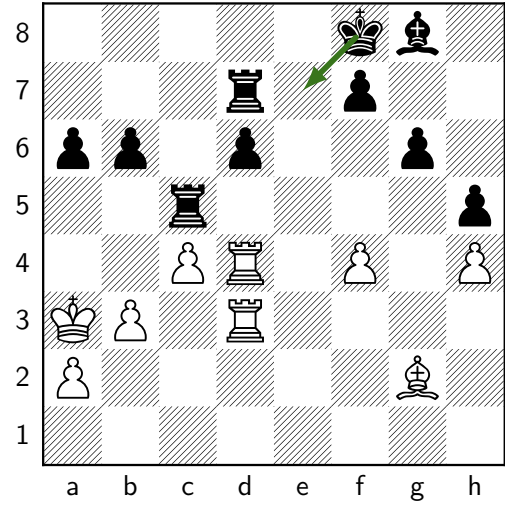
(a)



(b)

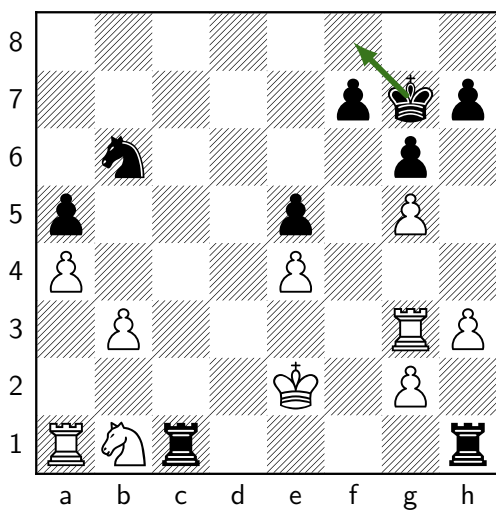


(c)

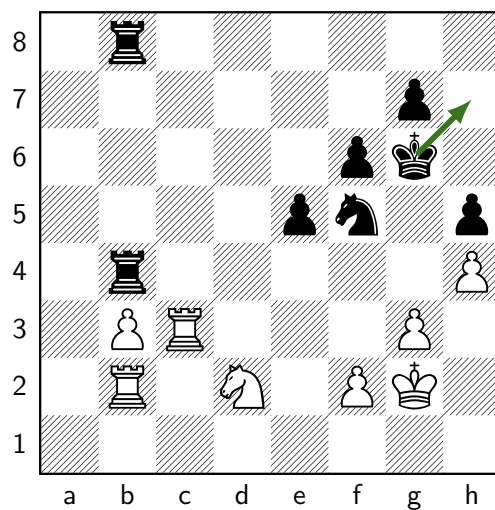


(d)

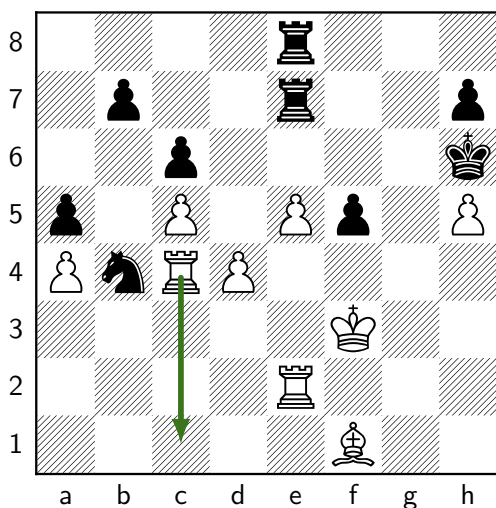
Figure 44: **Move 50, Black: Rh1**. (a) Our position; Melanie plays Rh1 (d1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#26` (ply 99) [structural distance  $d=62$ , QAP obj=4]: the master played Na4; under  $\mu$  our Rd8 plays the role of that game’s Nb3 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Rf8, our Nb6 as their Pg5; transferred move d1h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3406` (ply 88) [structural distance  $d=50$ , QAP obj=11]: the master played Re4; under  $\mu$  our Rc8 plays the role of that game’s Rc4; cross-type roles: our Nb6 as their Pb3, our nb1 as their pc5; transferred move c1c5. (d) Retrieved from `ccr1_top_p0002#2329` (ply 95) [structural distance  $d=62$ , QAP obj=10]: the master played Ke7; under  $\mu$  our Kg2 plays the role of that game’s Kf1; cross-type roles: our pe4 as their Bg8, our pb3 as their Rd7; transferred move g7g8. Synthesis: 1 retrieved analog transferred to Rh1 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



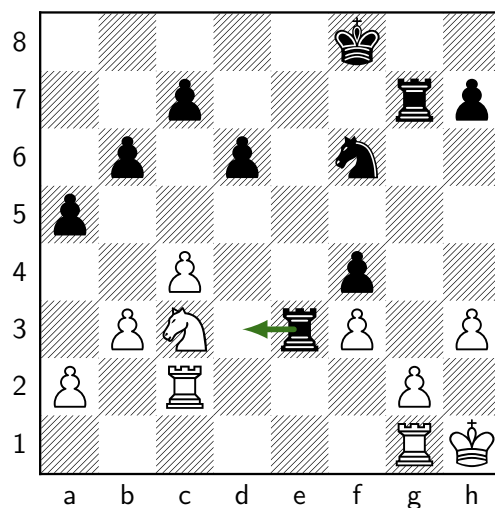
(a)



(b)

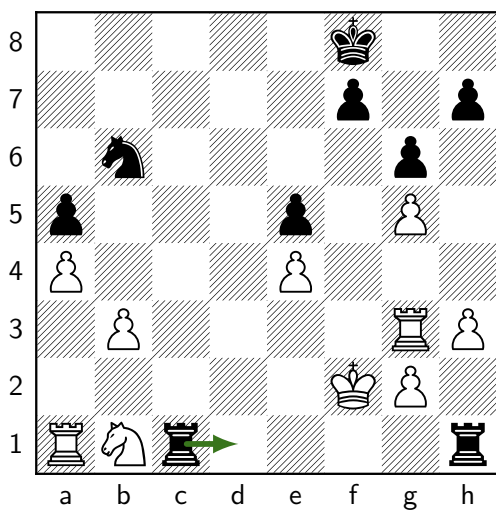


(c)

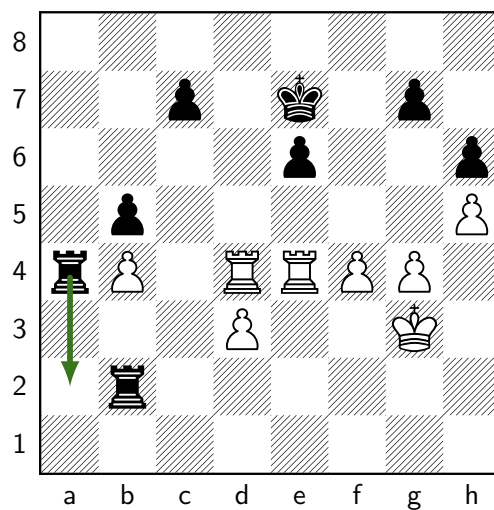


(d)

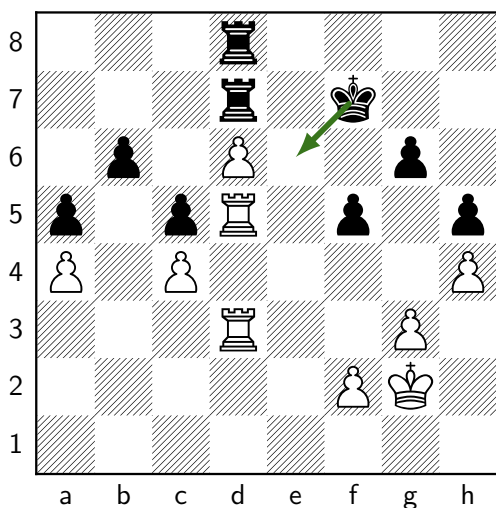
Figure 45: **Move 51, Black: Kf8**. (a) Our position; Melanie plays Kf8 (g7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2408` (ply 93) [structural distance  $d=64$ , QAP obj=24]: the master played Kh7; under  $\mu$  our Kg2 plays the role of that game’s Kg3; cross-type roles: our nb1 as their pb3, our pg2 as their rc3; transferred move g7f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3038` (ply 100) [structural distance  $d=65$ , QAP obj=23]: the master played Rc1; under  $\mu$  our Rc8 plays the role of that game’s Rc4; cross-type roles: our Pf7 as their Bf1, our Nb6 as their Pa4; transferred move c1c7. (d) Retrieved from `ccrl_top_p0001#833` (ply 117) [structural distance  $d=51$ , QAP obj=14]: the master played Rd3; under  $\mu$  our Rc8 plays the role of that game’s Re6; cross-type roles: our pg5 as their Rg7, our Nb6 as their Ph7; transferred move c1c5. Synthesis: 1 retrieved analog transferred to Kf8 (summed QAP objective 24, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



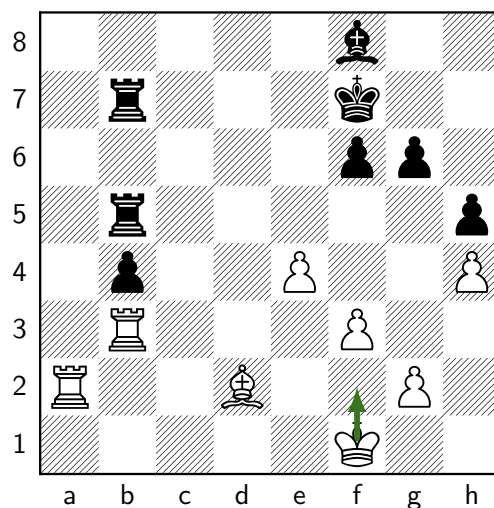
(a)



(b)

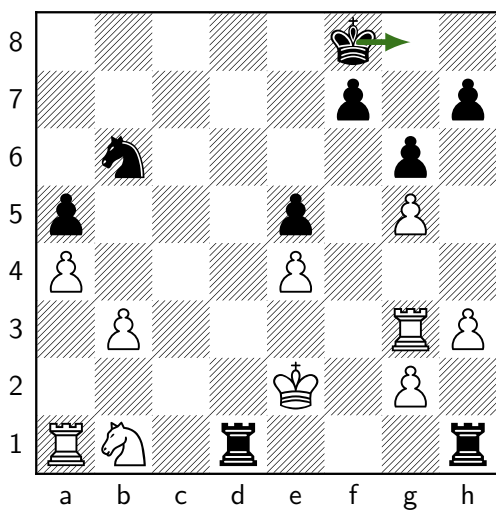


(c)

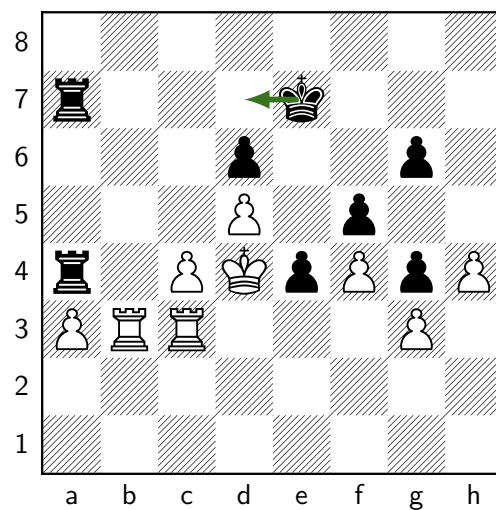


(d)

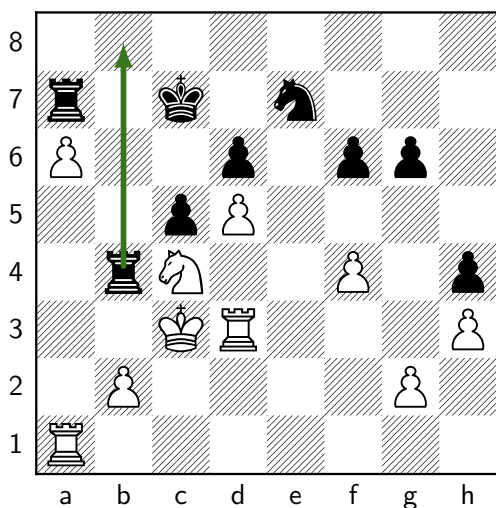
Figure 46: **Move 52, Black: Rcd1.** (a) Our position; Melanie plays Rcd1 (c1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#785` (ply 107) [structural distance  $d=60$ , QAP obj=20]: the master played Raa2; under  $\mu$  our Rc8 plays the role of that game’s Ra5; cross-type roles: our Rh1 as their Pb5, our nb1 as their pb4; transferred move c1d1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#634` (ply 91) [structural distance  $d=65$ , QAP obj=23]: the master played Ke6; under  $\mu$  our Kf1 plays the role of that game’s Kf2; cross-type roles: our nb1 as their pd6, our Nb6 as their Pc5; transferred move f8g7. (d) Retrieved from `ccrl_top_p0000#3728` (ply 88) [structural distance  $d=68$ , QAP obj=21]: the master played Kf2; under  $\mu$  our Kf1 plays the role of that game’s Kf1; cross-type roles: our Nb6 as their Bd2, our nb1 as their pb4; transferred move f8g7. Synthesis: 1 retrieved analog transferred to Rcd1 (summed QAP objective 20, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



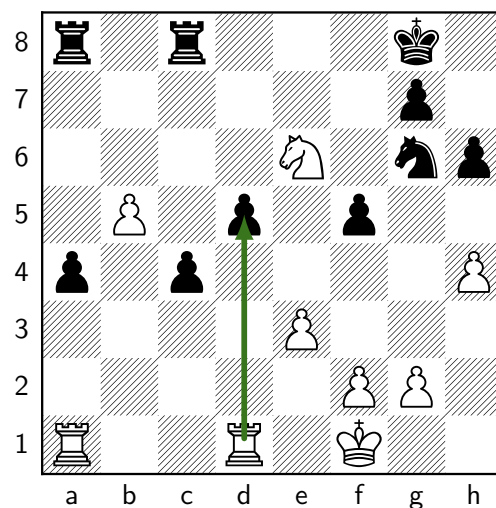
(a)



(b)

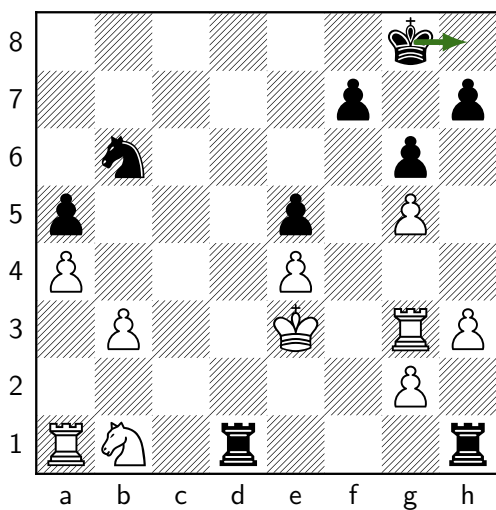


(c)

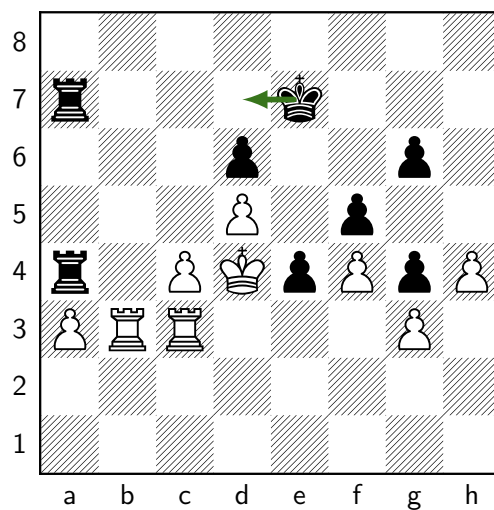


(d)

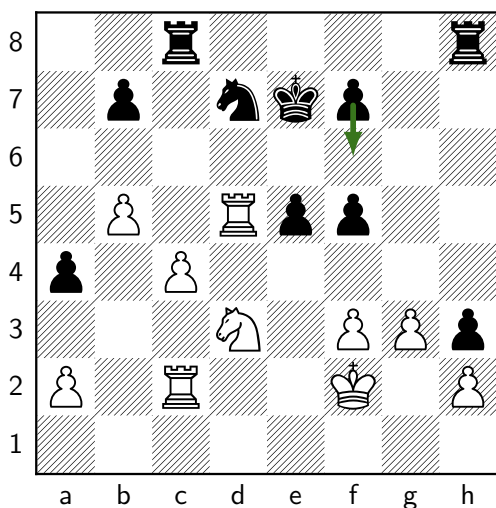
Figure 47: **Move 53, Black: Kg8.** (a) Our position; Melanie plays Kg8 (f8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#168` (ply 113) [structural distance  $d=71$ , QAP obj=14]: the master played Kd7; under  $\mu$  our Kf1 plays the role of that game’s Ke2; cross-type roles: our nb1 as their pc4, our Nb6 as their ph4; transferred move f8g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#852` (ply 105) [structural distance  $d=70$ , QAP obj=38]: the master played Rb8; under  $\mu$  our Rd8 plays the role of that game’s Rb5; cross-type roles: our Pg6 as their Ne7, our nb1 as their pa6; transferred move d1d7. (d) Retrieved from `ccrl_top_p0002#118` (ply 98) [structural distance  $d=57$ , QAP obj=12]: the master played Rxd5; under  $\mu$  our Rh8 plays the role of that game’s Rd1; cross-type roles: our Nb6 as their Pg2, our nb1 as their pa4; transferred move h1h2. Synthesis: 1 retrieved analog transferred to Kg8 (summed QAP objective 14, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



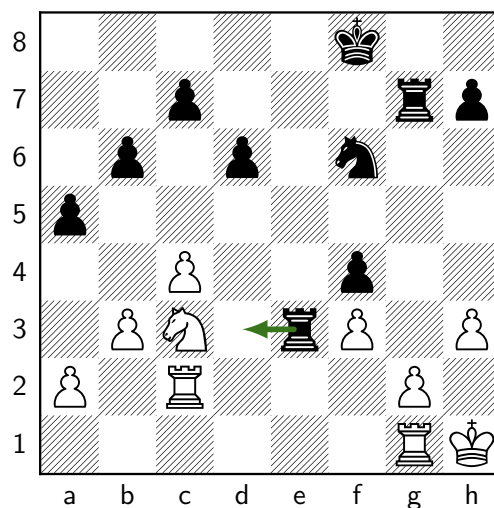
(a)



(b)

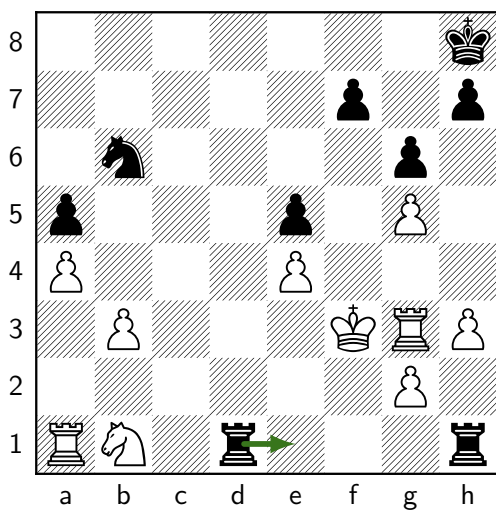


(c)

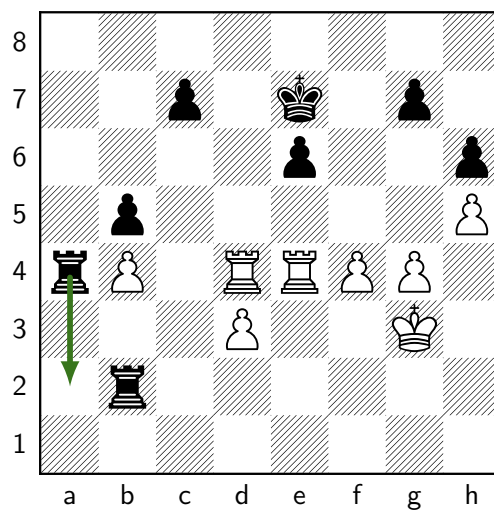


(d)

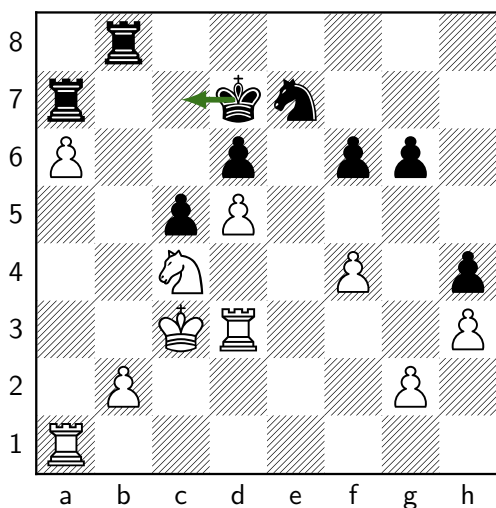
Figure 48: **Move 54, Black: Kh8**. (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#168` (ply 113) [structural distance  $d=58$ , QAP obj=30]: the master played Kd7; under  $\mu$  our Kg1 plays the role of that game’s Ke2; cross-type roles: our nb1 as their pc4, our nb1 as their pa3; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3151` (ply 93) [structural distance  $d=51$ , QAP obj=31]: the master played f6; under  $\mu$  our Pf2 plays the role of that game’s Pf2; cross-type roles: our ph3 as their Rc8, our pg5 as their Rh8; transferred move f7f5. (d) Retrieved from `ccrl_top_p0001#833` (ply 117) [structural distance  $d=47$ , QAP obj=30]: the master played Rd3; under  $\mu$  our Rd8 plays the role of that game’s Re6; cross-type roles: our Pe5 as their Rg7, our Rh1 as their Pf4; transferred move d1f1. Synthesis: 1 retrieved analog transferred to Kh8 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



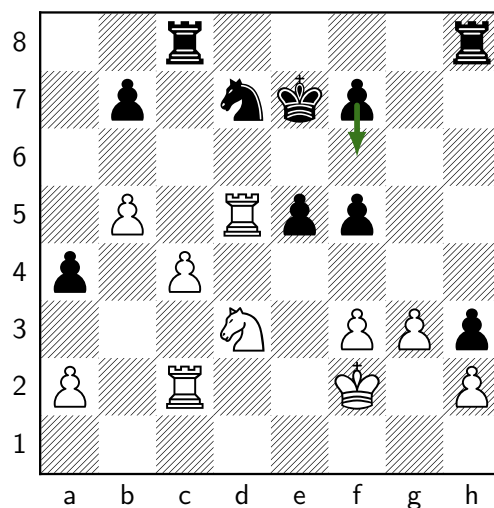
(a)



(b)

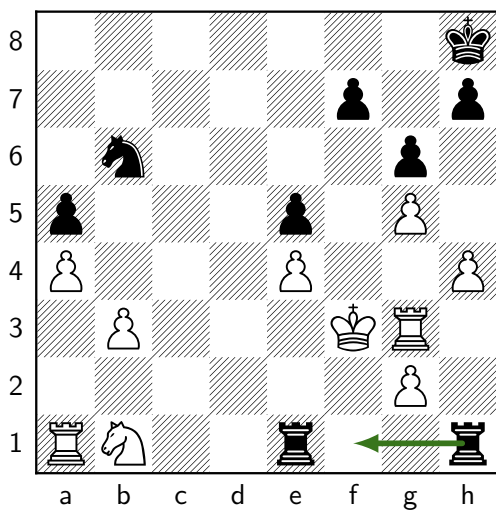


(c)

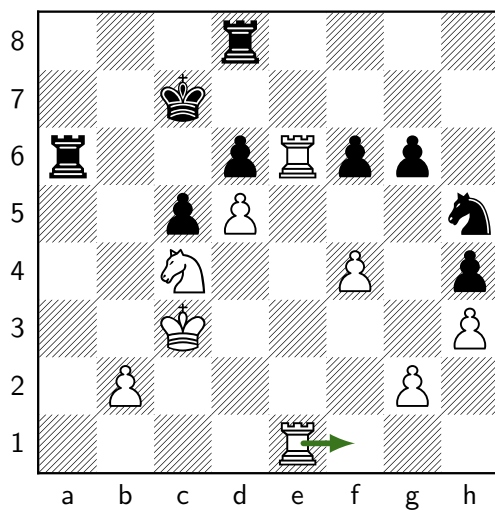


(d)

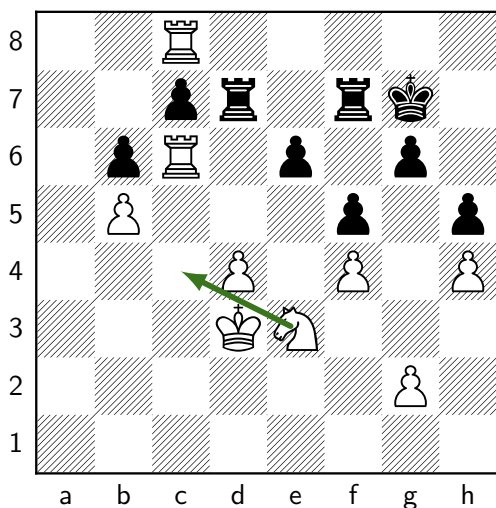
Figure 49: **Move 55, Black: Rde1.** (a) Our position; Melanie plays Rde1 (d1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#785` (ply 107) [structural distance  $d=66$ , QAP obj=40]: the master played Raa2; under  $\mu$  our Rd8 plays the role of that game’s Ra5; cross-type roles: our Rh1 as their Pb5, our nb1 as their pb4; transferred move d1e1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#852` (ply 109) [structural distance  $d=57$ , QAP obj=40]: the master played Kc7; under  $\mu$  our Kh1 plays the role of that game’s Kd2; cross-type roles: our pg5 as their Rb8, our Rh1 as their Ne7; transferred move h8g8. (d) Retrieved from `ccrl_top_p0001#3151` (ply 93) [structural distance  $d=56$ , QAP obj=33]: the master played f6; under  $\mu$  our Pf2 plays the role of that game’s Pf2; cross-type roles: our pb3 as their Rh8, our nb1 as their Pe5; transferred move f7f5. Synthesis: 1 retrieved analog transferred to Rde1 (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



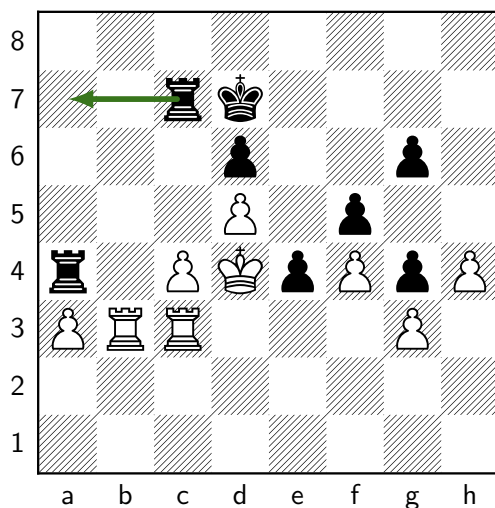
(a)



(b)

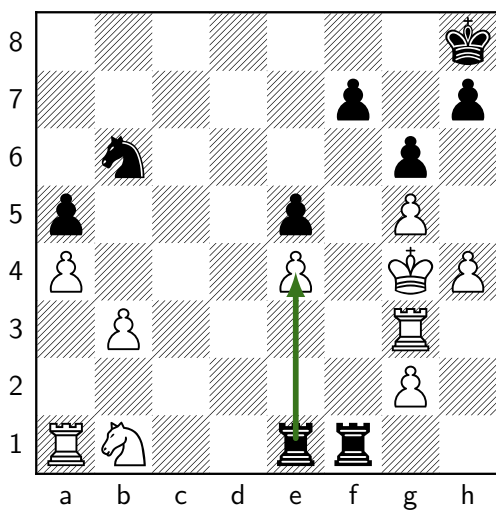


(c)

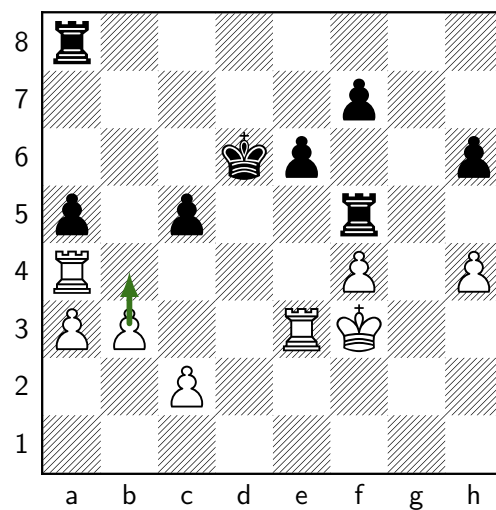


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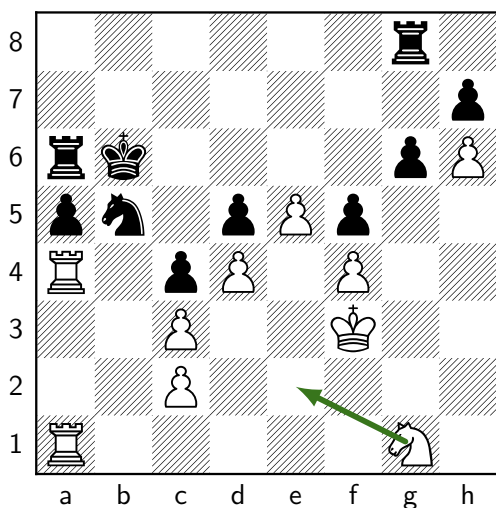
Figure 50: **Move 56, Black: Rhf1+.** (a) Our position; Melanie plays Rhf1+ (h1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#852` (ply 124) [structural distance  $d=61$ , QAP obj=74]: the master played Rf1; under  $\mu$  our Rh8 plays the role of that game’s Re1; cross-type roles: our Re1 as their Nc4, our nb1 as their pd6; transferred move h1h4. (c) Retrieved from `ccr1_top_p0000#1642` (ply 102) [structural distance  $d=61$ , QAP obj=70]: the master played Nc4; under  $\mu$  our Nb3 plays the role of that game’s Ne3; cross-type roles: our ra1 as their ph5, our nb1 as their pb6; transferred move b6d5. (d) Retrieved from `ccr1_top_p0001#168` (ply 117) [structural distance  $d=51$ , QAP obj=46]: the master played Rca7; under  $\mu$  our Re8 plays the role of that game’s Rc2; cross-type roles: our nb1 as their pc4, our Nb6 as their ph4; transferred move e1g1. Synthesis: no retrieved analog transferred to Rhf1+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



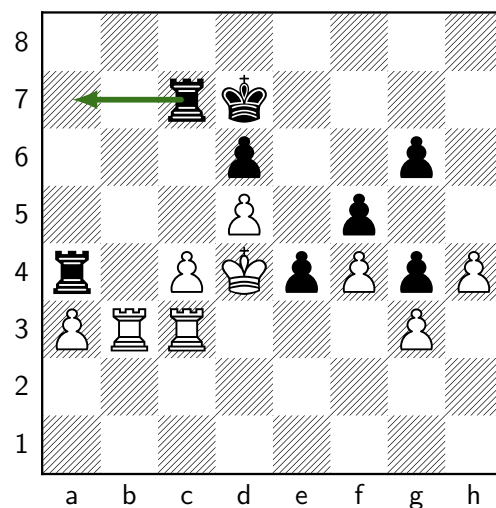
(a)



(b)



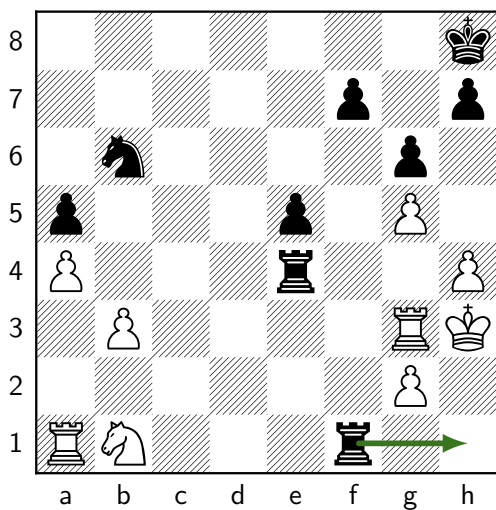
(c)



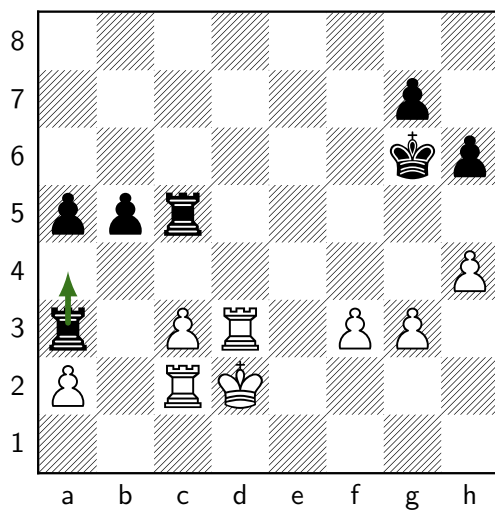
(d)

Figure 51: **Move 57, Black: Rxe4+.** (a) Our position; Melanie plays Rxe4+ (e1e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3406` (ply 96) [structural distance  $d=60$ , QAP obj=74]: the master played b4; under  $\mu$  our *Pf2* plays the role of that game’s *Pb3*; cross-type roles: our *Nb6* as their *Re3*, our *nb1* as their *Pf4*; transferred move f7f6. (c) Retrieved from `ccr1_top_p0002#3228` (ply 122) [structural distance  $d=58$ , QAP obj=67]: the master played Ne2; under  $\mu$  our *Pf2* plays the role of that game’s *Ng1* (a **cross-type** correspondence); cross-type roles: our *nb1* as their *pa5*; transferred move f7f6. (d) Retrieved from `ccr1_top_p0001#168` (ply 117) [structural distance  $d=57$ , QAP obj=46]: the master played Rca7; under  $\mu$  our *Re8* plays the role of that game’s *Rc2*; cross-type roles: our *Nb6* as their *Pf5*, our *nb1* as their *pc4*; transferred move e1d1. Synthesis: no retrieved analog transferred to Rxe4+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

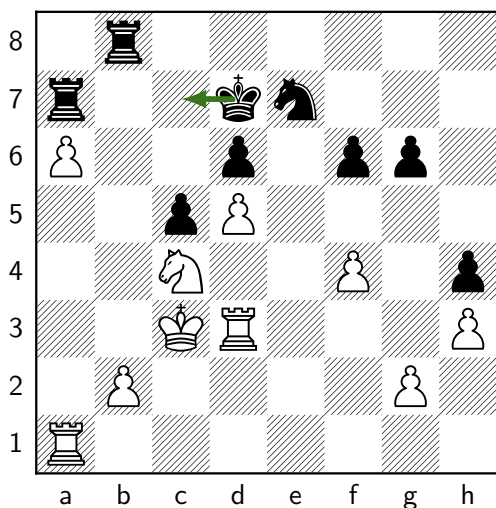




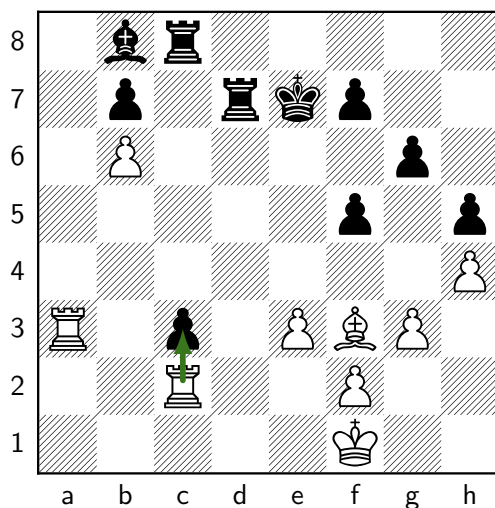
(a)



(b)



(c)



(d)

Figure 52: **Move 58, Black: Rh1#.** (a) Our position; Melanie plays Rh1# (f1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#2603 (ply 99) [structural distance  $d=62$ , QAP obj=57]: the master played Ra4; under  $\mu$  our Rf8 plays the role of that game’s Ra6; cross-type roles: our Nb6 as their Pg7, our nb1 as their pc3; transferred move f1f2. (c) Retrieved from ccr1\_top\_p0003#852 (ply 109) [structural distance  $d=58$ , QAP obj=49]: the master played Kc7; under  $\mu$  our Kh1 plays the role of that game’s Kd2; cross-type roles: our Pg6 as their Rb8, our Re4 as their Ne7; transferred move h8g8. (d) Retrieved from ccr1\_top\_p0002#1673 (ply 104) [structural distance  $d=68$ , QAP obj=49]: the master played Rxc3; under  $\mu$  our Rf8 plays the role of that game’s Rc2; cross-type roles: our nb1 as their pc3, our Nb6 as their Bf3; transferred move f1f6. Synthesis: no retrieved analog transferred to Rh1# – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.